SAFETY EVALUATION OF THE 10 MeV LINAC BUNKER AT RSUP Dr. HASAN SADIKIN USING THE OpenMC SIMULATION METHOD

UNDERGRADUATE THESIS (SKRIPSI)

Submitted in partial fulfillment of the requirements for the Degree of Bachelor in the Nuclear Engineering Study Program

Submitted by **MUHAMMAD ARYA HANIF** 19/443954/TK/49150

To the DEPARTMENT OF NUCLEAR ENGINEERING AND ENGINEERING PHYSICS FACULTY OF ENGINEERING UNIVERSITAS GADJAH MADA YOGYAKARTA 2024

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This work is dedicated to my late grandfather.

"We play with the cards we're dealt." - Stan Edgar

PREFACE (Acknowledgments)

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Muhammad Arya Hanif

LIST OF SYMBOLS AND ABBREVIATIONS

Roman Symbols

Symbol	Quantity	Unit
d_sec	Distance from isocenter to measurement point	cm
$\dot{D}$	Dose Rate	Sieverts/hour
Da	Dose per fluence	Sieverts cm^2
$\dot{D}max$	Maximum Dose Rate	Sieverts/hour
Do	Output Dose of the OpenMC function	Sieverts
Dosk	Output Dose of the Calibration Cell	Sieverts
$\dot{D}$	Dose Rate of the Linear Accelerator	MU/minute
DU	Dose based on angular utility	μSv/week
E	Energy	MeV
HVL	Half Value Layer	mm
S	Conversion Factor	-
SR	Source Rate	sources/ second
T	Occupancy Factor	-
TVL	Tenth Value Layer	cm
T	Time	seconds
tw	Irradiation time to reach the workload	hours
U	Utility Factor	%
V	Volume	cm^3

Symbol	Quantity	Unit
W	Workload	Gy/week
X	Material thickness	m

Greek Symbols

Symbol	Quantity	Unit
Φ	Simulation flux	Particle-cm / source

Subscripts

Symbol	Description
bo	Output
maks	Maximum
Sk	Calibration Cell

Abbreviations

Abbreviation	Meaning
AAPM TG	American Association of Physicists in Medicine Task Group
AC	Alternating Current
BAPETEN	Nuclear Energy Regulatory Agency (Badan Pengawas Tenaga Nuklir)
BNCT	Boron Neutron Capture Therapy
BPE	Boron Loaded Polyethylene
CPRG	Computational Reactor Physics Group
CSEWG	Cross Section Evaluation Working Group
Dr.	Doctor
ENDF	Evaluated Nuclear Data Library
FF	Flattening Filter
HVL	Half Value Layer
IAEA	International Atomic Energy Agency
KeV	Kilo electron Volt
LINAC	Linear Accelerator
MIT	Massachusetts Institute of Technology
MeV	Mega electron Volt
MV	Mega Volt

Abbreviation	Meaning
MU	Monitor Unit
NCRP	National Council on Radiation Protection & Measurements
NBD	Dose Limit Value (Nilai Batas Dosis)
RSUP	Central General Hospital (Rumah Sakit Umum Pusat)
SAD	Source to Axis Distance
TVL	Tenth Value Layer
UGM	Universitas Gadjah Mada

ABSTRACT (translated from the Indonesian INTISARI)

SAFETY EVALUATION OF THE 10 MeV LINAC BUNKER AT RSUP Dr. HASAN SADIKIN USING THE OpenMC SIMULATION METHOD

Muhammad Arya Hanif 19/443954/TK/49150

Submitted to the Department of Nuclear Engineering and Engineering Physics, Faculty of Engineering, Universitas Gadjah Mada on 12 July 2024, in partial fulfillment of the requirements for the Degree of Bachelor in the Nuclear Engineering Study Program.

This research was conducted to evaluate the safety of the LINAC Elekta Synergy radiotherapy bunker design built at RSUP Dr. Hasan Sadikin. Visualization of the dose distribution was performed to help interpret the dose distribution in the LINAC bunker.

The evaluation was carried out using Monte Carlo simulation through the OpenMC particle transport code version 0.13.1. Dose-rate measurements were performed at varying distances of 30, 100, and 200 cm from the bunker wall. The occupancy, utility, and workload factors were used to determine the applicable dose limit values at those points.

The dose rate around the room ranged from 8.28×10^{-3} to 0.208 μSv/week. This value is far smaller than the dose constraint enforced by BAPETEN in Regulation (Perka) No. 4 of 2013. This means that the bunker's safety already complies with the applicable standard. Dose-rate visualization was also carried out to help interpret the dose occurring at each dose-rate measurement point around the bunker.

Keywords: LINAC, OpenMC, Monte Carlo Simulation, Radiation Protection

Principal Supervisor: Ir. Anung Muharini, M.T., IPM. Co-Supervisor: Anisza Okselia, S.T., M.Si.

ABSTRACT (original English text)

SAFETY EVALUATION OF HASAN SADIKIN HOSPITAL'S 10 MeV LINAC BUNKER USING OpenMC SIMULATION METHOD

Muhammad Arya Hanif 19/443954/TK/49150

Submitted to the Department of Nuclear Engineering and Engineering Physics, Faculty of Engineering, Universitas Gadjah Mada on July 12, 2024, in partial fulfillment of the requirement for the Degree of Bachelor of Engineering in Nuclear Engineering.

This study will examine the safety of the LINAC Elekta Synergy bunker room at RSUP Dr. Hasan Sadikin. The safety examination will be conducted by comparing the radiation dose in the surrounding rooms with the dose limit enforced by BAPETEN. Visualization of the dose will be done to help the interpretation of the dose spread occurring in the bunker.

The examination is executed using Monte Carlo simulation provided by version 0.13.1 of the OpenMC particle transport code. This research was carried out by evaluating the surrounding rooms' dose at varying distances of 30, 100, and 200 cm to the bunker wall. Occupancy, utility, and workload factors will be used to determine the dose limit values that apply at each of these points.

The dose surrounding the room is detected in the range of 8.28×10^{-3} to 0.208 μSv/week. This is already much smaller than the implemented rule enforced by BAPETEN in Perka No. 3 of 2013. This means that the safety of the design is in accordance with the standard. Dose-rate visualization has also been carried out to help interpret the dose that occurs at each dose-rate measurement point around the bunker.

Keywords: LINAC, OpenMC, Monte Carlo Simulation, Radiation Protection

Supervisor: Ir. Anung Muharini, M.T., IPM. Co-supervisor: Anisza Okselia, S.T., M.Si.

CHAPTER I. INTRODUCTION

I.1. Background

Radiotherapy is one of the three main methods for treating cancer. The three main methods are surgery as the physical method, chemotherapy as the chemical method, and radiotherapy as the physics-based method. There are 66 radiotherapy units in 44 cancer-treatment centers spread across 16 provinces in Indonesia [1]. This means there was a radiotherapy-unit density of 0.247 per million Indonesian inhabitants in that year. This is quite far from the target of 7 radiotherapy units per million inhabitants recommended by the International Atomic Energy Agency (IAEA) [2]. It also means that a large number of radiotherapy bunkers must be built to reach that target.

A radiotherapy bunker is a room intended to protect the general public and hospital workers from the radiation emitted by a radiotherapy facility. The Linear Accelerator (LINAC) Synergy bunker at RSUP Dr. Hasan Sadikin has gone through a series of design and verification stages before becoming operational. The regulation that must be considered is Article 41 of BAPETEN Chairman Regulation No. 4 of 2013 concerning dose constraints for radiation workers and the public [3]. Then, to meet that standard, nuclear organizations such as the IAEA and the National Council on Radiation Protection & Measurements (NCRP) have developed calculation methods to satisfy the previously established standards [4]. In addition, there is the NCRP Report No. 151 document, which serves as a guide for building radiotherapy safety facilities [5]. This regulation was created to facilitate the optimization of benefit against risk in the use of radiotherapy facilities. The optimization method can also be carried out using computer simulation to ensure that the dose rate around the bunker room complies with the applicable standard. A radiotherapy bunker design formed based on that regulation can then be checked using analytical calculations or dose-rate measurements after installation.

Analytical calculations are performed at points behind the constructed bunker wall. The wall thickness between the LINAC and that point is then compared with the minimum wall-thickness value of the bunker so that the dose rate can be lower than the dose constraint. After the bunker is built and the LINAC is installed in the room, irradiation at maximum dose is performed so that the dose rate in the surrounding buildings can be checked. Based on this examination, the dose rate on the bunker roof exceeded 5 μSv/hour. This is assessed as greater than the dose constraint set by BAPETEN in Perka No. 3 of 2013, which is 200 μSv/week. Moreover, the measurement results do not always show a lower dose rate at greater distances. This contradicts the inverse square law applied in the dose-rate calculation. Interpretation of that dose rate can be carried out by visualizing the dose-rate distribution in the room using computer simulation.

Computer simulation is a method that can be used to design instruments and the supporting facilities for the use of instruments that involve radiation. This is because building a prototype often involves components that are not inexpensive to make. In addition, the probabilistic and invisible nature of radiation makes checking parameters during instrument fabrication not as easy as for instruments with visible parameters. This makes simulation, with its ability to visualize radiation, capable of providing more complete data and flexibility in managing the produced data. Simulation can be used for facility planning, design research, parameter optimization, and so on [6]. One advantage of simulation is the visualization of dose distribution and particle movement. That advantage is something that neither analytical calculation nor direct measurement possesses. Despite this advantage, simulation accuracy depends heavily on how detailed the geometry and parameters defined in the simulation are. The visualization produced by the simulation can help interpret the dose distribution that occurs in a radiotherapy facility.

I.2. Problem Formulation

After the bunker design is created based on the existing regulation, the design will be simulated to determine the estimated dose received in each room around the LINAC. Based on that dose rate, it is possible to estimate how much dose is received by workers each week. Based on the above, the problem formulation to solve this matter is as follows:

1. What is the dose rate around the bunker room according to the OpenMC simulation?
2. What is the dose distribution in the LINAC bunker room?

3. Does the simulated dose rate meet the dose constraint?

I.3. Research Objectives

1. To analyze the LINAC dose distribution in the bunker.
2. To determine the safety of the LINAC bunker.

I.4. Research Benefits

To add input for the management of RSUP Dr. Hasan Sadikin for the radiation-safety management of health workers and the general public in the Synergy bunker room.

I.5. Scope and Limitations

There are several limitations to this research, among them:

1. The detector uses a simple detection tally and does not replicate the actual detector design.
2. Irradiation uses a simplified LINAC geometry model with an energy of 10 MeV.
3. The Monte Carlo simulation uses OpenMC version 0.13.1.
4. The bunker design studied is the LINAC Synergy bunker design at RSUP Dr. Hasan Sadikin.
5. The simulation uses 1,000,000,000 particles with 3 batches.
6. The study is not performed on the top side or the southern side of the bunker.

CHAPTER II. LITERATURE REVIEW

A LINAC bunker room requires different specifications compared with ordinary hospital rooms. This is because the primary radiation, secondary radiation, and the neutronics produced by high-energy photons require different thicknesses and materials compared with ordinary hospital rooms. Therefore, the placement and construction are quite costly. This makes the reuse of a radiotherapy bunker room cut significant costs. Despite such cost reduction, safety remains the priority in realizing a radiotherapy facility. Based on a study conducted by A.S. Ferdiansyah, the repurposing of a 6 MeV LINAC bunker to 10 MeV can be done by adding 2 cm of iron-oxide nanoparticles and boron carbide to the waiting room, roof, and archive room of the LINAC bunker at RSUP Dr. Sardjito. Iron oxide was chosen because it has high attenuation of fast neutrons, while boron carbide has high attenuation of thermal to epithermal neutron radiation [7].

Radiation-dose measurement outside the LINAC bunker room had also previously been performed by the vendor at RSUP Dr. Sardjito. That study found that the radiation dose did not reach 2 µSv/h across the entire area around the radiotherapy room, except in front of the door at 4.7 µSv/h. The measurement was performed at seven different points around the bunker room. Based on that research, in his thesis Y. M. Dinata suggested that the bunker door be thickened using lead [8].

The use of simulation to examine the dose rate around a hot-lab room has also been performed to study the economic factors of combinations of bunker wall materials. In his research, A. Rafi combined concrete, concrete with a 1 cm lead layer, and concrete with a 2 cm lead layer. The use of a lead layer can reduce the wall thickness by up to 5 cm at certain points. Lead lining can cost up to seven times more than using only concrete. This research concluded that the most optimal design in terms of radiation protection and economics is the hot-lab design without lead lining on the bunker wall [9].

Simulation has also been performed by N. K. T. Kusnaedi for designing a bunker room for a 30 MeV radiotherapy facility. That energy is produced by a cyclotron used in a Boron Neutron Capture Therapy (BNCT) facility. The maximum radiation-dose standard enforced by BAPETEN for radiotherapy facilities was successfully met in that bunker design. The required wall thickness was 100 cm to 350 cm of concrete and 100 cm to 250 cm of boron concrete and barite concrete [10].

OpenMC is one of the open-source codes used in designing radioactive facilities. Bunker safety evaluation using this photon and neutron transport simulation code had previously been performed on the bunker of Jogja International Hospital. In that study, the analytical and simulation results were compared. That research successfully determined that the bunker complied with BAPETEN safety standards based on both the OpenMC simulation and the analytical calculation. It was also found that the bunker required the addition of at least 92 millimeters of Boron Loaded Polyethylene (BPE) thickness [11].

Research to validate the credibility of Monte Carlo simulation results on a LINAC profile has also been performed by replicating Quality Assurance (QA) on a TrueBeam STx variant LINAC at Songklanagarind Hospital. This research, conducted by M.A. Efendi, used the PRIMO code to simulate the Percentage Depth Dose (PDD) and lateral profile using LINAC geometry. The simulation results were consistent with the experimental results, the Analytical Anisotropic Algorithm (AAA) calculation, and the golden beam provided by the manufacturer. PRIMO is a Monte Carlo code intended specifically for radiotherapy [12].

CHAPTER III. THEORETICAL BACKGROUND

III.1. Radiation

Radiation is the propagation of energy that does not require a medium or space and is subsequently absorbed by another object. There are several ways to classify radiation, one of which is by its ability to ionize other atoms. In this category, there are two types of radiation: ionizing radiation and non-ionizing radiation [13]. Ionizing energy is generally in the range of kilo-electron volts and mega-electron volts [14]. For particle radiation, ionizing radiation is radiation that carries sufficiently high energy. The type of particle and the energy are the main factors in radiation protection and the main determinants of the effect of radiation on a material [15].

III.1.1. Interaction of Photons with Matter

A photon is a discrete packet of electromagnetic energy that can interact with other particles such as atoms [16]. For photon radiation, the wavelength of ionizing radiation is roughly less than 10 nm [13]. There are many reactions that can result from the interaction of photons with matter. However, the three main interactions are the photoelectric effect, Compton scattering, and pair production.

III.1.1.1. Photoelectric Effect

The photoelectric effect occurs when a photon and an electron bound to an atom collide. The energy of that photon then makes the electron in the atom break free as a free electron [13]. This photoelectric interaction only occurs if the binding energy of that electron is equal to or smaller than the energy carried by the photon [17].

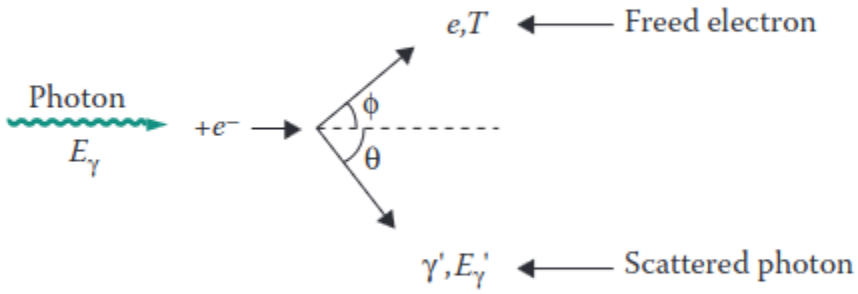

Figure 3.1. Illustration of the Photoelectric Effect [13]

III.1.1.2. Compton Scattering

Compton scattering occurs when a photon with energy in the keV range or higher strikes a free electron. A free electron has energy in the eV range. The electron that is struck will then be ejected in a different direction.

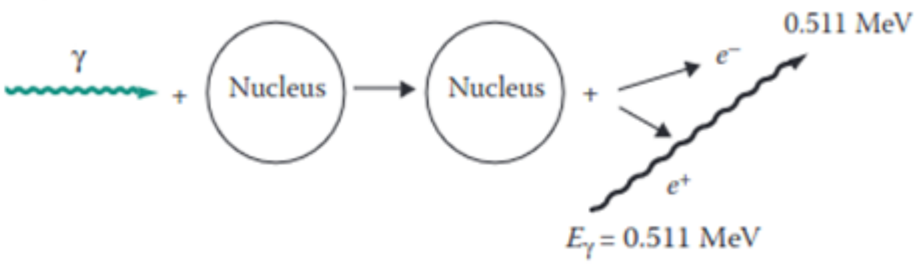

Figure 3.2. Illustration of Compton Scattering [13]

III.1.1.3. Pair Production

Pair production occurs when a high-energy photon strikes a nucleus. The energy of 1.022 MeV carried by that photon is then converted into an electron and a positron. The remaining energy is then distributed evenly between those two particles.

Figure III.1. Illustration of the Pair Production Effect [13]

III.1.1.4. Photodisintegration

Photodisintegration, or the photonuclear reaction, occurs when a nucleus captures a high-energy photon and ejects a neutron present in the nucleus [17]. The neutron produced can be absorbed by various materials, which then makes those materials radioactive. The photodisintegration reaction can only occur when the photon energy exceeds 7 MeV [15]. Photoneutrons usually have higher energy than thermal neutrons.

III.1.2. Interaction of Electrons with Matter

There are two types of photons that arise from the interaction of electrons with a material. An electron that loses energy due to the reduction of energy in the electromagnetic field of the nucleus releases the absorbed energy as a photon, which is called Bremsstrahlung [14]. The interaction that creates X-rays is generally the Bremsstrahlung interaction. The Bremsstrahlung reaction can also occur when that electron strikes an electron present in an atom. About 80% of the photon energy produced at 100 keV is

Bremsstrahlung radiation, and the remainder is characteristic X-rays. Meanwhile, for electrons on the order of MeV, the amount of Bremsstrahlung becomes so dominant that characteristic X-rays can be neglected. Characteristic X-rays occur when an electron strikes and dislodges another electron. The position of that electron is then taken over by another electron with a lower energy level, and the remaining energy is emitted as a characteristic X-ray.

III.1.3. Interaction of Neutrons with Matter

Neutrons can interact with a nucleus only through the nuclear force, because neutrons have no polarity [13]. When a neutron approaches a nucleus, the neutron cannot be influenced by the magnetic field of an atom. This makes nuclear interactions more likely to occur in the interaction of neutrons with matter compared with charged particles. There are two types of interactions that occur: scattering and absorption.

III.1.3.1. Scattering

In a scattering interaction, the neutron interacts with the nucleus. However, both particles reappear after the reaction occurs, with the same atomic number and mass number as before. There are two particle interactions that can occur: elastic and inelastic. In the elastic interaction, the kinetic energy of both particles is balanced without a change in the total kinetic energy. In the inelastic interaction, part of the kinetic energy is converted into excitation energy in the nucleus. That nucleus then undergoes de-excitation by releasing that energy as a photon.

III.1.3.2. Absorption

When absorption occurs, the neutron disappears. After the neutron disappears, one or more particles appear. The particles that can be produced include neutrons, protons, helium nuclei, gamma rays, and atoms different from the atom that was struck.

III.2. Dosimetry

Radiation dosimetry refers to the measurement, calculation, and estimation of the ionizing radiation received by an object, including the human body. This includes radiation that enters internally through ingestion or respiration, as well as externally, such as from a LINAC. The distinction of dose into absorbed dose and equivalent dose is needed because the biological effect on healthy human cells is strongly influenced by the type of radiation and the energy of that radiation.

III.2.1. Absorbed Dose

The energy absorbed by a mass during the use of radiation is measured by the quantity Gray (Gy). The energy absorbed by that mass is the energy delivered by the radiation. This value is also used in assessing the amount of energy received by a cancer. This is because the weighting factor of a particular radiation type for a cancer has a value different from the weighting factor for healthy tissue.

III.2.2. Equivalent Dose

Equivalent dose is the amount of radiation energy absorbed in healthy tissue. This value is commonly used in radiation protection of radiation workers in the use of radioactive facilities. The use of absorbed dose is done because radiation has different effects on human tissue depending on the type and energy of that radiation. This factor is called the weighting factor. To obtain the equivalent dose, the absorbed-dose value is multiplied by the weighting factor. Its international unit is still the same as for absorbed dose. The weighting factor is 1 for photon and electron radiation, 20 for alpha radiation, and 2-10 for neutrons depending on the amount of their energy [15].

III.3. Radiation Protection

Radiation protection is the action taken to reduce the influence of radiation exposure on radiation workers and the general public. In the use of a LINAC, the intensity of radiation is minimized through the process of attenuation [15]. The design of a radiotherapy bunker is guided by NCRP 151 and SRS 47. Those documents are used as guidelines in constructing the primary wall, secondary wall, and bunker door. Concepts such as the angular utility of the LINAC and occupancy are used to calculate the expected dose received by workers at the studied point.

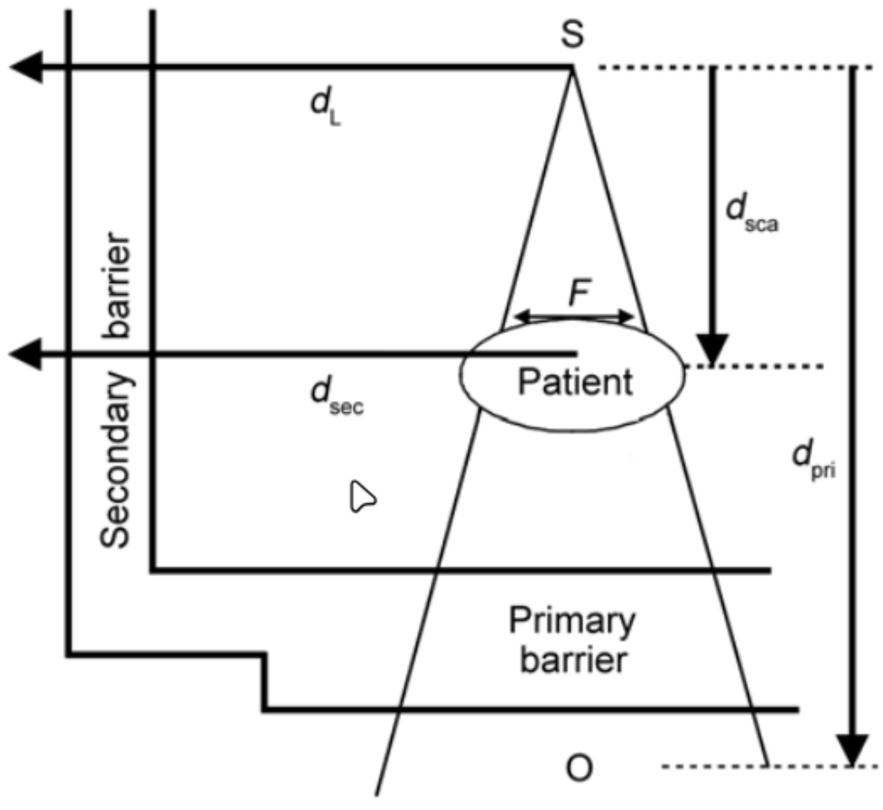

Figure III.2. Illustration of the types of walls and distances of a LINAC bunker room [5]

III.3.1. Dose Constraint (P)

The dose constraint is the upper limit of dose for radiation workers and members of the public. The dose constraint for radiation workers is half of the annual Dose Limit Value (NBD), or 10 mSv per year [3]. This value is then regulated in detail in BAPETEN Chairman Regulation No. 3 of 2013. In that regulation, the dose constraint for radiation workers is 0.2 mSv per week [18], while for the public it is 0.01 mSv per week. Those values are found by dividing the NBD into weeks, with 50 weeks per year. The dose constraint can then be achieved by considering the workload, occupancy factor, and utility factor of the LINAC operation.

III.3.2. Workload (W)

The workload, also called workload, is the total dose emitted by the LINAC during the irradiation fractions that are carried out. This value is obtained using the dose fraction per irradiation, the number of patients per day, and the number of working days of operating the instrument.

In the analytical calculation for the secondary wall, there is a value C that determines the workload of the treatment. This value is 4.5 for VMAT and IMRT, and 1 for conventional treatment as well as for Quality Assurance (QA).

III.3.3. Utility Factor (U)

The utility factor is the percentage of workload in which the beam is directed at a particular primary barrier. The smaller this value, the greater the tolerance of the dose-rate value at the point behind the wall.

Table III.1. LINAC Utility Factor [18]

Angle (90° interval)	U (%)
0° (bottom)	31.0
90° and 270°	21.3
180° (top)	26.3

III.3.4. Occupancy Factor (T)

The occupancy factor is the average amount of time a person is exposed to the LINAC when irradiation is performed. The greater the occupancy value, the greater the wall thickness required to reach the target attenuation value in the wall. The occupancy value is closely related to the categorization of the rooms around the bunker. Room categories in a nuclear facility are divided into two: controlled areas and uncontrolled areas [17]. All rooms around the LINAC Synergy bunker are controlled rooms accessible only by workers. Therefore, the applied dose constraint is the dose limit for radiation workers.

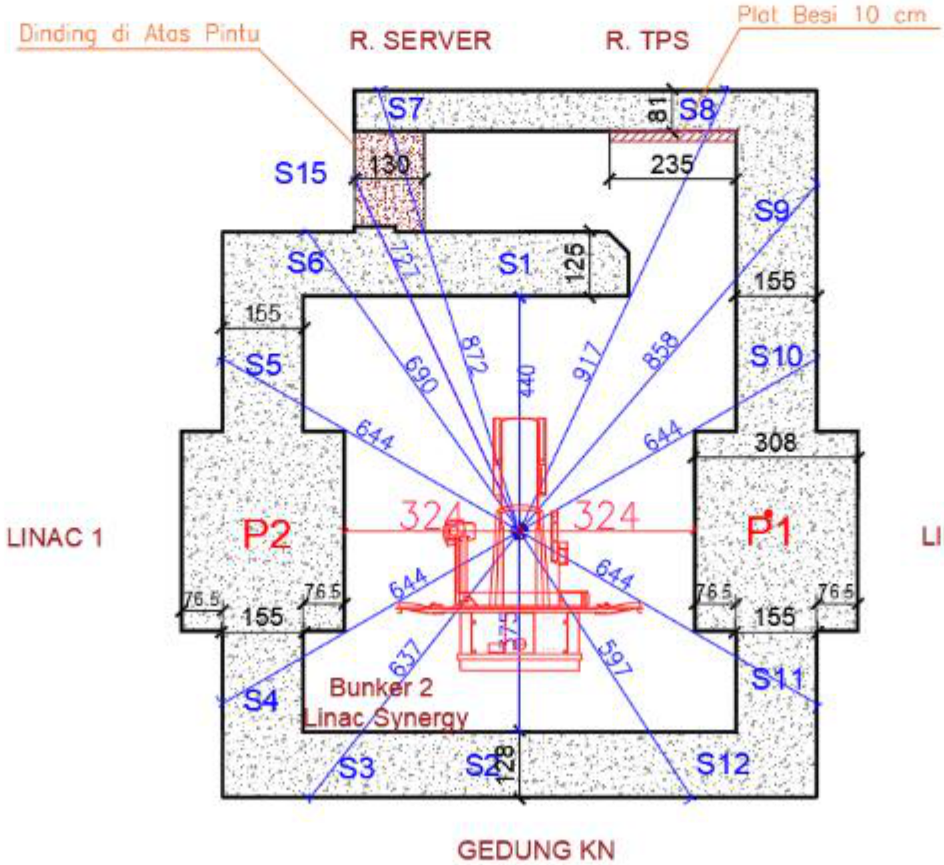

Figure III.3. Floor plan of the LINAC Synergy room and its surroundings [19]

Table III.2. Occupancy Factor for Rooms Around the Bunker [5]

Occupancy Factor Value	Type of Room
1	Treatment planning system room and operator room
0.5	Nuclear medicine building and bunker
0.2	Corridor
0.125	Room in front of the treatment door
0.025	Roof

It can be seen through Figure III.4 that there are 15 secondary walls and two primary walls that must be studied through analytical calculation. The distance of the primary wall is represented by the red line, while the secondary wall is represented by the blue line. The difference in occupancy value at each of those points is represented by the type of room located on the other side of that bunker. In the operator/server room and the treatment planning system (TPS) room, the occupancy value is calculated as a workers' room. Meanwhile, for LINAC bunker 3 and the Nuclear Medicine Building, the study is carried out as a treatment room.

III.4. Linear Accelerator

A linear accelerator is one type of accelerator. An accelerator is a device that can increase the energy of a charged particle such as an electron, proton, deuteron, triton, alpha particle, subatomic particles, or positive and negative ions [20].

Unlike telecobalt and brachytherapy, which require a raw material in the form of a radionuclide, a LINAC can operate as long as there is an electrical supply and the device is not damaged. The radiation produced by a LINAC ranges from 4 MeV to 25 MeV. That energy is higher than telecobalt, which has energies of 1.17 and 1.33 MeV. This variable energy also allows the LINAC to effectively reach tumors both at shallow depths and tumors buried under a large amount of fatty tissue.

III.4.1. LINAC Components

There are several main components of a linear accelerator, among them the X-ray tube, focusing coil, target, and collimator. The potential difference in the X-ray tube releases electrons, whose energy can then be increased by the focusing coil. Those electrons then strike a target, which is usually an element with a high atomic number. That target then creates Bremsstrahlung radiation, which is then used to irradiate the cancer.

III.4.2. LINAC Output Dose Calibration

There are two types of calibration performed in the operation of a LINAC: relative calibration and absolute calibration.

III.4.2.1. Relative Calibration

Relative calibration is the process of comparing the radiation-dose measurement at a particular point with the dose measurement at a reference point whose value is already known. Relative calibration on a LINAC aims to ensure that the radiation dose produced by this machine is accurate and consistent. This calibration process begins by determining the reference point. The reference point used on a LINAC is the dose at the maximum value of its depth dose. Dose measurement is performed so that it can then be compared with the reference point. From this, the correction factor for the LINAC profile can be created.

III.4.3. Absolute Calibration

Absolute calibration is the calibration of a LINAC performed through an ionization chamber located in the LINAC head. In that calibration, the Monitor Unit (MU) value is calibrated so that it is equivalent to 1 cGy at the depth with maximum output dose [21]. The MU value is the quantity of dose produced by the LINAC. This MU value is continuously checked for its uniformity and equivalence so that it is always ±1% of the value relative to cGy. This is in accordance with the guidelines provided by the American Association of Physicists in Medicine Task Group 40 (AAPM TG) [18].

There are several other proposed methods for this MU value. One of them is equalization not at the maximum output but at a certain depth, such as 10 cm from the target-to-patient distance [22]. In a journal written by F. Heuvel, he gives the opinion that this value would be more resistant to dose-distribution errors in the patient. However, in the same journal, he also states that this argument is not strong enough, especially to change the definition that has already been implemented in the regulations for building radiotherapy bunker facilities.

III.5. Monte Carlo

Monte Carlo is a method that uses samples of a phenomenon to estimate the population mean of that phenomenon [18]. The calculation principle and the database used as the particle-interaction probability data are explained as follows.

III.5.1. Monte Carlo Calculation Principle

Monte Carlo data is based on two principles: the law of large numbers and the central limit theorem. The principle of the law of large numbers is that the sample mean approaches the population mean as the number of samples increases. Meanwhile, the central limit theorem principle explains that the more samples there are, the more the distribution approaches a normal distribution.

Both of these principles still hold even if the samples initially have a distribution unlike a normal distribution. This can be seen if a number of samples are drawn from the initial distribution, and then those samples are displayed as a distribution. The more samples are drawn, the more the distribution approaches a normal distribution.

III.5.2. Empirical Database for Monte Carlo Particle Simulation

The basis of the particle-interaction probability is obtained using empirical results produced by experiments conducted by the IAEA or other nuclear institutions. The Evaluated Nuclear Data Library (ENDF) data is published by the Cross Section Evaluation Working Group (CSEWG). This document is continuously updated and supplemented to support other types of particle interactions. Monte Carlo simulation has an increasingly important role in modeling radiodiagnostic results and verifying patient treatment [15].

Previous research has proven that Monte Carlo simulation agrees with experimental measurements. This result is consistent across different energy ranges. That research compared the percentage depth dose curves produced at several different energies in simulation and direct measurement. This simulation was run at 10 MeV energy; the maximum dose Dmax is at a depth of 2.4 cm [23]. That distribution is described in Figure III.2. This result is also supported by other research comparing depth dose between linear accelerators made by Elekta and Siemens [24]. The comparison of real measurement with the Monte Carlo simulation method can be represented using Table III.1.

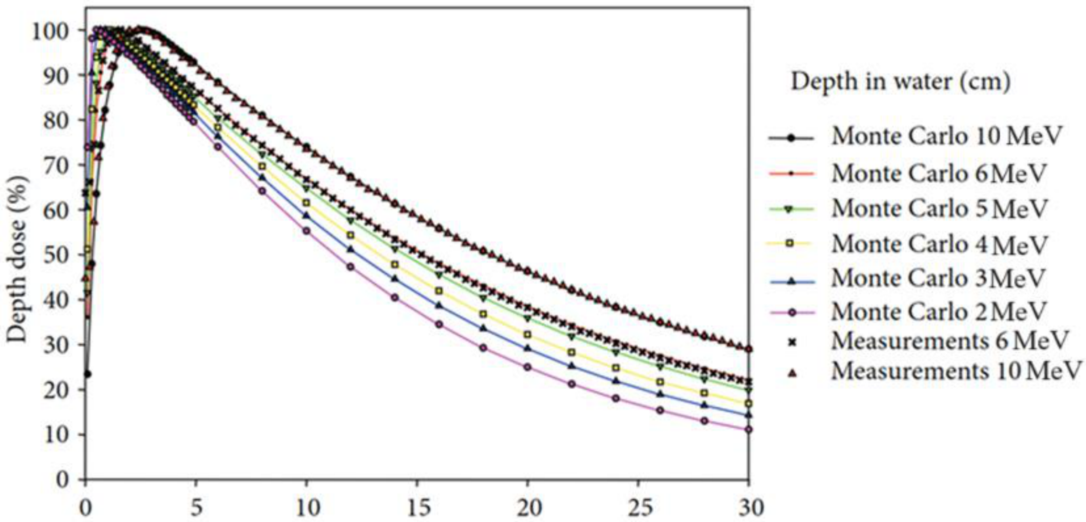

Figure III.4. Percentage Depth Dose of a water phantom, simulation and real [23]

Table III.3. Percentage depth dose in a water phantom [23]

Relative Dose (%)	10 MeV Simulation	10 MeV Experiment	6 MeV Simulation	6 MeV Experiment
100	2.4	2.4	1.5	1.6
90	5.6	5.5	4.2	4.2
80	8.2	8.2	6.6	6.6
70	11.1	11.1	9.2	9.2
60	14.5	14.4	12.0	12.0
50	18.3	18.3	15.0	15.2

III.5.3. OpenMC

OpenMC was created by the Computational Reactor Physics Group (CRPG) at the Massachusetts Institute of Technology (MIT) [20]. OpenMC uses the principle of Monte Carlo simulation for the transport of neutral particles. The numerical solution of the transport equation, also called deterministic calculation, discretizes time, energy, and angle, each of which then introduces systematic error [17]. Comparison of OpenMC simulation results with other particle transport codes such as PHITS and MCNP has also been performed and gives positive results [22]. OpenMC has also been used to design an energy modulator that was then implemented on a real LINAC [16].

III.5.3.1. Conversion of Flux to Equivalent Dose

The conversion of flux into equivalent dose is available in International Commission on Radiological Protection (ICRP) Publication 116, which is based on Monte Carlo calculations in the programs EGSnrc, FLUKA, GEANT4, MCNPX, and PHITS [25]. OpenMC produces a flux value in particle-cm/source. This value is then converted into μSv cm² based on the conversion provided by ICRP 116. Using the tally size mounted in the simulation, the pSv value is then found. This value is then equalized in the calibration to obtain dose per time. This dose rate is then converted into μSv/week, which is used as the dose constraint.

CHAPTER IV. RESEARCH METHODOLOGY

IV.1. Research Tools and Materials

In this research, the activity requiring research tools is the computer simulation. The computer simulation was carried out using the server available at the Department of Nuclear Engineering and Engineering Physics (DTNTF), Universitas Gadjah Mada (UGM).

IV.1.1. Simulation

The following are the tools used in this research:

1. Zephyrus G14 Laptop
    1. AMD Ryzen 7 5800HS @3.2 GHz
    2. 16 GB Random Access Memory
2. Intel(R) Xeon(R) Silver Server
    1. Intel Xeon Silver 4112 CPU @ 2.60 GHz
    2. 32 GB Random Access Memory

Both of these computers were used to run the simulation and to read the results obtained from that simulation. The server was provided by DTNTF UGM. The programs used have open-source licenses, among them:

1. OpenMC 0.13.1
2. Python 3.8.10
3. matplotlib 3.7.5

IV.2. Research Procedure

The general overview of this research can be described as the flow diagram below.

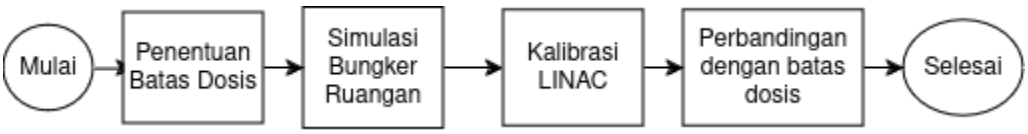

Figure IV.1. Illustration of the research workflow

IV.2.1.1. OpenMC Modeling

The OpenMC model was created based on the initial design, which represents the conditions when the direct measurement was performed. The tally was created without imitating the detector geometry. The output value obtained from OpenMC is actually not a dose in μSv/hour, but rather the number of particles passing through the created tally. Therefore, before running the room simulation, a calibration simulation was first performed to obtain the conversion factor so that the flux output from OpenMC could be converted into μSv/hour.

The cross-section data first needs to be downloaded from the page on the OpenMC website. There are 2 datasets available, namely ENDF and JEFF. This research was carried out using the ENDF/B-VII.1 data. There is a variety of empirical research data available, such as for neutrons, electrons, photons, deuterons, and others. This data is then imported through the Linux terminal.

IV.2.1.2. Absolute Calibration Simulation

The simulation produces particle types along with the energy carried by those particles. Based on the experimental results performed using organ phantoms by the International Commission on Radiological Protection (ICRP), the relationship between particle energy and dose per fluence is represented as Figure IV.2. The relationship between energy, flux, and dose per fluence can be written as Equation 4.1 [26]. Interpolation is performed using the data in Figure IV.2 so that particle energies not present in the table can also be converted.

> Equation (4.1) Da = Dose per fluence (pSv cm²) E = Particle energy (MeV) Φ = flux (particle-cm / source)

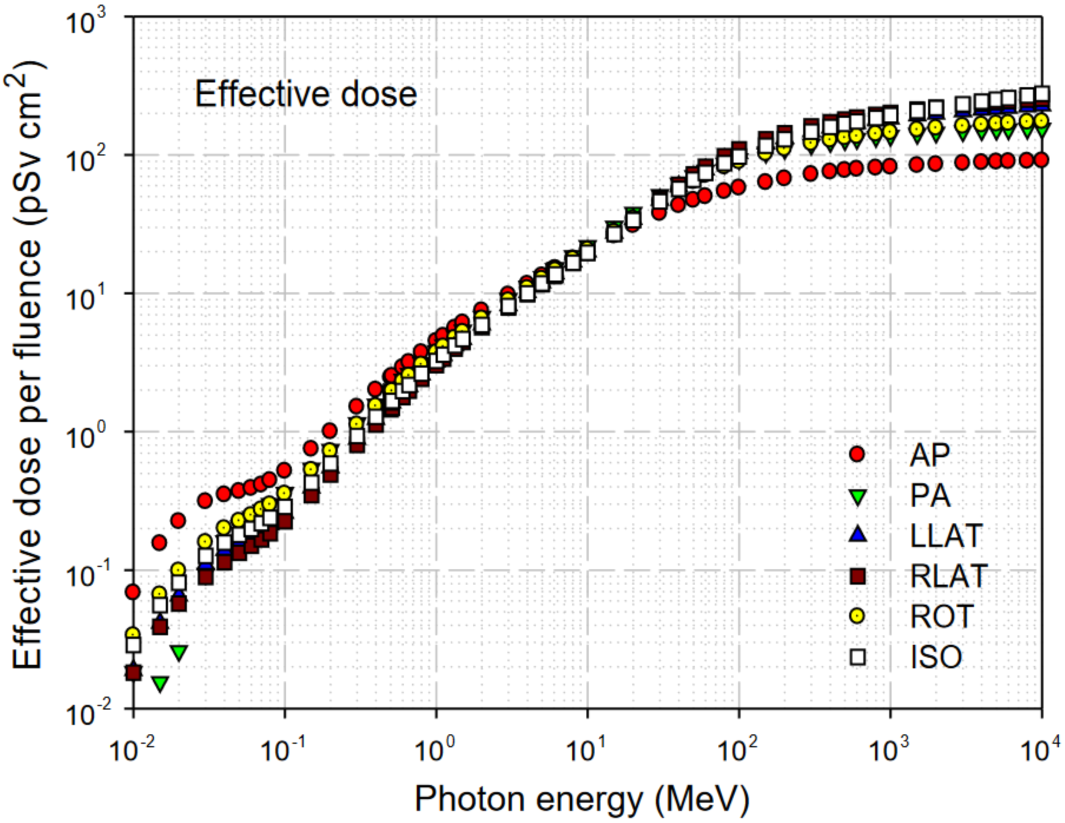

Figure IV.2. Conversion of particle energy values to effective dose per fluence using organ phantoms in the irradiation direction [25].

In Figure IV.2, the position of the irradiation direction relative to the phantom is described using the terms AP, antero-posterior; PA, postero-anterior; LLAT, left lateral; RLAT, right lateral; ROT, rotational; ISO, isotropic.

After the dose-per-fluence value is found, the flux value and the output dose are needed to determine the dose rate occurring in that simulation. This result is represented as Equation 4.2 [26].

> Equation (4.2) Dv = Total dose (pSv) SR = source / second t = time (seconds) V = volume (cm^3)

A simplification of Equation 4.2 is performed to facilitate the calculation of the dose rate at the tally detectors mounted in the room geometry. That simplification is performed using a value S that represents the relationship of dose per fluence with flux and source rate. To facilitate the calculation in the simulation, that S value is also used to convert pSv/s into (μSv/hour).

> Equation (4.3) $\dot{D}$ = Dose rate (μSv/hour) S = Conversion factor (particle-…)

For a radionuclide source, the Source Rate value is provided by the data sheet available from its manufacturer. Because for a LINAC the known data is in the form of Dose Rate, dose calibration is then performed to determine the source rate required to produce the LINAC Dose Rate, namely 360 MU/min or equivalent to 360 Gy per hour.

With the change in formula as shown in Equation 4.3 and the use of the conversion factor, the equation can be simplified as Equation 4.4.

> Equation (4.4)

Since the simulation output dose value is still in the form of pSv cm^3/source, the output value still needs to be divided by the volume of the calibration cell itself.

> Equation (4.5) Equation (4.6) $\dot{D}$maks = Maximum dose rate (μSv/hour) DOsk = Output dose of the calibration cell (pSv) S = Conversion factor Vsk = Calibration tally volume (cm^3)

The coefficient factor is then used to find the dose rate at each detector in the bunker room. The flux value along with its energy is produced by the OpenMC simulation. Then the function openmc.dose_coefficient is used to convert that energy into dose. This dose is converted into a dose rate using the conversion value found in the OpenMC calibration. The resulting value is still spread across the tally volume. Therefore, division by the tally volume used needs to be performed to obtain the dose-rate value at the measurement points.

In real conditions, the dose in a water phantom is influenced by dose scattering from the surroundings, the measurement-quality setup, the irradiation energy, and other environmental conditions such as temperature, pressure, and humidity. However, these are neglected because the simulation is performed under ideal conditions.

IV.2.1.3. Room Simulation at 4 Gantry Angles

The simulation was performed at four different angles with a 90° interval. These values were then used to estimate the total dose received in each room every week.

This simple program is used so that the rotation angle can be chosen before the simulation is run. This makes the execution of OpenMC easier to perform. The use of the source is a replication of LINAC irradiation. A box source is used and spreads until it reaches a FIELD SIZE of 40 x 40 cm.

The energy of the photon source is distributed discretely with an energy of 10 MeV. The distribution is performed discretely and spreads evenly across a spherical volume. That distribution only occurs over the area described as the Field Size area. The position of the irradiation direction is also formulated so that it can be located 100 cm from the isocenter according to its irradiation rotation angle. This is done with the function in Appendix 1, in the form of the sposi function in Appendix A.

A flux tally was installed with an energy filter. This type of tally is needed to determine the particle type and its energy so that it can then be converted into dose per fluence according to ICRP 116.

Then several points with varying distances from the wall are used as dose-rate measurement locations. These points are then studied to be compared with the applicable dose constraint.

The geometry display is then performed on 3 axis variations so that it can be checked whether it already conforms to the bunker design or not. This geometry can also be used to display the visualization of dose distribution. The dose distribution is obtained by creating a grid tally that spreads throughout the entire room.

The source is then defined through the functions source.space, source.angle, source.energy, and source.particle. Source.space is created as a box-shaped source at the position produced by the rotation function. Then source.angle is defined as polar azimuthal. This angle distribution occurs by forming something like a spherical surface, only to the widest Field Size, namely 40x40 cm. The energy distribution occurs discretely at 10 MeV energy. This is done with the description in the source.energy function. source.particle describes the type emitted by the source. In this research, that variable is set as photon. In Figure IV.3, the building floor plan is used to depict the measurement points using the simulation.

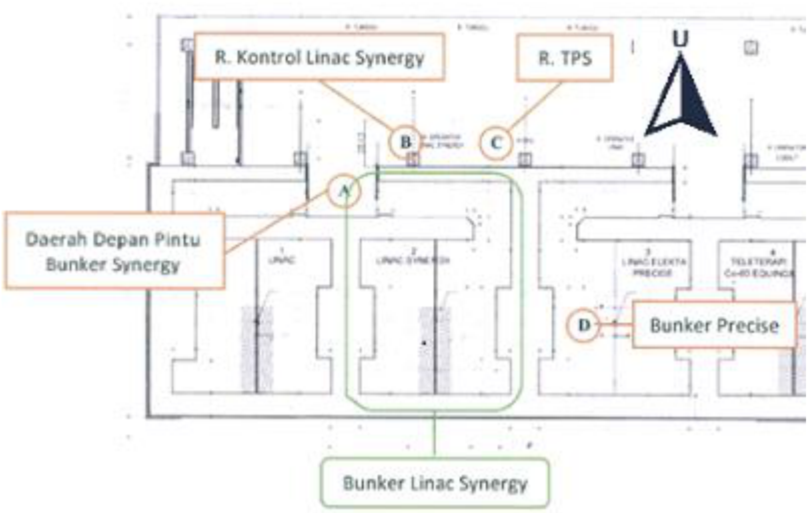

Figure IV.3. Floor plan of the dose-rate measurement points in the LINAC Synergy room [19]

IV.3. Analysis of Research Results

After the conversion value is obtained, the simulation at 4 gantry angles is performed, and then its output dose is converted into μSv/hour, which is the dose-rate quantity more commonly used in radiation detectors. The conversion using the calibration factor and the volume of the detector tally is written simply through Equation 4.4.

After the LINAC dose rate is known, this value is used to determine the dose rate of the measurement points at each irradiation angle. Before this is done, the duration of LINAC irradiation needs to be calculated so that it can then be multiplied by the dose rate. The total operating time is found using the assumptions used in the construction document and the assumption that each irradiation is performed using the maximum DR. That time is found by determining how long it takes to reach the workload at the DR used.

> Equation (4.7) Equation (4.8) tW = LINAC operating time to reach the workload at maximum DR (hours) W = Workload of LINAC operation per week (Gy/week) $\dot{D}$ = Dose Rate of LINAC operation (MU/minute) DU = Dose based on the specific angular utility and LINAC workload (μSv/week) U = Utility factor, the percentage of the LINAC irradiating that angle compared with other gantry angles (%)

Because the workload is met by varying the LINAC irradiation angle, the utility factor issued by NCRP 151 is used. That factor represents the use of LINAC irradiation at each 90° interval. The simulation dose rate at each angle is then multiplied by that percentage. The utility dose at each LINAC angle is then summed so that the expected dose rate at each point during each week of LINAC operation can be found. Because the measurement points are located in different rooms, the time spent by radiation workers at those points will differ. This is explained and used in the analytical calculation in designing the minimum bunker thickness. The relationship between dose rate and occupancy can then be represented through Equation 4.8.

> Equation (4.8) Occupancy Factor (T) = This value represents the frequency of people at that point during LINAC operation. DTO = Total Occupancy Dose at the measurement point (μSv/week) W = Workload of LINAC operation per week (Gy/week) $\dot{D}$ = Dose Rate of LINAC operation (MU/minute)

Equation 4.8 represents the expected dose rate received at one measurement point each week. For example, in the operator room, the total dose (DTO) comes from the dose rate when the LINAC operation is directed downward (DU0) and from the other irradiation angles, which are then multiplied by the occupancy factor (T) of the operator room.

IV.3.1. Interpretation of Radiation Geometry Distribution in the Simulation

The value produced by the simulation has the advantage that its distribution can be visualized. This will help interpret the dose distribution occurring in the LINAC room simulation. This visualization represents the dose-rate distribution occurring in that room.

CHAPTER V. RESULTS AND DISCUSSION

V.1. OpenMC Simulation Results

First, calibration was performed using a water-phantom replica. The Dmax value from that simulation was then used as the conversion of the OpenMC output dose into a dose that can be compared with the dose constraint.

V.1.1. Calibration of Dose to Simulation Dose Rate

Calibration was performed using a water-phantom replica located 100 cm from the source. The field size of the dose-calibration simulation was 30 cm x 30 cm.

The Dmax value in the calibration was then used as the equalization for LINAC irradiation, which has a value of 600 MU/minute or equivalent to 360 μSv/hour, which can be represented through Equation 4.4.

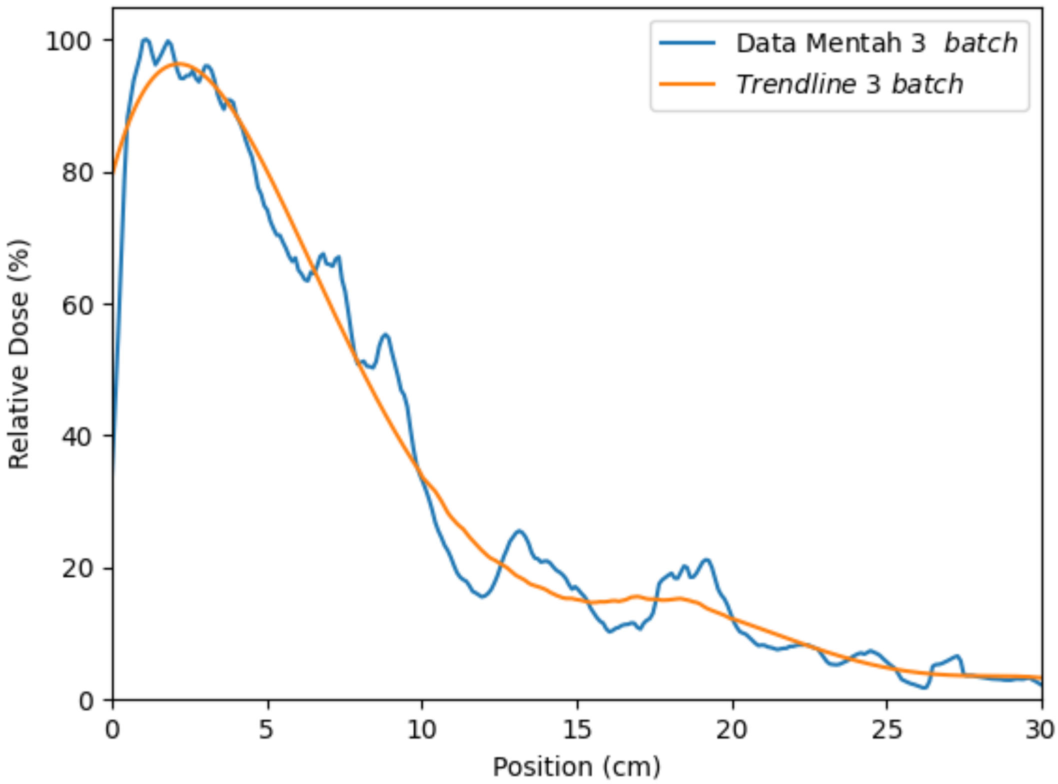

Figure V.1. Percent Depth Dose (PDD) of the Water Phantom Calibration Simulation

The PDD curve has a fairly significant difference compared with the shape in the simulation or real measurement, because the source used is a simple source. This source does not include components such as the Primary Collimator, Flattening Filter, Monitor Chamber, and Jaw as in a real LINAC head. This simulation also does not use a phase-space file in the form of a .phsp. The number of batches in this simulation is the same as the number of batches in the room simulation, namely 3 batches with 1,000,000,000 particles. It can be seen that the dose peak is at 1.1 and around 1.9 cm inside the water phantom, with a peak at 2.1 cm on the smoothed trendline curve.

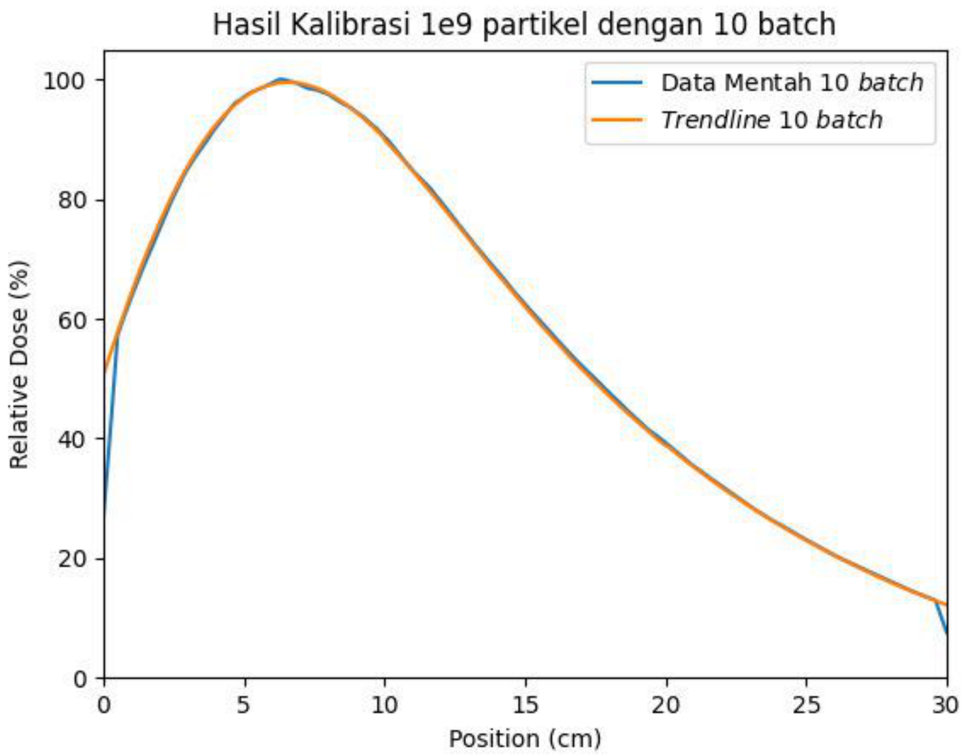

Figure V.2. Percent Depth Dose (PDD) of the Water Phantom Calibration Simulation at 10 batches

In the calibration measurement using a larger number of batches, that curve shows the dose peak at a different distance. This occurs because the geometry used to produce the LINAC beam does not resemble the real LINAC geometry. The peak in the batch is approximately at a depth of 6.1 cm. This is quite far when compared with the experimental results or Monte Carlo simulation with the actual geometry.

V.1.2. Geometry

The geometry was created using functions available in Python; the geometry was created on the xy, yz, and xz axes. Simply put, it can be described that the room consists of primary walls on both side faces and a corridor on one of the other sides. This is in accordance with the floor plan available in Figure V.3. In that geometry, the gray color means the concrete wall of the room. The yellow color represents the iron layer in the wall. There is also a tally detector illustrated in dark blue. The light blue color is the box source used to produce the photon radiation.

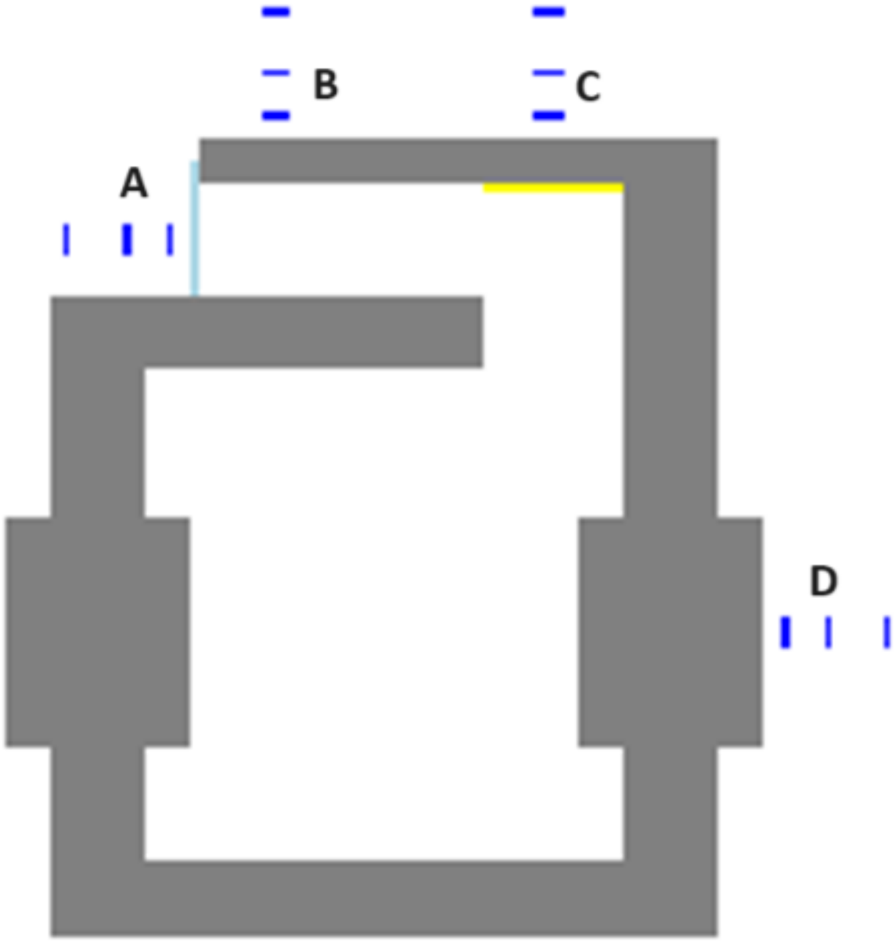

Figure V.3.
Geometry of the simulated LINAC room in the XY perception

Figure V.4. Geometry of the simulated LINAC room in the YZ perception

Figure V.5.
Geometry of the simulated LINAC room in the XZ perception

V.1.3. Per-Angle Dose Distribution Map

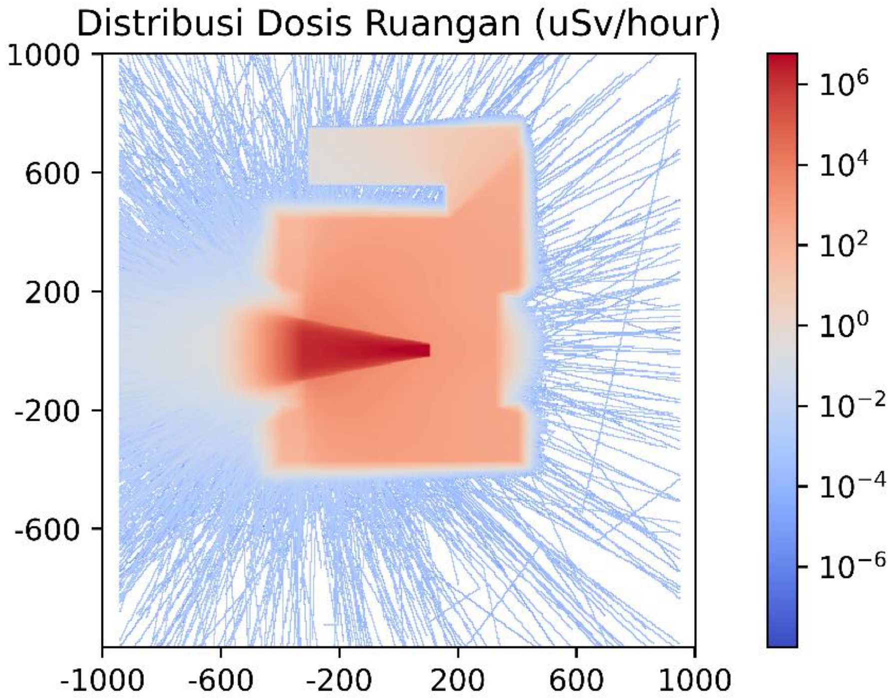

Figure V.6. Heatmap of the Default Simulation Dose Distribution

It can be seen that in the original heatmap output in Figure V.6, there is no significant radiation around the room except from the primary wall that is directly struck by the LINAC radiation. This dose distribution is formed using a grid function spread throughout that bunker. It can be seen that significant dose occurs in accordance with the direction of irradiation of the LINAC head. Meanwhile, in its surroundings, the detected dose is not very significant. The simulation was run with 1,000,000,000 particles in 3 batches. The white color in the room means that the number of particles in that simulation is still insufficient and needs to be increased. This is because a larger number of particles will be able to produce more accurate results in areas of the room with too small a dose. In Figures V.6 through V.9, the geometry is combined with the heatmap so that interpretation can be done more easily.

Using the geometry created previously and the dose heatmap from the simulation, a visualization of the bunker dose distribution can be made. At the 0° dose, with the LINAC beam directed toward the floor, the radiation reflection spreads throughout the room. This makes the dose rate at this angle significantly larger than at the other irradiation angles. The dose at this angle is also more accurate, because each point receives the dose rate from various angles until all points around the bunker are filled.

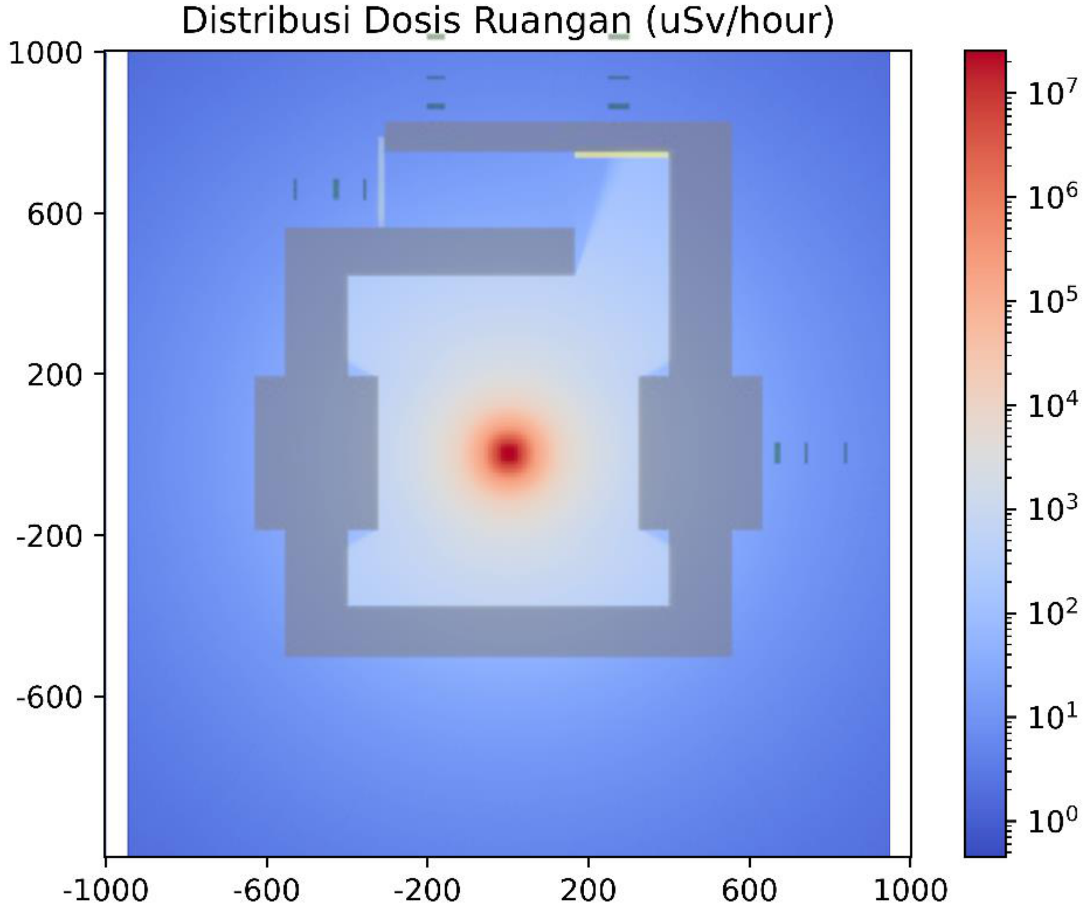

Figure V.7. Dose Distribution at gantry 0° (Bottom)

The same cannot be said for irradiation at the other angles. It can be seen that the dose for irradiation at the other angles is not yet sufficiently comprehensive to fill the grid tally in that angle's simulation. A simple analytical calculation can prove that every point around the bunker should have a radiation dose. Meanwhile, based on these irradiation images, it can be seen that there are tallies that do not receive any dose at all. An increase in the number of particles is greatly needed at the other angles. The dose rate at 270° and 90° is only significant toward the other bunker, while at 180° the distribution is directed toward the inaccessible roof.

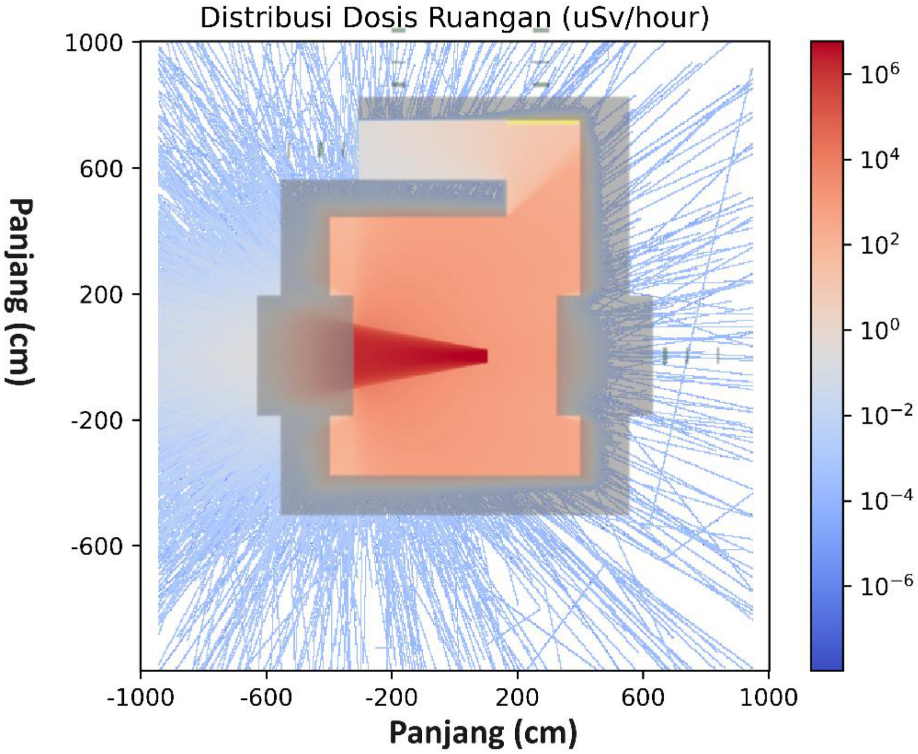

Figure V.8. Dose Distribution at gantry 90°

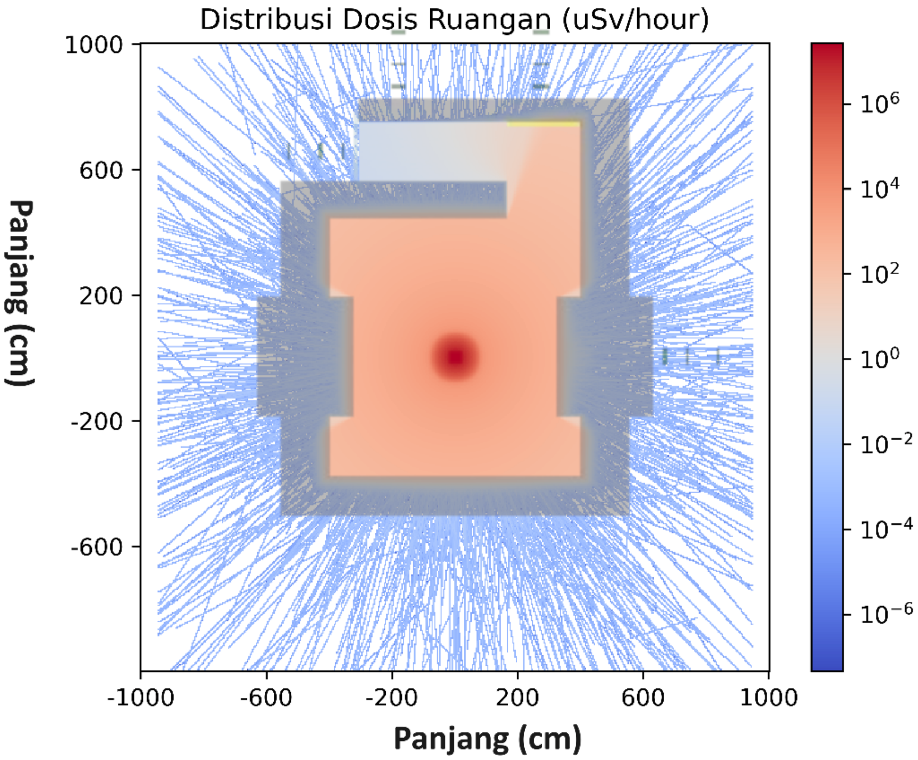

Figure V.9. Dose Distribution at gantry 180° (Top)

The dose rate at angles other than 0° shows empty points in that simulation. This can be remedied by increasing the number of particles by the amount of the order decrease in the dose rate around the bunker room. The dose rate around the room at those angles is on the order of -4, whereas at the order of -3 the radiation can already fill the simulation visualization. From this, it can be determined that the required increase in the number of particles is on the order of tens of times. This increase could be from $1x10^{10}$ up to $9x10^{10}$

particles. Using the computer specifications in this research, this could take 20 to 180 days. Therefore, an improvement in computer specifications needs to be made.

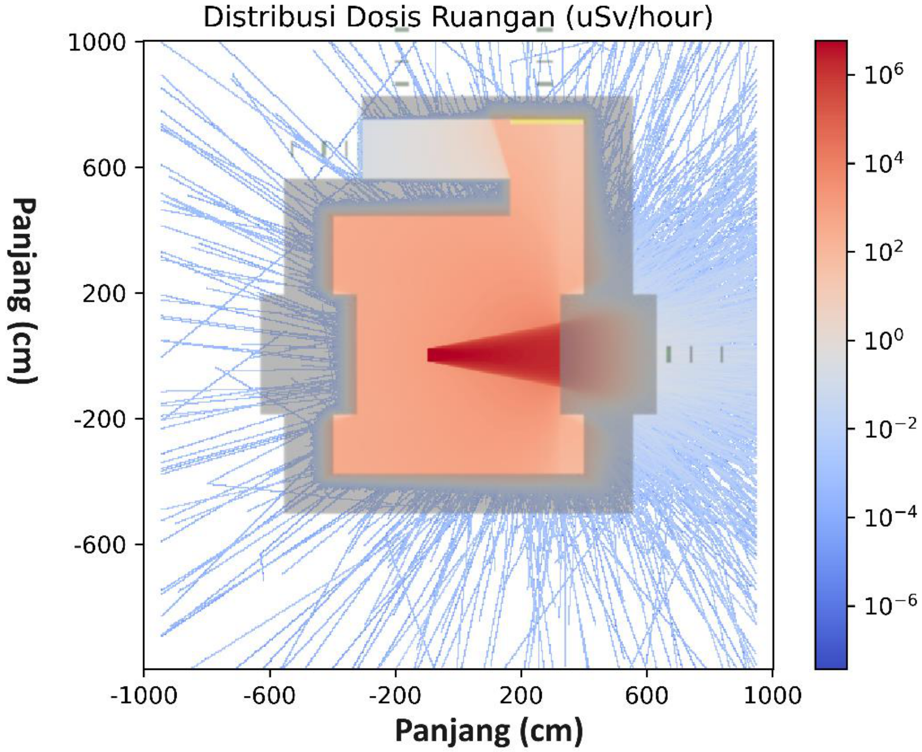

Figure V.10. Dose Distribution at gantry 270°

V.2. Comparison of Dose Limits and Simulation Results

The dose rate obtained at the mounted tallies yielded the results shown in Table V.1. It can be seen that the dose is significantly larger at the 0° angle. This is what makes the grid tally appear full at that angle.

Table V.1. Dose Rate Results Table

Point	Location	Distance to Wall (cm)	Simulation Dose Rate (μSv/hour) 0°	90°	180°	270°
A	Front of Bunker Door	30	5.71×10^{-3}	1.31×10^{-3}	2.15×10^{-3}	2.00×10^{-4}
A	Front of Bunker Door	100	3.35×10^{-3}	9.10×10^{-4}	0.00	6.37×10^{-4}
A	Front of Bunker Door	200	1.20×10^{-2}	3.83×10^{-4}	7.86×10^{-4}	5.63×10^{-4}
B	LINAC Control Room	30	2.07×10^{-2}	6.64×10^{-4}	1.67×10^{-3}	6.80×10^{-4}

Point	Location	Distance to Wall (cm)	Simulation Dose Rate (μSv/hour) 0°	90°	180°	270°
B	LINAC Control Room	100	4.00×10^{-2}	3.55×10^{-4}	1.93×10^{-4}	2.40×10^{-4}
B	LINAC Control Room	200	5.32×10^{-2}	2.68×10^{-4}	1.93×10^{-4}	2.04×10^{-4}
C	TPS Room	30	1.87×10^{-2}	6.64×10^{-4}	1.55×10^{-4}	0.00
C	TPS Room	100	2.90×10^{-2}	3.49×10^{-4}	1.22×10^{-3}	7.00×10^{-4}
C	TPS Room	200	4.81×10^{-2}	5.66×10^{-4}	1.23×10^{-3}	6.93×10^{-4}
D	Bunker LINAC Precise	30	5.47×10^{-2}	7.58×10^{-4}	2.51×10^{-3}	3.20×10^{-1}
D	Bunker LINAC Precise	100	6.37×10^{-2}	6.70×10^{-4}	3.30×10^{-2}	2.48×10^{-1}
D	Bunker LINAC Precise	200	9.93×10^{-2}	3.76×10^{-4}	1.81×10^{-4}	2.20×10^{-1}

Based on the dose-rate measurement in Table V.1, it can be seen that the radiation dose does not always decrease the farther the detection is performed. This is because the radiation passing through the farthest detector can be done by radiation that does not arrive perpendicular to the wall. At 180° and 270°, there are detectors not passed by radiation at all.

The dose produced is already very low, considering that the LINAC is only operated a few times each day. Using the workload assumption used to calculate the LINAC thickness, the total operating time of the LINAC can be calculated. This calculation is performed with the assumption that the maximum irradiation dose rate is used. The workload calculation can be computed comprehensively based on Table V.2. The workload calculation in the analytical calculation for the secondary wall is performed by considering the coefficient factor Q. This differs from the primary analytical calculation, which does not use that factor.

Table V.2. Weekly Workload Calculation

Type of Irradiation	Number of Patients (people)	Dose per patient (Gy/patient)	Operating Days (days/week)	Total Dose (Gy/week)
Conventional	20	4	5	400
IMRT	50	4	5	1000
VMAT	10	4	5	200
QA	12	2	6	144
Total Workload (Gy/week)				**1744**

Type of Irradiation	Number of Patients (people)	Dose per patient (Gy/ patient)	Operating Days (days/ week)	Total Dose (Gy/ week)
LINAC Operating Time (hours/week)				**4.84**

At each of those walls, the applicable dose limit is the workers' dose constraint. This is because every room around the bunker is a workers' room and cannot be entered by the general public. The dose constraint for radiation workers according to BAPETEN Regulation (Perka) No. 4 of 2013 is 200 µSv/week.

Table V.3. Dose Rate at LINAC Irradiation 0°

Point	Location	Distance to Wall (cm)	Dose Rate (µSv/ hour)	Dose Rate per week (µSv/ week)	Utility Dose Rate (µSv/ week)
A	Front of Bunker Door	30	5.71×10^{-3}	2.77×10^{-2}	8.58×10^{-3}
A	Front of Bunker Door	100	3.35×10^{-3}	1.62×10^{-2}	5.03×10^{-3}
A	Front of Bunker Door	200	1.20×10^{-2}	5.83×10^{-2}	1.81×10^{-2}
B	LINAC Control Room	30	2.07×10^{-2}	1.00×10^{-1}	3.11×10^{-2}
B	LINAC Control Room	100	4.00×10^{-2}	1.94×10^{-1}	6.00×10^{-2}
B	LINAC Control Room	200	5.32×10^{-2}	2.58×10^{-1}	8.00×10^{-2}
C	TPS Room	30	1.87×10^{-2}	9.07×10^{-2}	2.81×10^{-2}
C	TPS Room	100	2.90×10^{-2}	1.40×10^{-1}	4.35×10^{-2}
C	TPS Room	200	4.81×10^{-2}	2.33×10^{-1}	7.23×10^{-2}
D	Bunker LINAC Precise	30	5.47×10^{-2}	2.65×10^{-1}	8.21×10^{-2}
D	Bunker LINAC Precise	100	6.37×10^{-2}	3.08×10^{-1}	9.56×10^{-2}
D	Bunker LINAC Precise	200	9.93×10^{-2}	4.81×10^{-1}	1.49×10^{-1}

Table V.4. Dose Rate at LINAC Irradiation 90°

Point	Location	Distance to Wall (cm)	Dose Rate (μSv/hour)	Dose per week (μSv/week)	Utility Dose (μSv/week)
A	Front of Bunker Door	30	1.31×10^{-3}	6.34×10^{-3}	1.35×10^{-3}
A	Front of Bunker Door	100	9.10×10^{-4}	4.41×10^{-3}	9.39×10^{-4}
A	Front of Bunker Door	200	3.83×10^{-4}	1.85×10^{-3}	3.95×10^{-4}
B	LINAC Control Room	30	6.64×10^{-4}	3.22×10^{-3}	6.86×10^{-4}
B	LINAC Control Room	100	3.55×10^{-4}	1.72×10^{-3}	3.67×10^{-4}
B	LINAC Control Room	200	2.68×10^{-4}	1.30×10^{-3}	2.76×10^{-4}
C	TPS Room	30	6.64×10^{-4}	3.22×10^{-3}	6.86×10^{-4}
C	TPS Room	100	3.49×10^{-4}	1.69×10^{-3}	3.60×10^{-4}
C	TPS Room	200	5.66×10^{-4}	2.74×10^{-3}	5.84×10^{-4}
D	Bunker LINAC Precise	30	7.58×10^{-4}	3.67×10^{-3}	7.82×10^{-4}
D	Bunker LINAC Precise	100	6.70×10^{-4}	3.24×10^{-3}	6.91×10^{-4}
D	Bunker LINAC Precise	200	3.76×10^{-4}	1.82×10^{-3}	3.88×10^{-4}

Table V.5. Dose Rate at LINAC Irradiation 180° (Top)

Point	Location	Distance to Wall (cm)	Dose Rate (μSv/hour)	Dose per week (μSv/week)	Utility Dose (μSv/week)
A	Front of Bunker Door	30	2.15×10^{-3}	1.04×10^{-2}	2.74×10^{-3}
A	Front of Bunker Door	100	0.00	0.00	0.00
A	Front of Bunker Door	200	7.86×10^{-4}	3.81×10^{-3}	1.00×10^{-3}
B	LINAC Control Room	30	1.67×10^{-3}	8.09×10^{-3}	2.13×10^{-3}
B	LINAC Control Room	100	1.93×10^{-4}	9.33×10^{-4}	2.45×10^{-4}
B		200	1.93×10^{-4}	9.33×10^{-4}	2.45×10^{-4}

Point	Location	Distance to Wall (cm)	Dose Rate (μSv/hour)	Dose per week (μSv/week)	Utility Dose (μSv/week)
	LINAC Control Room				
C	TPS Room	30	1.55×10^{-4}	7.51×10^{-4}	1.97×10^{-4}
C	TPS Room	100	1.22×10^{-3}	5.92×10^{-3}	1.56×10^{-3}
C	TPS Room	200	1.23×10^{-3}	5.95×10^{-3}	1.56×10^{-3}
D	Bunker LINAC Precise	30	2.51×10^{-3}	1.21×10^{-2}	3.19×10^{-3}
D	Bunker LINAC Precise	100	3.30×10^{-2}	1.60×10^{-1}	4.20×10^{-2}
D	Bunker LINAC Precise	200	1.81×10^{-4}	8.76×10^{-4}	2.30×10^{-4}

Table V.6. Dose Rate at LINAC Irradiation 270°

Point	Location	Distance to Wall (cm)	Dose Rate (μSv/hour)	Dose per week (μSv/week)	Utility Dose (μSv/week)
A	Front of Bunker Door	30	2.00×10^{-4}	9.67×10^{-4}	2.06×10^{-4}
A	Front of Bunker Door	100	6.37×10^{-4}	3.09×10^{-3}	6.57×10^{-4}
A	Front of Bunker Door	200	5.63×10^{-4}	2.73×10^{-3}	5.81×10^{-4}
B	LINAC Control Room	30	6.80×10^{-4}	3.29×10^{-3}	7.02×10^{-4}
B	LINAC Control Room	100	2.40×10^{-4}	1.16×10^{-3}	2.48×10^{-4}
B	LINAC Control Room	200	2.04×10^{-4}	9.88×10^{-4}	2.11×10^{-4}
C	TPS Room	30	0.00	0.00	0.00
C	TPS Room	100	7.00×10^{-4}	3.39×10^{-3}	7.23×10^{-4}
C	TPS Room	200	6.93×10^{-4}	3.36×10^{-3}	7.16×10^{-4}
D	Bunker LINAC Precise	30	3.20×10^{-1}	1.55	3.30×10^{-1}
D	Bunker LINAC Precise	100	2.48×10^{-1}	1.20	2.56×10^{-1}
D		200	2.20×10^{-1}	1.07	2.27×10^{-1}

Point	Location	Distance to Wall (cm)	Dose Rate (μSv/hour)	Dose per week (μSv/week)	Utility Dose (μSv/week)
	Bunker LINAC Precise				

With the weekly LINAC operating time and the dose limit in accordance with the provisions of NCRP 151 and SRS 47, the safety of the bunker design can be assessed for compliance. That value is obtained by multiplying the total dose rate by the utility factor of each irradiation angle. Based on Table III.2, the utility is 31% at 0°, 21.3% at 90° and 270°, and 26.3% at 180°. The dose rate from the simulation at each angle is then multiplied by its respective utility so that it can then be summed to determine the expected worker dose at each studied point. This value is what can ultimately be compared to determine how small the dose received by workers at those points is compared with the applicable dose constraint. This is explained through Equation 4.7.

The dose rate from the measurement results at each angle is summed to find the dose rate on workers. The total of LINAC operation each week and the use of the occupancy factor to estimate the dose received by workers based on occupancy are shown in Table V.7.

Table V.7. Total Expected Dose Rate on Workers in the Rooms

Point	Location	Occupancy Factor (T)	Distance to Wall (cm)	Total Dose (μSv/week)	Total Occupancy Dose (μSv/week)
A	Front of Bunker Door	0.125	30	1.29×10^{-2}	1.61×10^{-3}
A	Front of Bunker Door	0.125	100	6.62×10^{-3}	8.28×10^{-4}
A	Front of Bunker Door	0.125	200	2.01×10^{-2}	2.51×10^{-3}
B	LINAC Control Room	1	30	3.46×10^{-2}	3.46×10^{-2}
B	LINAC Control Room	1	100	6.09×10^{-2}	6.09×10^{-2}
B	LINAC Control Room	1	200	8.07×10^{-2}	8.07×10^{-2}
C	TPS Room	1	30	2.90×10^{-2}	2.90×10^{-2}
C	TPS Room	1	100	4.62×10^{-2}	4.62×10^{-2}
C	TPS Room	1	200	7.52×10^{-2}	7.52×10^{-2}
D	Bunker LINAC Precise	0.5	30	4.16×10^{-1}	2.08×10^{-1}

Point	Location	Occupancy Factor (T)	Distance to Wall (cm)	Total Dose (µSv/ week)	Total Occupancy Dose (µSv/ week)
D	Bunker LINAC Precise	0.5	100	3.94×10^{-1}	1.97×10^{-1}
D	Bunker LINAC Precise	0.5	200	3.77×10^{-1}	1.88×10^{-1}

The dose delivered by the LINAC at those points is much smaller than its dose limit value of 200 µSv/week. This value is 961 to 242,000 times smaller than the applicable NBD. It should be noted that the dose target for each bunker is at the bunker wall, whereas the dose study is performed at distances of 30 cm, 100 cm, and 200 cm. These locations are the locations more likely to be occupied by radiation workers when irradiation is performed.

In the design of the bunker room, the addition of a thickness equal to the half value layer (HVL) is also performed after the minimum wall thickness is obtained. This is then followed by rounding up the dimensions of the concrete wall. Therefore, the inverse square law and the attenuation factor play a major role in the dose rate being many times smaller than its dose limit value.

In this simulation, the LINAC characteristics are a simplification of real LINAC irradiation. The accuracy of the simulation results can be improved by adding a flattening filter (FF), using a phase-space file, and increasing the number of particles.

CHAPTER VI. CONCLUSIONS AND SUGGESTIONS

VI.1. Conclusions

1. The simulated dose rate around the LINAC bunker ranges from 8.28×10^{-3} to 0.208 µSv/week. This value is far smaller than the dose constraint enforced by BAPETEN Regulation (Perka) No. 3 of 2013. This means that the bunker's safety already complies with the applicable standard.

2. A visualization of the dose-rate distribution in the bunker room was obtained to help interpret the dose occurring at each dose-rate measurement point around the bunker.

VI.2. Suggestions

There are things that can be developed in carrying out this research. An improvement in the accuracy of the simulation results can be achieved by adding a flattening filter (FF), using a phase-space file, and increasing the number of particles. The use of a computer with higher specifications can also be employed in order to simulate a larger number of particles.

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APPENDICES

The following appendices (LAMPIRAN A-G) contain the OpenMC program listings, commissioning dose-rate measurements, analytical bunker-design calculations, and the ICRP 116 flux-to-dose conversion tables. They are reproduced verbatim from the original thesis.

LAMPIRAN A

Listing Program OpenMC

Pemanggilan cross section ENDF

```
Export          OPENMC_CROSS_SECTIONS='/home/sena/Code/endfb-vii.1
hdf5/cross_sections.xml'

# export OPENMC_CROSS_SECTIONS=$var/cross_sections.xml

export HDF5_DISABLE_VERSION_CHECK=1
```

Program Kalibrasi

```
import openmc

# vim:foldmethod=marker

import openmc

import matplotlib.pyplot as plt

import openmc.stats, openmc.data

from math import cos, atan2, pi



batches = 10

inactive = 10

particles = int(input('Enter number of particle (It was 1e8)\n=
'))
#1_000_000_000




# Material

# {{{
```

```
air=openmc.Material(name='Air')

air.set_density('g/cm3',0.001205)

air.add_nuclide('N14',0.7)

air.add_nuclide('O16',0.3)

#air.add_s_alpha_beta('c_H_in_Air')

air.add_element('C',0.002)

air.add_element('Fe',0.001)

air.add_element('Si',0.001)

air.add_element('Mn',0.001)
```



```
water=openmc.Material(name='Water')

water.set_density('g/cm3',1.0)

water.add_nuclide('H1',2.0)

water.add_nuclide('O16',1.0)

water.add_s_alpha_beta('c_H_in_H2O')


water2=openmc.Material(name='Water')

water2.set_density('g/cm3',1.0)

water2.add_nuclide('H1',2.0)

water2.add_nuclide('O16',1.0)

water2.add_s_alpha_beta('c_H_in_H2O')


materials = openmc.Materials( [air,water,water2])

materials.export_to_xml()

# }}}


SSD = 100.0 #Source to Skin Distance
```

```
PHANTOM_WIDTH = 40

ld= 0.1# panjang dan lebar WP

d = 30.0 #kedalaman WP

padd = 10.0 #padding terhadap source dan detektor


# {{{


n = 300

phantom_cells = []

dx = d/n

for i in range(n):

  x0 = SSD + i*dx

  x1 = SSD +(i+1)*dx

  r_x = +openmc.XPlane(x0) & -openmc.XPlane(x1)
```



```
  r_y = +openmc.YPlane(-ld/2.0) & -openmc.YPlane(ld/2.0)

  r_z = +openmc.ZPlane(-ld/2.0) & -openmc.ZPlane(ld/2.0)


  cell = openmc.Cell(region=r_x & r_y & r_z)

  cell.fill = water2

  phantom_cells.append(cell)


r_x = +openmc.XPlane(SSD) & -openmc.XPlane(SSD+d)

r_y = +openmc.YPlane(-ld/2.0) & -openmc.YPlane(ld/2.0)

r_z = +openmc.ZPlane(-ld/2.0) & -openmc.ZPlane(ld/2.0)

r_phantally = r_x & r_y & r_z
```

```
r_y_wp= +openmc.YPlane(-PHANTOM_WIDTH/2.0)\

  & -openmc.YPlane(PHANTOM_WIDTH/2.0)

r_z_wp= +openmc.ZPlane(-PHANTOM_WIDTH/2.0)\

  & -openmc.ZPlane(PHANTOM_WIDTH/2.0)

r_wp= r_x & r_y_wp & r_z_wp

c_wp= openmc.Cell(region=r_wp& ~r_phantally)

c_wp.fill = water



r_x_air = +openmc.XPlane(-padd, boundary_type='vacuum')\

  & -openmc.XPlane(SSD+d+padd, boundary_type='vacuum')

r_y_air          =            +openmc.YPlane(-PHANTOM_WIDTH/2.0-
padd,
boundary_type='vacuum')\

  & -openmc.YPlane(PHANTOM_WIDTH/2.0+padd, boundary_type='vacuum')

r_z_air          =            +openmc.ZPlane(-PHANTOM_WIDTH/2.0-
padd,
boundary_type='vacuum')\

  & -openmc.ZPlane(PHANTOM_WIDTH/2.0+padd, boundary_type='vacuum')

r_air = r_x_air & r_y_air & r_z_air

c_air = openmc.Cell(region=r_air & ~r_wp)

c_air.fill = air



universe = openmc.Universe(cells= [c_air]+phantom_cells+ [c_wp])
```



```
geometry = openmc.Geometry()

geometry.root_universe = universe

geometry.export_to_xml()

# }}}
```

```
plot= openmc.Plot()

plot.basis = 'yz'

plot.filename='yz_cal'

plot.origin = (110, 0, 0)

plot.width = (50., 50.)

plot.pixels=(1000,1000)

plot.color_by='material'

plot.colors={

    water:'blue',

    air:'green',

    water2:'black'

}

plot_file=openmc.Plots( [plot])

plot_file.export_to_xml()

openmc.plot_geometry()

plot.to_ipython_image()


plot.basis='xz'

plot.filename='xz_cal'

plot.width=(100,100)

plot.pixels=(1000,1000)

plot.origin=(100,0,0)

plot_file=openmc.Plots( [plot])

plot_file.export_to_xml()

openmc.plot_geometry()

plot.to_ipython_image()
```

```
# {{{

batches= batches

inactive = inactive

particles = particles


particle_filter = openmc.ParticleFilter('photon')

# energy, dose = openmc.data.dose_coefficients('photon', 'AP')

energy, dose = openmc.data.dose_coefficients('photon', 'RLAT')

dose_filter = openmc.EnergyFunctionFilter(energy, dose)


#######    TALLIES      #########

tallies_file = openmc.Tallies()

cell_filter = openmc.CellFilter(phantom_cells)

tally = openmc.Tally(name='tally')

tally.filters = [cell_filter, particle_filter, dose_filter]

tally.scores = ['flux']


"""

#######    MESH TALLY       #########

tallysize=0.1

mesh = openmc.Mesh()

mesh.dimension = [1, 1, 400]

mesh.lower_left = [SSD, -tallysize/2, -FIELD_SIZE/2]#type: ignore

mesh.upper_right = [SSD+tallysize, tallysize/2, FIELD_SIZE/
2]#type:
ignore

mesh_filter = openmc.MeshFilter(mesh)
```

```
tally01 = openmc.Tally(name="0 depth tally")

tally01.filters = [mesh_filter, particle_filter, dose_filter]

tally01.scores = ['flux']



meshdpp01 = openmc.Mesh()
```



```
meshdpp01.dimension = [300, 1, 1]

meshdpp01.lower_left = [SSD, -tallysize/2,
-tallysize/2]#type:
ignore

meshdpp01.upper_right = [SSD+d, tallysize/2,
tallysize/2]#type:
ignore

meshdpp01_filter = openmc.MeshFilter(meshdpp01)



tallydpp01 = openmc.Tally(name="profile depth dose tally")

tallydpp01.filters = [meshdpp01_filter,
particle_filter,
dose_filter]

tallydpp01.scores = ['flux']



meshdpp5 = openmc.Mesh()

meshdpp5.dimension = [300, 1, 1]

meshdpp5.lower_left = [SSD, -tallysize/2,
-tallysize/2]#type:
ignore

meshdpp5.upper_right = [SSD+d, tallysize/2,
tallysize/2]#type:
ignore

meshdpp5_filter = openmc.MeshFilter(meshdpp5)
```

```
tallydpp5 = openmc.Tally(name="profile depth dose tally")

tallydpp5.filters = [meshdpp5_filter, particle_filter,
dose_filter]

tallydpp5.scores = ['flux']



tallies_file.append(tallydpp01)

tallies_file.append(tallydpp5)

tallies_file.append(tally01)

"""

tallies_file.append(tally)

tallies_file.export_to_xml()



source = openmc.Source() #type: ignore

source.space = openmc.stats.Point((0,0,0))

phi = openmc.stats.Uniform(0, 2*pi)
```



```
#mu = openmc.stats.Uniform(cos(atan2(l/2, SSD)), 1)

mu = openmc.stats.Uniform(cos(atan2(40/2, SSD)), 1)

source.angle                                                    =
openmc.stats.PolarAzimuthal(mu,phi,reference_uvw=(1,0,0))

source.energy = openmc.stats.Discrete( [10e6], [1]) #10MeV

source.particle = 'photon'



settings = openmc.Settings()

settings.batches = batches

settings.inactive = inactive

settings.particles = particles
```

```
settings.run_mode = 'fixed source'

settings.photon_transport = True

settings.export_to_xml()

# }}}


openmc.run()
```

Program Pembaca Kalibrasi

```
import openmc

import matplotlib.pyplot as plt

import numpy as np

from scipy.signal import savgol_filter

import matplotlib.ticker as mtick

##


sp = openmc.StatePoint('statepoint.5.h5')

##

print(sp.n_particles)


n = 300

d = 30
```



```
tal = sp.tallies [1]

datay = tal.mean

# datax is linspace between 0 and 50

datax = np.linspace(0,d,n)

datay.shape = (n,)

# smooth datay
```

```
datay = np.convolve(datay, np.ones(10)/10, mode='same')

# find the index of the maximum value in datay

max_index = np.argmax(datay)

x_max = datax [max_index]

y_max = datay [datay.argmax()]

print(x_max,max_index,y_max)


y_max=datay [max_index]

plt.axvline(x=x_max, color='r')

plt.legend(loc='best')

plt.ylabel("Relative Dose (%)")

plt.xlabel("Position (cm)")

plt.xlim(0,30)

plt.ylim(0,105)

datay=datay/y_max*100

plt.plot(datax, datay)


tal = sp.tallies [1]

datay = tal.mean

datay=datay/y_max*100

n = len(datay)


x_smooth=datax

y_smooth = np.interp(x_smooth, datax, datay)

# print (f"datay =\n {datay}")

# print (f"ysmooth=\n{y_smooth}")
```

```
# Apply Savitzky-Golay filter for smoothing

window_size = 200

poly_order =5

y_smooth = savgol_filter(y_smooth, window_size, poly_order)


max_index=np.argmax(y_smooth)

x_maxs = x_smooth [max_index]

y_maxs=y_smooth [max_index]

plt.plot(x_smooth, y_smooth)

plt.title(f'x_max = {x_max} cm \n xsmooth_max = {x_maxs} cm')

plt.savefig('pdd.png')

plt.show()

plt.savefig('pdd.png')

"""

tal = sp.tallies [2]

datay = tal.mean

# datax is linspace between 0 and 50

datax = np.linspace(0,d,n)

datay.shape = (n,)

# smooth datay

datay = np.convolve(datay, np.ones(10)/10, mode='same')

# find the index of the maximum value in datay

max_index = np.argmax(datay)

x_max = datax [max_index]

y_max = datay [datay.argmax()]

print(x_max,max_index,y_max)
```

```
y_max=datay [max_index]

#plt.axvline(x=x_max, color='r')

plt.legend(loc='best')

plt.ylabel("Relative Dose (%)")
```



```
plt.xlabel("Position (cm)")

plt.xlim(-0,30)

plt.ylim(0,105)

datay=datay/y_max*100

plt.plot(datax, datay)


tal = sp.tallies [1]

datay = tal.mean

datay=datay/y_max*100

n = len(datay)


x_smooth=datax

y_smooth = np.interp(x_smooth, datax, datay)

# print (f"datay =\n {datay}")

# print (f"ysmooth=\n{y_smooth}")


# Apply Savitzky-Golay filter for smoothing

window_size = 200

poly_order =5

y_smooth = savgol_filter(y_smooth, window_size, poly_order)
```

```
max_index=np.argmax(y_smooth)

x_maxs = x_smooth [max_index]

y_maxs=y_smooth [max_index]

plt.plot(x_smooth, y_smooth)

# draw line at x_max

#plt.axvline(x=x_maxs, color='r')

# write x_max value on plot

plt.title(f'x_max = {x_max} cm \n xsmooth_max = {x_maxs} cm')

#print(f"Y Value on xmax={_max}")

plt.show()

plt.savefig('2.5 depth.png')
```



```
def rounde(unrounded):

  roundede="{:.4e}".format(unrounded)

  return roundede


print(f"\nDosis = {rounde(y_max)}pSvcm3/src")

print(f"Dosis smoothen = {rounde(y_maxs)}pSvcm3/src\n")


print(f"s_rate=mu*volume_wp/phandosevalues") # mu=

mu=1e11#600 MU/min = 10 cGy/s = 1e11 pSv/s

volume_wp=(d/n)*6*6

wpdose=y_max

s_rate=mu*volume_wp/wpdose


print(f"s_rate={mu}*{volume_wp}/{rounde(wpdose)}={s_rate}")
```

```
#600cGy/min=36000cGy/h=36uSv/h

print(f"600 cgy/min = 36uSv/h")

print(f"\n36uSv/h = k * {y_max} pSvcm3/src/{volume_wp}cm3\n")

k=volume_wp/y_max*36

print(f"k      *      {y_max}          pSvcm3/src/{volume_wp}cm3
=
{k*y_max/volume_wp}uSv/h")

print(f"36 uSv/h = {k*y_max/volume_wp} uSv/h")

print(f"k = {k}")



print(f"k smoothen = {volume_wp/y_maxs*36}")



tally = sp.tallies [3]#2.5 depth {{{

import matplotlib.pyplot as plt



flux = tally.get_slice(scores= ['flux'])

data = tally.mean.flatten()
```



```
x = np.linspace(-25, 25, len(data))



max_index=np.argmax(data)

x_max = x [max_index]

y_max = np.max(data)

print("ymax = ", y_max)

plt.axvline(x=x_max,color='r',label='peak')

yflat=data/y_max*100

#print(yflat)
```

```
fmt='%.0f%%' #changing into format

xticks = mtick.FormatStrFormatter(fmt)



#plt.yflat.set_major_formatter(FuncFormatter(lambda     y,     _:
'{:.0%}'.format(yflat)))

plt.plot(x, yflat,label="raw")



ysmooth=np.interp(x,x,yflat)

window_size=300

poly_order=10

start=100

end=400

area_tofilter=yflat [start:end]

filtered=savgol_filter(area_tofilter,window_size,poly_order)

ysmooth [start:end]=filtered

plt.plot(x,ysmooth,label="savgol filter")

plt.legend(loc='best')

plt.ylabel("Relative Dose (%)")

plt.xlabel("Position (cm)")

plt.xlim(-25,25)

plt.ylim(0,105)
```



```
plt.title('2.5 depth.png')

plt.savefig('2.5 depth.png')

plt.show()

"""
```

Program Ruangan

```
import openmc #type: ignore

import matplotlib.pyplot as plt

from math import * #type: ignore


particle=int(input('Particle number (\'twas 1e7)\n= '))

SOURCE_SIZE=20

FIELD_SIZE=40

SSD=100


air=openmc.Material(name='Air')

air.set_density('g/cm3',0.001205)

air.add_nuclide('N14',0.7)

air.add_nuclide('O16',0.3)

#air.add_s_alpha_beta('c_H_in_Air')

air.add_element('C',0.002)

air.add_element('Fe',0.001)

air.add_element('Si',0.001)

air.add_element('Mn',0.001)


air2=openmc.Material(name='Air')

air2.set_density('g/cm3',0.001205)

air2.add_nuclide('N14',0.7)

air2.add_nuclide('O16',0.3)

#air.add_s_alpha_beta('c_H_in_Air')

air2.add_element('C',0.002)
```

```
air2.add_element('Fe',0.001)

air2.add_element('Si',0.001)

air2.add_element('Mn',0.001)


water=openmc.Material(name='Water')

water.set_density('g/cm3',1.0)

water.add_nuclide('H1',2.0)

water.add_nuclide('O16',1.0)

water.add_s_alpha_beta('c_H_in_H2O')


soft=openmc.Material(name='Soft Tissue')

soft.set_density('g/cm3',1.0)

soft.add_element('H', 10.4472, percent_type='ao')#1

soft.add_element('C', 23.219, percent_type='ao')#6

soft.add_element('N', 2.388, percent_type='ao')#7

soft.add_element('O', 63.0238, percent_type='ao')#8

soft.add_element('Na', 0.113, percent_type='ao')#11

soft.add_element('Mg', 0.013, percent_type='ao')#12

soft.add_element('S', 0.199, percent_type='ao')#16

soft.add_element('Cl', 0.134, percent_type='ao')#17

soft.add_element('K', 0.199, percent_type='ao')#19

soft.add_element('Ca', 0.023, percent_type='ao')#20

#https://physics.nist.gov/cgi-bin/Star/compos.pl?matno=261


bpe=openmc.Material(name='Borate Polyethylene')
```

```
bpe.set_density('g/cm3',1.0)

bpe.add_element('H', 0.111, percent_type='ao')#1

bpe.add_element('C', 0.856, percent_type='ao')#6

bpe.add_element('B', 0.143, percent_type='ao')#5

bpe.add_element('O', 0.889, percent_type='ao')#8


lead=openmc.Material(name='Lead')
```



```
lead.set_density('g/cm3',11.35)

lead.add_element('Pb',1.0)


concrete=openmc.Material(name='Concrete')

concrete.set_density('g/cm3',2.3)      #Harus   disesuaikan
dengan   uji
beton

concrete.add_element('H', 0.01, percent_type='ao')#1

concrete.add_element('C', 0.01, percent_type='ao')#6

concrete.add_element('O', 0.52, percent_type='ao')#8

concrete.add_element('Na', 0.01, percent_type='ao')#11

concrete.add_element('Mg', 0.01, percent_type='ao')#12

concrete.add_element('Al', 0.01, percent_type='ao')#13

concrete.add_element('Si', 0.25, percent_type='ao')#14

concrete.add_element('S', 0.01, percent_type='ao')#16

concrete.add_element('K', 0.01, percent_type='ao')#19

concrete.add_element('Ca', 0.01, percent_type='ao')#20

concrete.add_element('Fe', 0.01, percent_type='ao')#26

concrete.add_element('Pb', 0.01, percent_type='ao')#82
```

```
iron = openmc.Material(name="Iron")

iron.set_density('g/cm3',7.87) #Harus disesuaikan dengan uji beton

iron.add_element('Fe', 1.0)#1


materials=openmc.Materials(
[air,air2,air3,water,soft,bpe,lead,concrete,iron])

materials.export_to_xml()


################################################

rotationDegree=int(input("Harap masukkan sudut rotasi LINAC\n= "))

############# Geometry ########################

#x

t1=openmc.XPlane(632)

t2=openmc.XPlane(632-76.5)
```



```
t3=openmc.XPlane(632-76.5-155)

t3fe=openmc.XPlane(632-76.5-155-235)

t4=openmc.XPlane(632-76.5-155-76.5)

t5=openmc.XPlane(632-76.5-155-76.5)

t6=openmc.XPlane(632-76.5-155-235)


b1=openmc.XPlane(-632)

b2=openmc.XPlane(-632+76.5)

b3=openmc.XPlane(-632+76.5+155.0)

b4=openmc.XPlane(-632+76.5+155.0+76.5)

b5=openmc.XPlane(-632+76.5+250.5)
```

```
#y #Di berkas 125, tapi ga make sense jadi diubah ke 1200 biar
masuk
angkanya

u1=openmc.YPlane(190.0+250.0+120.0+185.0+81.0)

u2=openmc.YPlane(190.0+250.0+120.0+185.0)

ufe=openmc.YPlane(190.0+250.0+120.0+185.0-10)

u3=openmc.YPlane(190.0+250.0+120.0)

u4=openmc.YPlane(190.0+250.0)

u5=openmc.YPlane(190.0)



s1=openmc.YPlane(-190.0-185.0-128.0)

s2=openmc.YPlane(-190.0-185.0)

s3=openmc.YPlane(-190.0)



#z total tinggi = 6000, lantai 1 setebal 480



zm1=openmc.ZPlane(-300.0)

zmax=openmc.ZPlane(300.0)

z3=openmc.ZPlane(-300.0+48.0+124.0+186.0+117.0+125.0)#1250
ATO
2500???

z2=openmc.ZPlane(-300.0+48.0+124.0+186.0+117.0)

z1=openmc.ZPlane(-300.0+48.0+124.0+186.0)
```



```
z0=openmc.ZPlane(-300.0+48.0)



#pintu utara, pintu barat, pintu selatan geometri nya

pu=openmc.YPlane(190.0+250.0+120.0+185.0+40.0)

#                   ^^^ asumsi pintu lebih lebar 40cm dibandingkan
lubang pintunya
```

```
pb0=b5

pb1=openmc.XPlane(-632.0+76.5+250.5-1) #Angkanya ini masih ngarang
karena gatau tebal pintu, ada kemungkinan formulanya di
RHSPintu.py
salah

pb2=openmc.XPlane(-632.0+76.5+250.5-1-15)

pb3=openmc.XPlane(-632.0+76.5+250.5-1-15-1)

ps=u3



###############################################

dt1 = -t1 & +t2 & +s3 & -u5 & +z0 & -z2

dt2 = -t2 & +t3 & +s1 & -u1 & +z0 & -z2

dt3 = -t3 & +t4 & +s3 & -u5 & +z0 & -z2


db1 = +b1 & -b2 & +s3 & -u5 & +z0 & -z2

db2 = +b2 & -b3 & +s1 & -u3 & +z0 & -z2

db3 = +b3 & -b4 & +s3 & -u5 & +z0 & -z2


du1 = +b5 & -t3 & -u1 & +u2 & +z0 & -z2

du2 = +b3 & -t6 & -u3 & +u4 & +z0 & -z2


ds1 = +b3 & -t3 & +s1 & -s2 & +z0 & -z2


fe1 = -t3 & +t3fe & -u1 & +ufe & +z0 & -z2

datas= +b1 & -t1 & +s1 & -u1 & +z2 & -z3 #celing

datte= +b4 & -t4 & -u5 & +s3 & +z1 & -z2 #linac's middle wall

dbaw = +b1 & -t1 & +s1 & -u1 & +zm1 & -z0 #flooring
```

```
#############################################

#pintu

ppb = -pu & +ps & -pb0 & +pb1 & +z0 & -z2#pintu Pb

pbpe= -pu & +ps & -pb1 & +pb2 & +z0 & -z2#pintu BPE

ppb2= -pu & +ps & -pb2 & +pb3 & +z0 & -z2#pintu Pb



#Udara

#void1= -dt1 & +dt2 & -dt3 & +db1 & -db2 & +db3 & -du1 & +du2 & -
ds1 & +ppb & -pbpe & +ppb2 & +datas & -dbaw

#void1cell=openmc.Cell(fill=air,region=void1)



#Cell =



dt1cell=openmc.Cell(fill=concrete,region=dt1)

dt2cell=openmc.Cell(fill=concrete,region=dt2)

dt3cell=openmc.Cell(fill=concrete,region=dt3)

db1cell=openmc.Cell(fill=concrete,region=db1)

db2cell=openmc.Cell(fill=concrete,region=db2)

db3cell=openmc.Cell(fill=concrete,region=db3)

du1cell=openmc.Cell(fill=concrete,region=du1)

du2cell=openmc.Cell(fill=concrete,region=du2)

ds1cell=openmc.Cell(fill=concrete,region=ds1)


ppbcell=openmc.Cell(fill=lead,region=ppb)

pbpecell=openmc.Cell(fill=bpe,region=pbpe)

ppb2cell=openmc.Cell(fill=lead,region=ppb2)
```

```
datascell=openmc.Cell(fill=concrete,region=datas)

dattecell=openmc.Cell(fill=concrete,region=datte)

dbawcell=openmc.Cell(fill=concrete,region=dbaw)


fecell=openmc.Cell(fill=iron, region=fe1)
```



```
###############################################

#           Detektor/Tally       #


detd= 4.5

detde=2

#                Utara       #

deu1=openmc.YPlane (190.0+250.0+120.0+185.0+81.0+30.0)

deu1t=openmc.YPlane(190.0+250.0+120.0+185.0+81.0+30.0+10.8)
#+tebal detektor

deu1ts=openmc.YPlane(190.0+250.0+120.0+185.0+81.0+30.0+detde)
#+tebal detektor

deu2=openmc.YPlane (190.0+250.0+120.0+185.0+81.0+100.0)

deu2t=openmc.YPlane(190.0+250.0+120.0+185.0+81.0+100.0+10.8)

b5=openmc.XPlane(-632+76.5+250.5)

deu2ts=openmc.YPlane(190.0+250.0+120.0+185.0+81.0+100.0+detde)

deu3=openmc.YPlane (190.0+250.0+120.0+185.0+81.0+200.0)

deu3t=openmc.YPlane(190.0+250.0+120.0+185.0+81.0+200.0+10.8)

deu3ts=openmc.YPlane(190.0+250.0+120.0+185.0+81.0+200.0+detde)


deuz0=openmc.ZPlane(-300.0+48.0+100.0)#Tinggi     detektor,
default
untuk semua detektor kecuali atas
```

```
deuz1=openmc.ZPlane(-300.0+48.0+100.0+200.0)

deuz0s=openmc.ZPlane(-300.0+48.0+100.0-detd/2)#Tinggi
detektor,
default untuk semua detektor kecuali atas

deuz1s=openmc.ZPlane(-300.0+48.0+100.0+detd/2)

#deuz0s=openmc.ZPlane(-300.0+48.0+100.0-(150/2))#Tinggi
detektor,
default untuk semua detektor kecuali atas

#deuz1s=openmc.ZPlane(-300.0+48.0+100.0+(150/2))



deubb=openmc.XPlane(-632.0+76.5+250.5+100.0) #Koordinat x nya
masih
ngasal

deubt=openmc.XPlane(-632.0+76.5+250.5+100.0+50.0)

deubbs=openmc.XPlane(-632.0+76.5+250.5+100.0-detd/2) #Koordinat x
nya masih ngasal
```



```
deubts=openmc.XPlane(-632.0+76.5+250.5+100.0+detd/2)



deucb=openmc.XPlane(250.0) #koordinat x nya masih ngasal

deuct=openmc.XPlane(250.0+50.0)

deucbs=openmc.XPlane(250.0-detd/2) #koordinat x nya masih ngasal

deucts=openmc.XPlane(250.0+detd/2)



detb1= +deu1 & -deu1t & +deuz0 & -deuz1 & +deubb & -deubt
#detektor
utara barat, x nya ngasal

detb2= +deu2 & -deu2t & +deuz0 & -deuz1 & +deubb & -deubt

detb3= +deu3 & -deu3t & +deuz0 & -deuz1 & +deubb & -deubt

dett1= +deu1 & -deu1t & +deuz0 & -deuz1 & +deucb & -deuct
#detektor
utara timur, x nya ngasal

dett2= +deu2 & -deu2t & +deuz0 & -deuz1 & +deucb & -deuct
```

```
dett3= +deu3 & -deu3t & +deuz0 & -deuz1 & +deucb & -deuct


detb1s= +deu1 & -deu1ts & +deuz0s & -deuz1s & +deubbs & -deubts
#detektor utara barat, x nya ngasal

detb2s= +deu2 & -deu2ts & +deuz0s & -deuz1s & +deubbs & -deubts

detb3s= +deu3 & -deu3ts & +deuz0s & -deuz1s & +deubbs & -deubts

dett1s= +deu1 & -deu1ts & +deuz0s & -deuz1s & +deucbs & -deucts
#detektor utara timur, x nya ngasal

dett2s= +deu2 & -deu2ts & +deuz0s & -deuz1s & +deucbs & -deucts

dett3s= +deu3 & -deu3ts & +deuz0s & -deuz1s & +deucbs & -deucts


#Detektor

detub1cell=openmc.Cell(fill=air2,region=detb1) #sel detektor barat
1

detub2cell=openmc.Cell(fill=air2,region=detb2)

detub3cell=openmc.Cell(fill=air2,region=detb3)

detut1cell=openmc.Cell(fill=air2,region=dett1)

detut2cell=openmc.Cell(fill=air2,region=dett2)

detut3cell=openmc.Cell(fill=air2,region=dett3)
```



```
detub1scell=openmc.Cell(fill=air2,region=detb1s)   #sel
detektor
barat 1

detub2scell=openmc.Cell(fill=air2,region=detb2s)

detub3scell=openmc.Cell(fill=air2,region=detb3s)

detut1scell=openmc.Cell(fill=air2,region=dett1s)

detut2scell=openmc.Cell(fill=air2,region=dett2s)

detut3scell=openmc.Cell(fill=air2,region=dett3s)
```

```
#             Timur         #

det1=openmc.XPlane(632.0+30.0)

det1t=openmc.XPlane(632.0+30.0+10.8)

det1ts=openmc.XPlane(632.0+30.0+detde)

det2=openmc.XPlane(632.0+100.0)

det2t=openmc.XPlane(632.0+100.0+10.8)

det2ts=openmc.XPlane(632.0+100.0+detde)

det3=openmc.XPlane(632.0+200.0)

det3t=openmc.XPlane(632.0+200.0+10.8)

det3ts=openmc.XPlane(632.0+200.0+detde)


detu=openmc.YPlane(25.0)

dets=openmc.YPlane(-25.0)

detus=openmc.YPlane(detd/2)

detss=openmc.YPlane(-detd/2)


dett1= +det1 & -det1t & +deuz0 & -deuz1 & -detu & +dets

dett2= +det2 & -det2t & +deuz0 & -deuz1 & -detu & +dets

dett3= +det3 & -det3t & +deuz0 & -deuz1 & -detu & +dets

dett3s= +det3 & -det3t & +deuz0 & -deuz1 & -detu & +dets

dett1s= +det1 & -det1ts & +deuz0s & -deuz1s & -detus & +detss

dett2s= +det2 & -det2ts & +deuz0s & -deuz1s & -detus & +detss

dett3s= +det3 & -det3ts & +deuz0s & -deuz1s & -detus & +detss
```



```
dett1cell=openmc.Cell(fill=air2,region=dett1)
```

```
dett2cell=openmc.Cell(fill=air2,region=dett2)

dett3cell=openmc.Cell(fill=air2,region=dett3)

dett1scell=openmc.Cell(fill=air2,region=dett1)

dett2scell=openmc.Cell(fill=air2,region=dett2)

dett3scell=openmc.Cell(fill=air2,region=dett3)


#              Barat       #


deb1=openmc.XPlane(-632.0+76.5+250.5-1-15-30.0)

deb1t=openmc.XPlane(-632.0+76.5+250.5-1-15-30.0-10.8)

deb1ts=openmc.XPlane(-632.0+76.5+250.5-1-15-30.0-detde)

deb2=openmc.XPlane(-632.0+76.5+250.5-1-15-100.0)

deb2t=openmc.XPlane(-632.0+76.5+250.5-1-15-100.0-10.8)

deb2ts=openmc.XPlane(-632.0+76.5+250.5-1-15-100.0-detd)

deb3=openmc.XPlane(-632.0+76.5+250.5-1-15-200.0)

deb3t=openmc.XPlane(-632.0+76.5+250.5-1-15-200.0-10.8)

deb3ts=openmc.XPlane(-632.0+76.5+250.5-1-15-200.0-detde)


debu=openmc.YPlane(190.0+250.0+120.0+185.0-67.5)

debs=openmc.YPlane(190.0+250.0+120.0+185.0-67.5-50.0)

debus=openmc.YPlane(190.0+250.0+120.0+185.0-(67.5+detd/2))

debss=openmc.YPlane(190.0+250.0+120.0+185.0-(67.5-detd/2))


detb1= -deb1 & +deb1t & +deuz0 & -deuz1 & -debu & +debs

detb2= -deb2 & +deb2t & +deuz0 & -deuz1 & -debu & +debs

detb3= -deb3 & +deb3t & +deuz0 & -deuz1 & -debu & +debs

detb1s= -deb1 & +deb1ts & +deuz0s & -deuz1s & -debus & +debss
```

```
detb2s= -deb2 & +deb2ts & +deuz0s & -deuz1s & -debus & +debss

detb3s= -deb3 & +deb3ts & +deuz0s & -deuz1s & -debus & +debss
```



```
detb1cell=openmc.Cell(fill=air2,region=detb1)

detb2cell=openmc.Cell(fill=air2,region=detb2)

detb3cell=openmc.Cell(fill=air2,region=detb3)

detb1scell=openmc.Cell(fill=air2,region=detb1s)

detb2scell=openmc.Cell(fill=air2,region=detb2s)

detb3scell=openmc.Cell(fill=air2,region=detb3s)


#             Atas          #

deau=openmc.XPlane(25.0)

deas=openmc.XPlane(-25.0)

deaus=openmc.XPlane(detd/2)

deass=openmc.XPlane(-detd/2)


deat=openmc.YPlane(25.0)

deab=openmc.YPlane(-25.0)

deats=openmc.YPlane(detd/2)

deabs=openmc.YPlane(-detd/2)


dea1=openmc.ZPlane(300.0+50.0) #1250 ATO 2500???

dea1t=openmc.ZPlane(300.0+50.0+10.8) #1250 ATO 2500???

dea1ts=openmc.ZPlane(300.0+30.0+detde) #1250 ATO 2500???

dea2=openmc.ZPlane(300.0+100.0)

dea2t=openmc.ZPlane(300.0+100.0+10.8)
```

```
dea2ts=openmc.ZPlane(300.0+100.0+detde)

dea3=openmc.ZPlane(300.0+200.0)

dea3t=openmc.ZPlane(300.0+200.0+10.8)

dea3ts=openmc.ZPlane(300.0+200.0+detde)


deta1= +dea1 & -dea1t & -deau & +deas & +deab & -deat

deta2= +dea2 & -dea2t & -deau & +deas & +deab & -deat

deta3= +dea3 & -dea3t & -deau & +deas & +deab & -deat

deta1s= +dea1 & -dea1ts & -deaus & +deass & +deabs & -deats
```



```
deta2s= +dea2 & -dea2ts & -deaus & +deass & +deabs & -deats

deta3s= +dea3 & -dea3ts & -deaus & +deass & +deabs & -deats


deta1cell=openmc.Cell(fill=air2,region=deta1)

deta2cell=openmc.Cell(fill=air2,region=deta2)

deta3cell=openmc.Cell(fill=air2,region=deta3)

deta1scell=openmc.Cell(fill=air2,region=deta1s)

deta2scell=openmc.Cell(fill=air2,region=deta2s)

deta3scell=openmc.Cell(fill=air2,region=deta3s)


#Water Phantom 10x10x5

#TODO: Water phantom dpp and lateral tallies, to get flux value
and
relative comparison to the dose.

#make tally, search for the dose, search the corresponding flux,
use it as the conversion rate for searching dose(sv/h)


phantom_rotation=270 #phantom rotation
```

```
pr=phantom_rotation

detaxu=openmc.YPlane(5)

detaxs=openmc.YPlane(-5)


if pr==0 or pr==180:

  detaxt=openmc.XPlane(5)

  detaxb=openmc.XPlane(5)

  detaxza=openmc.ZPlane(-128+100+2.5)

  detaxzb=openmc.ZPlane(-128+100-2.5)

if pr==180:

  detaxt=openmc.XPlane(5)

  detaxb=openmc.XPlane(-5)

  detaxza=openmc.ZPlane(-128-100+2.5)

  detaxzb=openmc.ZPlane(-128-100-2.5)

elif pr==90 or pr==270:

  detaxza=openmc.XPlane(2.5)
```



```
  detaxzb=openmc.XPlane(-2.5)

  detaxt=openmc.ZPlane(-128+5)

  detaxb=openmc.ZPlane(-128-5)


else:

  print('Phantom Rotation Error, benarkan kode pr')


detax= -detaxu & +detaxs & -detaxt & +detaxb & -detaxza & +detaxzb
# type: ignore


detaxcell=openmc.Cell(fill=water,region=detax)
```

```
#Kotak Udara Pembatas

ymax=openmc.YPlane(1100,boundary_type='vacuum')

ymin=openmc.YPlane(-1100,boundary_type='vacuum')

xmax=openmc.XPlane(945,boundary_type='vacuum')

xmin=openmc.XPlane(-945,boundary_type='vacuum')

zmin=openmc.ZPlane(-550,boundary_type='vacuum')

zmaxx=openmc.ZPlane(550,boundary_type='vacuum')



#void1cell = openmc.Cell(fill=air, region= (-datascell.region) &
(-
dt1cell.region) & (-dt2cell.region) & (-dt3cell.region) & (-
db1cell.region) & (-db2cell.region) & (-db3cell.region) & (-
du1cell.region) & (-du2cell.region) & (-ds1cell.region))



void1= +zmin & -zmaxx \

  & +ymin & -ymax & +xmin & -xmax\

  & ~dt1cell.region & ~dt2cell.region & ~dt3cell.region \

    & ~db1cell.region & ~db2cell.region & ~db3cell.region \

        & ~du1cell.region & ~du2cell.region & ~ds1cell.region\

        & ~ppbcell.region & ~pbpecell.region & ~ppb2cell.region\

        & ~datascell.region & ~dbawcell.region &
~dattecell.region\

        & ~detb1cell.region & ~detb2cell.region &
~detb3cell.region\
```



```
      &     ~detub1cell.region          &   ~detub2cell.region   &
~detub3cell.region\

      &     ~detut1cell.region          &   ~detut2cell.region   &
~detut3cell.region\
```

```
      & ~dett1cell.region & ~dett2cell.region & ~dett3cell.region\
      & ~deta1cell.region & ~deta2cell.region & ~deta3cell.region\

void1cell = openmc.Cell(fill=air, region=void1)

univ=openmc.Universe(cells= [dt1cell,dt2cell,dt3cell,
              db1cell,db2cell,db3cell,
              du1cell,du2cell,ds1cell,
              ppbcell,pbpecell,ppb2cell,
              datascell,dbawcell,dattecell,
              void1cell,
              detb1cell,detb2cell,detb3cell,
              detub1cell,detub2cell,detub3cell,
              detut1cell,detut2cell,detut3cell,
              dett1cell,dett2cell,dett3cell,
              deta1cell,deta2cell,deta3cell,
              detb1scell,detb2scell,detb3scell,
              detub1scell,detub2scell,detub3scell,
              detut1scell,detut2scell,detut3scell,
              dett1scell,dett2scell,dett3scell,
              deta1scell,deta2scell,deta3scell,
              detaxcell,fecell])
geometry=openmc.Geometry(univ)
geometry.export_to_xml()

colors= {}
colors [lead]='black'
```

```
colors [bpe]='lightblue'

colors [concrete]='grey'
```



```
colors [air]='white'

colors [air2]='blue'

ssh=SOURCE_SIZE/2

###############################################

#            Rotation

###############################################

def sposi(d,rot):

     u= (-sin(radians(rot)))

     v= 0

     w= (-cos(radians(rot))) #source position

     uvw = (u,v,w)

     xyz = ( d*(sin(radians(rot))) ), 0, -128+
( d*(cos(radians(rot))
))

     xyz = xyz

     if rot==0 or rot==180:

    xyz1=   (   d*(sin(radians(rot)))         )+ssh,     ssh,
-128+   (
d*(cos(radians(rot)) )+0.1)

    xyz2=   (   d*(sin(radians(rot)))         )-ssh,    -ssh,
-128+   (
d*(cos(radians(rot)) ))

     else:

    xyz1=   (   d*(sin(radians(rot)))         )+0.1,     ssh,
-128+   (
d*(cos(radians(rot)) )+ssh)

    xyz2=    (   d*(sin(radians(rot)))          ),     -ssh,
-128+   (
d*(cos(radians(rot)) )-ssh)
```

```
     return uvw, xyz,xyz1,xyz2

     #asumsi tinggi pasien 75

###############################################

#       Input (linac distance,rotation)   #

linacuvw, linacxyz,linacxyzn1,linacxyzn2=sposi(SSD,rotationDegree)

###############################################

print("linacuvw,linacxyz,linacuvwn1,linacuvwn2",linacuvw,
linacxyz,linacxyzn1,linacxyzn2)



###############################################
```



```
#       Penampil Geometri         #

###############################################

plot=openmc.Plot()

plot.basis='xy'

plot.origin=(0,200,0)

plot.width=(2000,2000)

plot.pixels=(2000,2000)

plot.filename='xy_room'

plot.colors={

    lead:'black',

    bpe:'lightblue',

    concrete:'grey',

    air:'white',

    air2:'blue'

}

plot.color_by='material'
```

```
plot_file=openmc.Plots( [plot])

plot_file.export_to_xml()

openmc.plot_geometry()

#!convert something.ppm something.png


plot.to_ipython_image()


plot.basis='yz'

plot.filename='yz_room'

plot.width=(2000,2000)

plot_file=openmc.Plots( [plot])

plot_file.export_to_xml()

openmc.plot_geometry()

plot.to_ipython_image()


plot.basis='xz'
```



```
plot.filename='xz_room'

plot.pixels=(2000,2000)

plot.width=(2000,2000)

plot.origin=(0,0,0)

plot_file=openmc.Plots( [plot])

plot_file.export_to_xml()

openmc.plot_geometry()

plot.to_ipython_image()

"""
```

```
plt.rcParams.update({'font.size': 5})

univ.plot(width=(2500,2700),basis='xy',color_by='material',colors=
colors)

plt.savefig('xyRSHS.png',dpi=500, bbox_inches='tight')

plt.show()

univ.plot(width=(1400,1040),basis='xz',color_by='material',colors=
colors)

plt.savefig('xzRSHS.png',dpi=500, bbox_inches='tight')

plt.show()

univ.plot(width=(1800,1040),basis='yz',color_by='material',colors=
colors)

plt.savefig('yzRSHS.png',dpi=500, bbox_inches='tight')

plt.show()

"""



#Tidak terdapat library matplotlib pada server, sehingga penampil
geometri harus dimatikan untuk running server



###############################################

#         Plot Grid          #

###############################################

#ax.set_title('Distribusi Dosis Ruangan (uSv/hour)') #type: ignore

#univ.plot(width=(2000,2000),basis='xy',color_by='material',colors
=colors)
```



```
#plt.savefig('DoseDistributionMap.png',dpi=500,
bbox_inches='tight')



###############################################
```

```
#        Setting           #

###############################################

settings=openmc.Settings()

source =openmc.Source()

"""

#source.space=openmc.stats.Points(xyz=)

source.space=openmc.stats.Point(xyz=linacxyz) # type: ignore

#phi2=openmc.stats.Isotropic() #isotropic ato uniform?

#phi1=openmc.stats.Monodirectional((0,0,1))

phi     =openmc.stats.Uniform(0.0,2*pi)     #     type:
ignore
#phi=distribution of the azimuthal angle in radians

#mu= distribution of the cosine of the polar angle



#tan theta = r/SAD=20/1000; theta = atan(20/100)=0.19739555984988;
cos theta=0.98058

mu=openmc.stats.Uniform(0.98058,1) # type: ignore #mu=
distribution
of the cosine of the polar angle



source.angle                                                     =
openmc.stats.PolarAzimuthal(mu,phi,reference_uvw=linacuvw) # type:
ignore

source.energy = openmc.stats.Discrete( [10e6], [1]) #10MeV # type:
ignore

"""

#source.space=openmc.stats.Point(xyz=linacxyz) # type: ignore

#source.space = openmc.stats.Box((-FIELD_SIZE/2-d, -SOURCE_SIZE/2,
-SOURCE_SIZE/2),        (-FIELD_SIZE/2-d-t,         SOURCE_SIZE/2,
SOURCE_SIZE/2))

phi     =openmc.stats.Uniform(0.0,2*pi)     #     type:
ignore
#phi=distribution of the azimuthal angle in radians
```

```
mu = openmc.stats.Uniform(cos(atan2(((FIELD_SIZE-SOURCE_SIZE)/2),SSD)), 1)
```



```
source.space = openmc.stats.Box((linacxyzn1), (linacxyzn2))

##source.angle = openmc.stats.Monodirectional(linacuvw)

source.angle = openmc.stats.PolarAzimuthal(mu,phi,reference_uvw=linacuvw) # type: ignore

source.energy = openmc.stats.Discrete( [10e6], [1])



source.particle = 'photon'

#source.particle = 'neutron'

settings.source = source

settings.batches= 3

settings.particles = particle

#Asumsi 36e7 partikel pada mula, pada 600MU=600cGy/m=6Gy/m=6Sv/m=6e6uSv/m=360e6uSv/h

settings.run_mode = 'fixed source'

settings.photon_transport = True

settings.export_to_xml()



###############################################

#          Tallies               #

###############################################

tally = openmc.Tallies()

#TODO: Change Tally size to smaller actual tally size value

#Tally Dose Distribution

mesh = openmc.RegularMesh() # type: ignore

mesh.dimension = [500, 500]
```

```
xlen = 2000;ylen = 2000

mesh.lower_left = [-xlen/2, -xlen/2]

mesh.upper_right = [ylen/2, ylen/2]

mesh_filter = openmc.MeshFilter(mesh)


tally1 = openmc.Tally(name = 'Room Dose Distribution')

tally1.scores = ['flux']
```



```
particle1 = openmc.ParticleFilter('photon')


energy, dose = openmc.data.dose_coefficients('photon',     'RLAT')
#Data konve # type: ignore

dose_filter = openmc.EnergyFunctionFilter(energy, dose) #konvert
partikel energi tertentu ke deskripsi icrp116 partikelcm/src-
>pSvcm3/srcwphantom_cell,particle3,dose_filter


tally1.filters= [mesh_filter,particle1,dose_filter]

tally.append(tally1)


#Tally Detektor

filter_cell = openmc.CellFilter((detb1cell,detb2cell,detb3cell,\

              detub1cell,detub2cell,detub3cell,\

              detut1cell,detut2cell,detut3cell,\

              dett1cell,dett2cell,dett3cell,\

              deta1cell,deta2cell,deta3cell))

tally2 = openmc.Tally(name = 'flux')

particle2 = openmc.ParticleFilter('photon')

tally2.filters = [filter_cell, particle2, dose_filter]
```

```
tally2.scores = ['flux']

tally.append(tally2)



filter_cell_small
=
openmc.CellFilter((detb1scell,detb2scell,detb3scell,\

              detub1scell,detub2scell,detub3scell,\

              detut1scell,detut2scell,detut3scell,\

              dett1scell,dett2scell,dett3scell,\

              deta1scell,deta2scell,deta3scell))

tally4 = openmc.Tally(name = 'flux detektor small')

tally4.filters = [filter_cell_small, particle2, dose_filter]

tally4.scores = ['flux']

tally.append(tally4)
```



```
energy2, dose2 = openmc.data.dose_coefficients('neutron',    'AP')
#Data konve # type: ignore

dose_filter = openmc.EnergyFunctionFilter(energy2, dose2) #konvert
partikel energi tertentu ke deskripsi icrp116 partikelcm/src-
>pSvcm3/srcwphantom_cell,particle3,dose_filter

particle3=openmc.ParticleFilter('neutron')

neutroncell = openmc.CellFilter((detb1cell,detb2cell,detb3cell))

tally3= openmc.Tally(name='neutron')

tally3.filters= [neutroncell,particle3,dose_filter]

tally3.scores= ['flux']

tally.append(tally3)



""""
```

```
#Tally Water Phantom

wphantom_cell=openmc.CellFilter(detaxcell)

tally3=openmc.Tally(name='wphantom')

particle3= openmc.ParticleFilter('photon')

#tally3.filters= [wphantom_cell,particle3]

#Energy_filter = openmc.EnergyFilter( [1e-3, 1e13])

tally3.filters=   [wphantom_cell,particle3,dose_filter]    #output
pSvcm3/src

tally3.scores = ['flux']

tally.append(tally3)

"""



tally.export_to_xml()

openmc.run()
```

Program Pembaca Ruangan

```
import openmc # type: ignore

import matplotlib.pyplot as plt

from matplotlib.colors import LogNorm
```



```
import pandas as pd



sp = openmc.StatePoint('./statepoint.3.h5')



f=open("output.txt","w")

f.write(str(sp.tallies))

f.close()

print(sp.n_particles)
```

```
print(sp.tallies)

meshtally = sp.tallies [1]

print(sp.tallies [1])

print(sp.tallies [2])

#print(sp.tallies [3])



x=500#harus sama dengan resolusi pada file utama

y=500



conversion = 16368352439.27938

# dose * flux * src_rate * t / V = [pSv cm²] [p-cm/src] [src/sec]
[sec]/ [cm³]= [pSv]

# dose * cfac *t/V=pSv

# dose*cfac*60/V




v=(2000/x)*(2000/y)*5 #volume of room dose distribution



# {{{ ######## Pembuatan Visualisasi Distribusi Dosis
#################

dosevalues=meshtally.get_values() #Nilai perpixel dari grid
500x500

dosevalues.shape=(x,y)

#pSvcm3/src*(src/s)/cm3=pSv/s
```



```
dosevalues = dosevalues*conversion/v #pSv/s = (pSv*cm3/src) *src/s
/cm3

dose=(dosevalues/1_000_000)*3600 #pSv/s -> uSv/hour
```

```
dose=dose [::-1,:]


GD=pd.DataFrame(dose)


GD.to_csv("gridTallyDose.csv")


fig, ax = plt.subplots()

cs = ax.imshow(dose, cmap='coolwarm', norm=LogNorm()) # type:
ignore

cb = plt.colorbar(cs)

ax.set_title('Distribusi Dosis Ruangan (uSv/hour)') #type: ignore

plt.savefig('RoomDoseDistribution.png',dpi=900 )

plt.axis('off')

#################################################################

# }}}


celltally = sp.tallies [2]

celldosevalues = celltally.get_values() #pSvcm3/src;dosevolume per
source

celldosestddev = celltally.std_dev

print(celltally)

celldosevalues.shape = celldosevalues.shape [0]

celldosestddev.shape = celldosestddev.shape [0]


smalltally = sp.tallies [2]

smalltallydose = smalltally.get_values()

smalltallydose.shape = smalltallydose.shape [0]

print("===========",smalltallydose,"===========")
```

```
vcell=10.8*50*200 #cm3
#conversion = 4092088109819844.5
```



```
"""
dose_psvs=celldosevalues * conversion/ vcell
usvh_dose=(dose_psvs/1e6)*3600

usvh_stddev=(stddev_psvs/1e6)*3600

dose=usvh_dose
dosestddev=usvh_stddev
"""
dose= celldosevalues*conversion/vcell
dosestddev=celldosestddev*conversion/vcell

DF=pd.DataFrame(dose,dosestddev)

DF.to_csv("cellTallyDose.csv")

#dose=(dose/1e6)*3600 #pSv/s -> uSv/h 3.6e9
#dosestddev = (celldosestddev * s_rate / vcell) *(1e6/3600)
#k=2457897.1324533457
#dose=k*celldosevalues*600/vcell
#dosestddev = (k*celldosevalues*600/vcell) *(1e6/3600)

print(dose)
for v,s in zip(dose,dosestddev):
```

```
  f=open("output.txt","a")

  print(f'{v:.7e} +- {s:.7e}uSv/h')

  f.write(str(f'\n{v} +- {s} uSv/h'))

  f.close()


plt.show()

#dosevalues = dosevalues*s_rate/v #picosieverts/s
```



```
"""

# plt.show()

plt.savefig('figlogdoseplot.png')

plt.show()

# plt.close()

# plt.clf()

"""


# plt.clf()
```


LAMPIRAN B

Hasil Pengukuran Laju Dosis Commisioning

Dosis Background : 0,095 µSv/jam

LAMPIRAN C

Perhitungan Analitik Desain Bunker

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LAMPIRAN D
Contoh Penggunaan Variasi Sudut Gantry LINAC

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LAMPIRAN E
Program Pengubahan Fluks Ke Dosis

```
from pathlib import Path

import numpy as np

_FILES = (
    ('electron', 'electrons.txt'),
    ('helium', 'helium_ions.txt'),
    ('mu-', 'negative_muons.txt'),
    ('pi-', 'negative_pions.txt'),
    ('neutron', 'neutrons.txt'),
    ('photon', 'photons.txt'),
    ('photon kerma', 'photons_kerma.txt'),
    ('mu+', 'positive_muons.txt'),
    ('pi+', 'positive_pions.txt'),
    ('positron', 'positrons.txt'),
    ('proton', 'protons.txt')
)
```

```
_DOSE_ICRP116 = {}



def _load_dose_icrp116():

    """Load effective dose tables from text files"""

    for particle, filename in _FILES:

      path = Path(__file__).parent / filename

      data = np.loadtxt(path, skiprows=3, encoding='utf-8')

      data [:, 0]*= 1e6   # Change energies to eV

      _DOSE_ICRP116 [particle]= data
```



```
 [docs]def dose_coefficients(particle, geometry='AP'):

  """Return effective dose conversion coefficients from ICRP-116



  This function provides fluence (and air kerma) to effective dose
conversion

  coefficients for various types of external exposures based on
values in

  `ICRP
Publication                                116
<https://doi.org/10.1016/j.icrp.2011.10.001>`_.

  Corrected values found in a correigendum are used rather than
the
values in

  theoriginal report.



  Parameters

  ----------

  particle :    {'neutron',   'photon',   'photon   kerma',
'electron',
'positron'}
```

```
        Incident particle

    geometry : {'AP', 'PA', 'LLAT', 'RLAT', 'ROT', 'ISO'}

      Irradiation geometry assumed. Refer to ICRP-116 (Section 3.2)
for the

        meaning of the options here.



    Returns

    -------

    energy : numpy.ndarray

        Energies at which dose conversion coefficients are given

    dose_coeffs : numpy.ndarray

        Effective dose coefficients in [pSv cm^2]at provided
energies.
For

        'photon kerma', the coefficients are given in [Sv/Gy].



    """

    if not _DOSE_ICRP116:
```



```
     _load_dose_icrp116()



  # Get all data for selected particle

  data = _DOSE_ICRP116.get(particle)

  if data is None:

    raise ValueError(f"{particle} has no effective dose data")



  # Determine index for selected geometry

  if particle in ('neutron', 'photon', 'proton', 'photon kerma'):
```

```
    index = ('AP', 'PA', 'LLAT', 'RLAT', 'ROT', 'ISO').index(geometry)
  else:
    index = ('AP', 'PA', 'ISO').index(geometry)


  # Pull out energy and dose from table
  energy = data [:, 0].copy()
  dose_coeffs = data [:, index + 1].copy()
  return energy, dose_coeffs
```


LAMPIRAN F

Contoh Tabel Konversi Fluks Dosis ICRP 116

LAMPIRAN G

Program Pengubahan Fluks Ke Dosis

No.	NSUB	Sublibrary Name	Short Name	VII.1	VII.0	VI.8
1	0	Photonuclear	g	163	163	-
2	3	Photo-atomic	photo	100	100	100
3	4	Radioactive decay	decay	3817	3838	979
4	5	Spont. fis. yields	s/fpy	9	9	9
5	6	Atomic relaxation	ard	100	100	100
6	10	Neutron	n	423	393	328
7	11	Neutron fis.yields	n/fpy	31		

31 31

8 12 Thermal scattering tsl 21
20 15

9 19 Standards std 8
8 8

10 113 Electro-atomic e 100
100 100

11 10010 Proton p 48
48 35

12 10020 Deuteron d 5
5 2

13 10030 Triton t 3
3 1

14 20030 3He he3 2
2 1

1. The photonuclear sublibrary was carried over unchanged from ENDF/B-VII.0.
   It contains evaluated cross sections for 163 materials (all isotopes) mostly up to
   140 MeV. The sublibrary was supplied by Los Alamos National Laboratory
   (LANL) and it is largely based on the IAEA-coordinated collaboration
   completed in 2000.
2. The photo-atomic sublibrary was taken over from ENDF/B-VII.0=ENDF/B-
   VI.8. It contains data for photons from 10 eV up to 100 GeV interacting with
   atoms for 100 materials (all elements). The sublibrary was supplied by Lawrence
   Livermore National Laboratory (LLNL).
3. The decay data sublibrary has been re-evaluated and considerably improved by
   the Brookhaven National Laboratory (BNL).

4. The spontaneous fission yields were taken over from ENDF/B-VII.0=ENDF/B-
   VI.8. The data were supplied by LANL.
5. The atomic relaxation sublibrary was taken over from ENDF/B-VII.0=ENDF/B-

VI.8. It contains data for 100 materials (all elements)
supplied by LLNL.
6. The neutron reaction sublibrary represents the heart of the
ENDF/B-VII.1
   library. The sublibrary has been updated and extended, it
contains 423 materials,
   including 422 isotopic and 1 elemental evaluation.. Altogether
234 materials
   have been changed in ENDF/B-VII.1 compared to ENDF/B-VII.0.
7. Neutron fission yields were reevaluated for 239Pu (fast and 14
MeV) by Los
   Alamos; others were taken over from ENDF/B-VII.0=ENDF/B-VI.8,
having
   been supplied by LANL.
8. The thermal neutron scattering sublibrary was carried over
unchanged from
   ENDF/B-VII.0. It contains thermal scattering-law data, largely
supplied by
   LANL, with the addition of SiO2 in ENDF/B-VII.1 from ORNL.
9. The neutron cross section standards sublibrary was carried over
from ENDF/B-
   VII.0 unchanged. Therefore, as for VII.0, the VII.0 standard
cross sections were
   completely adopted by the VII.1 neutron reaction sublibrary
except for the
   thermal cross section for 235U(n,f) where a slight difference
occurs to satisfy
   thermal data testing, and some very small differences for
235U(n,f) and 238U(n,γ)
   in the keV–MeV region.
10. The electro-atomic sublibrary was taken over from ENDF/B-
VII.0=ENDF/B-
    VI.8. It contains data for 100 materials (all elements)
supplied by LLNL.
11. The proton-induced reactions were carried over from ENDF/B-
VII.0 with only
    minor corrections. These were supplied by LANL and the data
being mostly to
    150 MeV.
12. The deuteron-induced reactions were supplied by LANL, carried
over
    unchanged from ENDF/B-VII.0. This sublibrary contains 5
evaluations.
13. The triton-induced reactions were supplied by LANL, carried
over unchanged
    from ENDF/B-VII.0. This sublibrary contains 3 evaluations.
14. Reactions induced with 3He were supplied by LANL, carried over
unchanged
    from ENDF/B-VII.0. This sublibrary contains 2 evaluations.
https://www.nndc.bnl.gov/endf-b7.1/
https://openmc.org/official-data-libraries/

Lampiran: Histori alur persetujuan

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2	Ketua Program Studi Sarjana Teknik Nuklir	Dr. Ing. Ir. Sihana	Paraf	Selasa, 30 Juli 2024 16:27
3	Ketua Departemen Teknik Nuklir dan Teknik Fisika	Dr. Ir. Alexander Agung, S.T., M.Sc., IPU.	Tanda Tangan	Selasa, 30 Juli 2024 17:59

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EVALUASI KESELAMATAN BUNKER LINAC 10 MEV RSUP DR. HASAN SADIKIN MENGGUNAKAN METODE SIMULASI OPENMC

SKRIPSI

untuk memenuhi sebagian persyaratan
untuk memperoleh derajat Sarjana
Program Studi Teknik Nuklir

Diajukan oleh
MUHAMMAD ARYA HANIF
19/443954/TK/49150

Kepada

DEPARTEMEN TEKNIK NUKLIR DAN TEKNIK FISIKA
FAKULTAS TEKNIK
UNIVERSITAS GADJAH MADA
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SKRIPSI

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Karya ini kupersembahkan untuk almarhum kakek saya

“We play with the cards we're dealt.” – Stan Edgar

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Yogyakarta, 19 Juli 2023

Muhammad Arya Hanif

DAFTAR ISI

DAFTAR TABEL

DAFTAR GAMBAR

DAFTAR LAMBANG DAN SINGKATAN

Lambang Romawi

Lambang	*Kuantitas*	*Satuan*
d_{sec}	Jarak isosenter terhadap titik pengukuran	cm
$\dot{D}$	Laju Dosis	Sieverts/jam
Da	Dosis per fluens	Sieverts cm^2
$\dot{D}_{max}$	Laju Dosis Maksimum	Sieverts/jam
Do	Dosis Keluaran Fungsi OpenMC	Sieverts
Do_{sk}	Dosis Keluaran Sel Kalibrasi	Sieverts
$\dot{D}$	Laju Dosis *Linear Accelerator*	MU/menit
D_U	Dosis berdasarkan utilitas sudut	µSv/minggu
E	Energi	MeV
HVL	*Half Value Layer*	mm
S	Faktor Konversi	-
SR	*Source Rate*	sumber/detik
T	Faktor Okupansi	-
TVL	*Tenth Value Layer*	cm
T	Waktu	detik
t_w	Waktu penyinaran untuk mencapai beban kerja	jam
U	Faktor Utilitas	%
V	Volume	cm^3
W	Beban Kerja / *Workload*	Gy/minggu

X	Ketebalan materi	m

Lambang Yunani

Lambang	*Kuantitas*	*Satuan*
Φ	Fluks simulasi	Partikel-cm /sumber

Subskrip

bo	*Deskripsi*
maks	Maksimum
Sk	Sel Kalibrasi

Singkatan

AAPM TG	*American Association of Physicists in Medicine Task Group*
AC	*Alternating Current*
BAPETEN	Badan Pengawas Tenaga Nuklir
BNCT	*Boron Neutron Capture Therapy*
BPE	*Boron Loaded Polyethylene*
CPRG	*Computational Reactor Physics Group*
CSEWG	*Cross Section Evaluation Working Group*
Dr.	Dokter
ENDF	Evaluated Nuclear Data Library
FF	*Flattening Filter*
HVL	*Half Value Layer*
IAEA	*International Atomic Energy Agency*
KeV	*Kilo electron Volt*

LINAC	*Linear Accelerator*
MIT	*Massachusetts Institute of Technology*
MeV	*Mega electron Volt*
MV	*Mega Volt*
MU	*Monitor Unit*
NCRP	*National Council on Radiation Protection & Measurements*
NBD	Nilai Batas Dosis
RSUP	Rumah Sakit Umum Pusat
SAD	*Source to Axis Distance*
TVL	*Tenth Value Layer*
UGM	Universitas Gadjah Mada

EVALUASI KESELAMATAN BUNKER LINAC 10 MEV RSUP DR. HASAN SADIKIN MENGGUNAKAN METODE SIMULASI OPENMC

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INTISARI

Penelitian ini dilakukan untuk mengkaji keselamatan pada desain bunker radioaterapi LINAC Elekta Synergy yang dibangun di RSUP Dr. Hasan Sadikin. Visualisasi persebaran dosis akan dilakukan demi membantu interpretasi persebaran dosis pada bunker LINAC tersebut.

Pengkajian dilakukan dengan menggunakan simulasi Monte Carlo melalui kode transportasi partikel OpenMC versi 0.13.1. Pengukuran laju dosis dilakukan dengan variasi jarak 30, 100 dan 200 cm terhadap dinding bunker. Faktor okupansi, utilitas serta beban kerja digunakan untuk mengetahui nilai batas dosis yang berlaku pada titik-titik tersebut.

Laju dosis pada sekitar ruangan berada pada rentang $8{,}28 \times 10^{-3}$ hingga 0,208 μSv/minggu. Nilai ini jauh kali lebih kecil dibandingkan dengan pembatas dosis yang diberlakukan oleh BAPETEN pada Perka No. 4 tahun 2013. Hal ini berarti keselamatan bunker sudah sesuai dengan standar yang berlaku. Visualisasi laju dosis juga telah dilakukan untuk membantu interpretasi dosis yang terjadi pada masing-masing titik pengukuran laju dosis pada sekitar bunker.

Kata kunci: LINAC, OpenMC, Simulasi Monte Carlo, Proteksi Radiasi

Pembimbing Utama : Ir. Anung Muharini, M.T., IPM.
Pembimbing Pendamping : Anisza Okselia, S.T., M.Si.

SAFETY EVALUATION OF HASAN SADIKIN HOSPITAL'S 10 MEV LINAC BUNKER USING OPENMC SIMULATION METHOD

Muhammad Arya Hanif

19/443954/TK/49150

Submitted to the Department of Nuclear Engineering and Engineering Physics
Faculty of Engineering Universitas Gadjah Mada on July 12, 2024
in partial fulfillment of the requirement for the Degree of
Bachelor of Engineering in Nuclear Engineering

ABSTRACT

This study will examine the safety of LINAC Elekta Synergy bunker room at RSUP Dr. Hasan Sadikin. Safety examination will be conducted by comparing the radiation dose on the surrounding room with the dose limit enforced by BAPETEN. Visualization of fdose will be done to help the interpretation of dose spread happening on the bunker

The examination is executed using Monte Carlo simulation provided by 0.13.1 version of OpenMC particle transportation code. This research was carried out by evaluation the surrounding room's dose at varying distances of 30, 100 and 200 cm to the bunker wall. Occupancy, utility and workload factors will be used to determine the dose limit values that apply at each of these points.

The dose surrounding the room is detected on the range of 8,28 × 10^{-3} to 0.208 µSv/week. This is already much smaller than the implemented rule enforced by BAPETEN on Perka No. 3 2013. This means that the safety of the design is in accordance with the standard. Dose rate visualization has also been carried out to help interpret the dose that occurs at each dose rate measurement point around the bunker.

Keywords: LINAC, OpenMC, Monte Carlo Simulation, Radiation Protection

Supervisor : Ir. Anung Muharini, M.T., IPM.
Co-supervisor : Anisza Okselia, S.T., M.Si

BAB I
PENDAHULUAN

I.1. Latar Belakang

Radioterapi adalah satu dari tiga metode utama pada pengobatan penyakit kanker. Tiga metode utama tersebut adalah bedah sebagai metode fisik, kemoterapi sebagai metode kimia dan radioterapi sebagai metode fisika. Terdapat 66 alat radioterapi di 44 pusat pengobatan kangker yang tersebar di 16 provinsi di Indonesia [1]. Ini berarti terdapat densitas alat radioterapi sejumlah 0,247 pada tiap juta penduduk Indonesia pada tahun tersebut. Hal tersebut cukup jauh dibandingkan target 7 alat radioterapi per juta penduduk yang disarankan oleh *International Atomic Energy Agency* (IAEA) [2]. Hal itu juga berarti terdapat banyak jumlah bunker radioterapi yang perlu dibuat untuk mencapai target tersebut.

Bunker radioterapi adalah ruangan yang diperuntungkan untuk melindungi masyarakat umum dan pekerja rumah sakit dari radiasi yang dipancarkan oleh fasilitas radioterapi. Bunker *Linear Accelerator* (LINAC) Synergy pada RSUP Dr. Hasan Sadikin telah melalui serangkaian tahap perancangan serta verifikasi hingga dapat beroperasi. Regulasi yang perlu dipertimbangkan adalah pasal 41 Peraturan Kepala BAPETEN nomor 4 tahun 2013 mengenai pembatas dosis pada pekerja radiasi dan masyarakat [3]. Kemudian untuk memenuhi standar tersebut, organisasi kenukliran seperti IAEA dan *National Council on Radiation Protection & Measurements* (NCRP) telah membuat metode perhitungan agar dapat memenuhi standar yang sebelumnya telah ditetapkan [4]. Selain itu, terdapat pula berkas *National Council on Radiation Protection & Measurements* (NCRP *Report* nomor 151 yang merupakan panduan dalam pembuatan fasilitas keselamatan radioterapi [5]. Regulasi ini dibuat untuk mempermudah optimasi manfaat terhadap risiko pada pemanfaatan fasilitas radioterapi. Metode optimasi juga dapat dilakukan dengan menggunakan simulasi komputer untuk dapat memastikan bahwa laju dosis yang ada pada sekitar ruangan bunker sudah sesuai dengan standar yang berlaku. Desain bunker radioterapi yang dibentuk berdasarkan regulasi tersebut kemudian dapat diperiksa menggunakan perhitungan analitik maupun pengukuran laju dosis setelah instalasi dilakukan.

Perhitungan analitik dilakukan pada titik-titik di balik dinding bunker yang dibangun. Ketebalan dinding antara LINAC dengan titik tersebut kemudian dibandingkan dengan nilai minimum ketebalan dinding bungker agar laju dosis dapat lebih rendah dibandingkan nilai pembatas dosis. Setelah bunker terbangun dan LINAC dipasang pada ruangan tersebut, penyinaran dengan dosis maksimum dilakukan untuk kemudian laju dosis pada bangunan disekitarnya dapat diperiksa. Berdasarkan pemeriksaan tersebut, laju dosis pada atap bungker melebihi 5 μSv/jam. Hal ini dinilai sebagai lebih besar dibandingkan pembatas dosis yang ditetapkan oleh BAPETEN pada perka no 3 tahun 2013 yang sejumlah 200 μSv/minggu. Selain itu, hasil pengukuran juga tidak selalu menunjukkan laju dosis yang lebih kecil pada jarak yang lebih jauh. Hal ini berlawanan dengan *inverse square law* yang diterapkan pada perhitungan laju dosis. Interpretasi laju dosis tersebut dapat dilakukan dengan visualisasi dari persebaran laju dosis pada ruangan tersebut dengan menggunakan simulasi komputer.

Simulasi komputer adalah metode yang dapat digunakan untuk mendesain alat maupun fasilitas pendukung pemanfaatan alat yang memiliki radiasi. Hal ini dikarenakan pembuatan prototipe seringkali melibatkan komponen-komponen yang tidak murah untuk dibuat. Selain itu sifat radiasi yang probabilistik dan tidak kasat mata juga membuat pengecekan parameter pada pembuatan alat tidak semudah alat-alat yang memiliki parameter yang kasat mata. Hal ini membuat simulasi dengan kemampuan memvisualisasi radiasi untuk memberikan data yang lebih lengkap serta fleksibilitas dalam mengelola data yang dihasilkan. Simulasi dapat dimanfaatkan untuk perencanaan fasilitas, penelitian desain, optimasi parameter dan lain-lain [6] . Salah satu kelebihan dari simulasi adalah visualisasi distribusi dosis serta pergerakan partikel. Kelebihan itu merupakan hal yang tidak dimiliki baik oleh perhitungan analitik pengukuran langsung. Terlepas dari kelebihan tersebut, akurasi simulasi sangat bergantung pada seberapa detail geometri serta parameter-parameter yang ditetapkan pada simulasi tersebut. Visualisasi yang dihasilkan oleh simulasi dapat membantu interpretasi distribusi dosis yang terjadi pada fasilitas radioterapi.

I.2. Perumusan Masalah

Setelah desain bunker dibuat berdasarkan regulasi yang ada. Desain tersebut akan disimulasikan demi mengetahui perkiraan dosis yang diterima pada masing-masing ruangan disekitar LINAC. Berdasarkan laju dosis tersebut maka dapat diperkirakan berapa banyak dosis yang diterima oleh para pekerja pada tiap minggunya. Berdasarkan hal diatas, perumusan masalah untuk menyelesaikan persolan tersebut adalah sebagai berikut:

1. Berapa hasil laju dosis pada sekitar ruangan bunker tersebut menurut simulasi OpenMC?
2. Bagaimana distribusi dosis pada ruangan Bunker LINAC?
3. Apakah laju dosis simulasi memenuhi pembatas dosis?

I.3. Tujuan Penelitian

1. Menganalisis distribusi dosis LINAC pada bunker.
2. Mengetahui keselamatan pada bunker LINAC.

I.4. Manfaat Penelitian

Menambah masukan kepada manajemen RSUP Dr. Hasan Sadikin untuk manajemen keselamatan radiasi tenaga kesehatan dan Masyarakat umum pada ruangan bunker Synergy.

I.5. Batasan Masalah

Terdapat beberapa batasan masalah pada penelitian ini di antaranya:

1. Detektor menggunakan *tally* deteksi sederhana dan tidak mereplikasi desain detektor asli.
2. Penyinaran menggunakan model sederhana geometri LINAC dengan energi 10 MeV.
3. Simulasi Monte Carlo menggunakan OpenMC versi 0.13.1.
4. Desain Bunker yang dikaji adalah desain bunker LINAC Synergy RSUP Dr. Hasan Sadikin.
5. Simulasi menggunakan 1.000.000.000 partikel dengan 3 *batch.*
6. Pengkajian tidak dilakukan pada bagian atas maupun sisi selatan dari bunker.

BAB II

TINJAUAN PUSTAKA

Ruangan bunker LINAC memerlukan spesifikasi yang berbeda dibandingkan ruangan rumah sakit pada umumnya. Hal ini dikarenakan radiasi primer, radiasi sekunder serta neutronik yang dihasilkan oleh foton berenergi tinggi memerlukan ketebalan serta bahan yang berbeda dibandingkan ruangan rumah sakit pada umumnya. Oleh karena itu, posisi serta pembuatan merupakan hal yang memakan biaya yang cukup besar. Hal ini membuat penggunaan ulang ruangan bunker radioterapi akan memangkas banyak biaya. Terlepas dari pemangkasan biaya tersebut, keselamatan tetap menjadi prioritas dari perwujudan fasilitas radioterapi. Berdasarkan kajian yang dilakukan oleh A.S. Ferdiansyah, peralihfungsian bunker LINAC 6 MeV ke 10 MeV dapat dilakukan dengan penambahan nano besi oksida serta boron karbida setebal 2 cm pada ruang tunggu, atap dan ruang arsip pada bunker LINAC RSUP Dr. Sardjito. Besi oksida dipilih karena memiliki atenuasi tinggi terhadap neutron cepat, sedangkan boron karbida memiliki nilai atenuasi tinggi terhadap radiasi neutron termal hingga epitermal [7].

Pengukuran dosis radiasi pada luar ruangan bunker LINAC juga pernah dilakukan oleh pihak vendor pada RSUP Dr. Sardjito sebelumnya. Penelitian ini berhasil menemukan bahwa dosis radiasi tidak mencapai 2 μSv/h pada seluruh daerah sekitar ruang radioterapi kecuali pada depan pintu sebesar 4,7 μSv/h. Pengukuran dilakukan pada tujuh titik yang berbeda di sekitar ruangan bunker. Berdasarkan penelitian tersebut. Pada skripsinya Y. M. Dinata menyarankan bahwa penebalan pintu bunker dilakukan dengan menggunakan timah [8].

Penggunaan simulasi untuk memeriksa laju dosis pada sekitar ruangan *hot lab* juga telah dilakukan demi mengkaji faktor ekonomi dari kombinasi bahan dinding bunker. Pada penelitiannya, A. Rafi melakukan kombinasi berupa beton, beton dengan 1 cm lapisan timbal dan beton dengan 2 cm lapisan timbal. Penggunaan lapisan timbal dapat mengurangi ketebalan dinding hingga 5 cm pada titik tertentu. Pelapisan timbal dapat mengeluarkan biaya hingga tujuh kali lipat dibandingkan hanya menggunakan beton. Penelitian ini menyimpulkan bahwa desain yang paling optimal dari segi proteksi radiasi dan ekonomis adalah desain *hot lab* tanpa pelapisan timbal pada dinding bunker [9].

Simulasi juga telah dilakukan oleh N. K. T. Kusnaedi untuk pembuatan desain ruangan bunker fasilitas radioterapi dengan energi 30 MeV. Energi tersebut dihasilkan oleh siklotron yang digunakan pada fasilitas *Boron Neutron Capture Therapy* (BNCT). Standar dosis radiasi maksimum yang diberlakukan oleh BAPETEN pada fasilitas radioterapi berhasil dipenuhi pada desain bunker tersebut. Ketebalan dinding yang diperlukan adalah 100 cm hingga 350 cm pada beton dan 100 cm hingga 250 cm pada beton boron dan beton barit [10]

OpenMC adalah salah satu *opensource* yang dipakai dalam pembuatan desain fasilitas radioaktif. Pengkajian keselamatan bunker mengunakan kode simulasi transportasi foton dan neutron ini telah dilakukan sebelumnya dilakukan pada bunker Rumah Sakit Jogja International Hospital. Pada penelitian tersebut, hasil analitik serta simulasi dibandingkan. Penelitian ini telah berhasil mengetahui bahwa bunker tersebut telah sesuai dengan standar keselamatan BAPETEN berdasarkan simulasi OpenMC maupun perhitungan analitik. Diketahui juga bahwa bunker tersebut memerlukan penambahan ketebalan *Boron Loaded Polyethylene* BPE sejumlah minimal 92 milimeter [11].

Penelitian untuk memvalidasi kredibilitas hasil simulasi Monte Carlo pada profil LINAC juga telah dilakukan dengan mereplikasi *Quality Assurance* (*QA*) pada LINAC varian seri TrueBeam STx di Rumah Sakit Songklanagarind. Penelitian yang dilakukan oleh M.A. Efendi ini menggunakan kode PRIMO untuk mensimulasikan *Precentage Depth Dose* (PDD) serta *lateral profile* menggunakan geometri LINAC. Hasil simulasi sudah sesuai dengan hasil ekperimental, perhitungan *Analitical Anisotropic Algorithm* (AAA) serta *golden beam* yang disediakan oleh pihak manufaktur. PRIMO adalah kode Monte Carlo yang diperuntungkan secara spesifik untuk radioterapi[12].

BAB III
DASAR TEORI

III.1. Radiasi

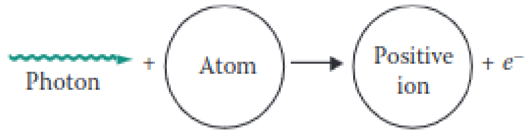
Radiasi adalah perambatan energi yang tidak melalui media ataupun ruang yang kemudian diserap oleh benda lain. Terdapat beberapa jenis pembagian radiasi, salah satunya adalah dengan kemampuannya dalam mengionisasi atom lain. Dalam kategori ini, terdapat dua jenis radiasi yaitu radiasi pengion dan non pengion [13]. Energi pengion pada umumnya berada pada rentang kiloelektron volt serta megaelektron volt[14]. Sedangkan pada radiasi partikel, radiasi pengion adalah radiasi yang memiliki muatan energi yang cukup tinggi. Jenis partikel serta energi merupakan faktor utama dari proteksi radiasi merupakan faktor utama penentu efek radiasi terhadap suatu materi [15].

III.1.1. Interaksi Foton dengan Materi

Foton adalah paket diskrit dari energi elektromagnetik yang dapat berinteraksi dengan partikel lain seperti atom [16]. Pada radiasi foton, panjang gelombang radiasi pengion kira-kira kurang dari 10 nm[13]. Terdapat banyak reaksi yang dapat dihasilkan oleh interaksi foton dengan materi. Akan tetapi, tiga interaksi utama yaitu fotolistrik, hamburan compton serta *pair production*.

III.1.1.1. Fotolistrik

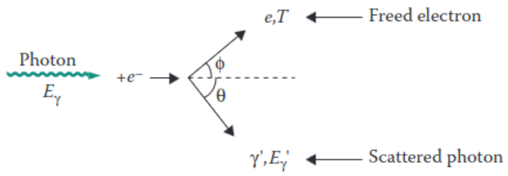
Efek fotolistrik terjadi ketika foton dan elektron yang terikat dengan atom bertubrukan. Energi dari foton tersebut kemudian membuat elektron pada atom lepas sebagai elektron bebas[13]. Interaksi fotolistrik ini hanya terjadi jika energi ikat pada elektron tersebut sejumlah maupun lebih kecil dibandingkan energi yang dimiliki oleh foton [17].



Gambar 3.1. Ilustrasi Efek Fotolistrik [13]

III.1.1.2. Hamburan Compton

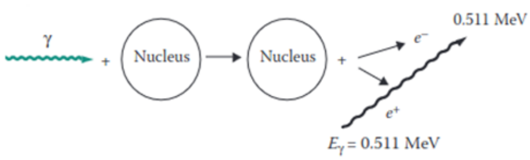
Hamburan compton terjadi ketika foton dengan energi keV atau lebih menabrak elektron bebas. Elektron bebas memiliki energi dalam kisaran eV. Elektron yang bertumbukan kemudian akan terhempas ke arah yang berbeda.



Gambar 3.2. Ilustrasi Efek Fotolistrik [13]

III.1.1.3. *Pair Production*

Pair production terjadi ketika foton berenergi tinggi menubruk nukleus. Energi sejumlah 1,022 MeV yang dimiliki foton tersebut kemudian akan terkonversi menjadi elektron dan positron. Energi yang tersisa kemudian akan terdistribusi merata pada dua partikel tersebut.



Gambar III.1 Ilustrasi Efek *Pair Production* [13]

III.1.1.4. Fotodisintegrasi

Fotodisintegrasi atau reaksi *photonuclear* terjadi ketika nukleus menangkap foton energi tinggi menghempaskan neutron yang ada pada sebuah nukleus[17]. Neutron yang dihasilkan tersebut dapat diabsorbsi oleh berbagai material yang kemudian membuat material tersebut menjadi radioaktif. Reaksi fotodisintegrasi hanya dapat terjadi ketika energi foton melebihi 7 MeV [15]. Fotoneutron biasanya memiliki energi lebih tinggi dibandingkan neutron termal.

III.1.2. Interaksi Elektron dengan Materi

Terdapat dua jenis foton yang akan muncul pada interaksi elektron terhadap suatu materi. Elektron yang kehilangan energi dikarenakan pengurangan energi pada medan elektromagnetik yang dimiliki nukleus akan mengeluarkan energi yang terserap sebagai foton yang disebut dengan Bremsstrahlung[14]. Interaksi yang menciptakan sinar X pada umumnya adalah interaksi Bremsstrahlung. Reaksi Bremsstrahlung juga dapat terjadi ketika elektron tersebut menghantam elektron yang ada pada atom. Sekitar 80% dari energi foton yang dihasilkan pada 100 keV adalah radiasi bremsstrahlung dan sisanya adalah sinar X karakteristik. Sedangkan pada elektron dengan orde MeV, jumlah Bremsstrahlung akan sangat dominan hingga sinar-x karakteristik dapat diabaikan. Sinar X karakteristik terjadi ketika elektron menabrak dan mementalkan elektron lain. Posisi elektron tersebut kemudian akan digantikan oleh elektron lain dengan tingkat energi yang lebih rendah dan energi sisanya akan terpental sebagai sinar x karakteristik.

III.1.3. Interaksi Neutron dengan Materi

Neutron dapat berinteraksi dengan nukleus hanya melalui gaya nuklir karena neutron tidak memiliki polaritas [13]. Ketika neutron mendekati nukleus, neutron tidak bisa dipengaruhi oleh medan magnet pada suatu atom. Hal ini membuat interaksi nuklir lebih memungkinkan untuk terjadi pada interaksi neutron dengan materi dibandingkan dengan partikel bermuatan. Terdapat dua jenis interaksi yang terjadi yaitu hamburan dan serapan.

III.1.3.1. Hamburan

Pada interaksi hamburan, neutron berinteraksi dengan nukleus. Akan tetapi, kedua partikel tersebut muncul kembali setelah reaksi terjadi dengan nomor atom serta nomor massa yang sama seperti semula. Terdapat dua interaksi partikel yang dapat terjadi yaitu elastis dan tidak elastis. Energi kinetik pada kedua partikel akan disetarakan tanpa perubahan energi kinetik total pada interaksi elastis. Pada interaksi inelastik, sebagian energi kinetik berubah menjadi energi eksitasi pada nucleus. Nukleus tersebut kemudian akan mengalami deeksitasi dengan mengeluarkan energi tersebut sebagai foton.

III.1.3.2. Serapan

Ketika serapan terjadi, neutron menghilang. Sesudah neutron menghilang, satu partikel atau lebih muncul. Partikel yang dapat dihasilkan di antaranya adalah neutron, proton, inti helium, gamma serta atom yang berbeda dengan atom yang bertabrakan.

III.2. Dosimetri

Dosimetri radiasi mengacu pada pengukuran, perhitungan serta penaksiran dari radiasi pengion yang diterima oleh suatu objek termasuk pada tubuh manusia. Hal ini termasuk pada radiasi yang masuk secara internal melalui pencernaan maupun pernapasan serta eksternal seperti LINAC. Pembedaan dosis sebagai dosis serap dan ekivalen diperlukan karena efek biologis pada sel manusia sehat sangat dipengaruhi oleh jenis radiasi serta energi dari radiasi tersebut.

III.2.1. Dosis Serap

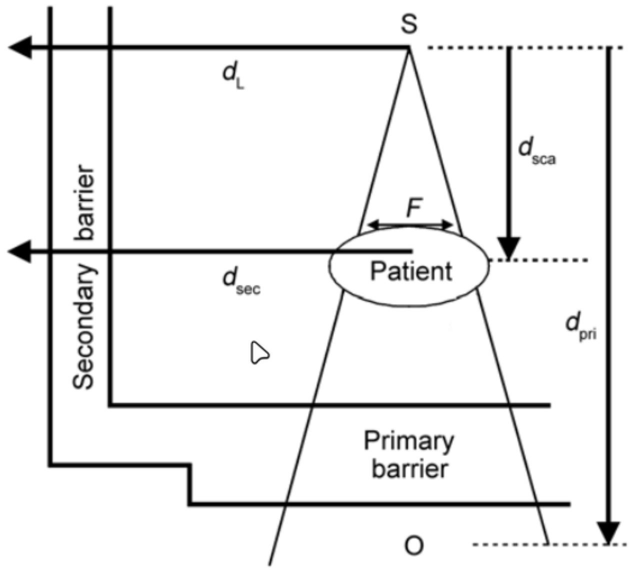
Energi yang terserap pada suatu massa pada pemanfaatan radiasi diukur dengan besaran Gray (Gy). Energi yang diserap oleh suatu massa tersebut adalah energi yang diberikan oleh sebuah radiasi. Nilai ini juga dimanfaatkan dalam pengkajian besar energi yang diterima oleh suatu kanker. Hal ini dikarenakan faktor bobot yang dimiliki jenis radiasi tertentu terhadap suatu kanker memiliki nilai yang berbeda dengan faktor bobot pada jaringan sehat.

III.2.2. Dosis Ekivalen

Dosis ekivalen adalah jumlah energi radiasi yang terserap pada suatu jaringan sehat. Nilai ini biasa digunakan dalam proteksi radiasi pada pekerja radiasi terhadap pemanfaatan fasilitas radioaktif. Penggunaan dosis serap dilakukan karena radiasi memiliki efek yang berbeda pada jaringan manusia tergantung pada jenis serta energi dari radiasi tersebut. Faktor ini disebut sebagai faktor bobot. Untuk mendapatkan dosis ekivalen, maka nilai dosis serap dikalikan faktor bobot. Besaran satuan internasionalnya masih sama dengan dosis serap. Faktor bobot adalah senilai 1 pada radiasi foton maupun elektron, 20 pada radiasi alfa dan 2-10 untuk neutron berdasarkan jumlah energinya [15].

III.3. Proteksi Radiasi

Proteksi radiasi adalah tindakan yang dilakukan untuk mengurangi pengaruh paparan radiasi pada pekerja radiasi maupun masyarakat umum. Pada pemanfaatan LINAC, Intensitas radiasi diminimalisasi melalui proses atenuasi[15]. Perancangan desain bunker radioterapi berpedoman pada NCRP 151 dan SRS 47. Dokumen tersebut digunakan sebagai pedoman dalam pembuatan dinding primer, sekunder maupun pintu bunker. konsep-konsep seperti utilitas sudut LINAC dan okupansi untuk menghitung ekspektasi dosis yang diterima oleh pekerja pada titik yang dikaji.



Gambar III.2 Ilustrasi jenis dinding dan jarak ruang bunker LINAC[5]

III.3.1. Pembatas Dosis (P)

Pembatas dosis adalah batas atas dosis pekerja radiasi dan anggota masyarakat. Pembatas dosis pada pekerja radiasi adalah sebesar setengah dari NBD pertahun atau senilai 10 mSv per tahun [3]. Nilai ini kemudian diatur kembali secara detail pada Peraturan Kepala BAPETEN nomor 3 tahun 2013. Pada peraturan tersebut pembatas dosis bagi pekerja radiasi adalah sebesar 0,2 mSv perminggu [18]. Sedangkan pada masyarakat senilai 0,01 mSv perminggu. Nilai tersebut ditemukan dengan membagi NBD menjadi minggu dengan 50 minggu pada tiap tahunnya. Kemudian pembatas dosis dapat dicapai dengan mempertimbangkan beban kerja, faktor okupansi dan faktor utilitas dari pengoperasian LINAC.

III.3.2. Beban Kerja (W)

Beban kerja atau disebut juga *workload* adalah total dosis yang dipancarkan oleh LINAC selama fraksi-fraksi penyinaran dijalankan. Nilai ini didapatkan dengan menggunakan fraksi dosis pada tiap penyinaran, jumlah pasien perhari serta jumlah hari kerja pengoperasian alat.

Pada perhitungan analitik pada dinding sekunder, terdapat nilai C yang menentukan beban kerja pada *treatment*. Nilai ini adalah sebesar 4,5 pada VMAT dan IMRT serta 1 pada konvensional maupun pada *Quality Assurance* (QA).

III.3.3. Faktor Utilitas (U)

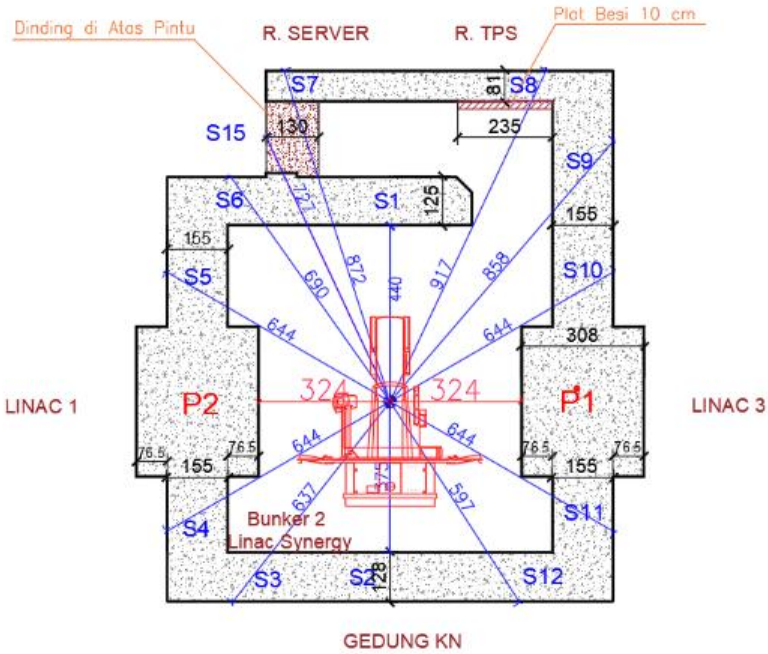
Faktor utilitas adalah presentase beban kerja di mana sinar diarahkan pada penghalang primer tertentu. Semakin kecil nilai ini, semakin besar toleransi nilai laju dosis pada titik di balik dinding.

Tabel III.1 Faktor Utilitas LINAC[18]

Sudut (interval 90°)	U (%)
0° (bawah)	31,0
90° dan 270°	21,3
180° (atas)	26,3

III.3.4. Faktor Okupansi (T)

Faktor okupansi adalah rata-rata waktu seseorang untuk terekspos dari LINAC ketika penyinaran dilakukan. Semakin besar nilai okupansi, akan semakin besar pula ketebalan dinding yang diperlukan untuk mencapai target nilai atenuasi pada dinding. Nilai okupansi sangat berhubungan dalam pengkategorian ruangan di sekitar bunker. Kategori ruangan pada fasilitas nuklir terbagi menjadi dua yaitu daerah terkontrol dan daerah tidak terkontrol [17]. Semua ruang pada sekitar bunker ruangan LINAC Synergy merupakan ruangan terkontrol yang hanya diakses oleh pekerja. Maka dari itu nilai pembatas dosis yang diterapkan merupakan batas dosis pekerja radiasi.



Gambar III.3 Denah ruang LINAC Synergy dan sekitarnya [19]

Tabel III.2 Faktor Okupansi Pada Ruangan Sekitar Bunker [5]

Nilai Faktor Okupansi	Jenis Ruangan
1	Ruang *treatment planning system* dan ruang operator
0,5	Gedung kedokteran nuklir dan bunker
0,2	Koridor
0,125	Ruang di depan pintu *treatment*
0,025	Atap

Dapat dilihat melalui Gambar III.4 bahwa terdapat 15 dinding sekunder serta dua dinding primer yang harus dikaji melalui perhitungan analitik. Jarak dinding primer direpresentasikan melalui garis berwarna merah, sedangkan dinding sekunder direpresentasikan melalui garis biru. Perbedaan nilai okupansi pada masing-masing titik tersebut direpresentasikan melalui jenis ruangan yang ada pada sisi lain dari bunker tersebut. Pada ruang operator/server dan ruang *treatment planning system* (TPS), nilai okupansi dihitung sebagai ruangan pekerja. Sedangkan pada bunker LINAC 3 dan Gedung Kedokteran Nuklir, pengkajian dilakukan sebagai ruangan *treatment*.

III.4. Akselerator Linear

Akselerator linear adalah salah satu jenis akselerator. Akselerator adalah sebuah alat yang dapat meningkatkan energi dari suatu partikel bermuatan seperti elektron, proton, deutron, triton, partikel alfa, partikel-partikel subatom atau ion positif dan negatif[20].

Berbeda dengan telekobalt dan brakhiterapi yang memerlukan bahan baku berupa radionuklida, LINAC dapat berjalan selama ada asupan listrik serta alat tersebut tidak rusak. Radiasi yang dihasilkan oleh LINAC berkisar antara 4 MeV hingga 25 MeV. Energi tersebut lebih tinggi daripada telekobalt yang memiliki energi 1,17 dan 1,33 MeV. Energi yang variatif ini juga memungkinkan LINAC untuk secara efektif mengenai tumor baik pada kedalaman yang dangkal maupun tumor yang tertimbun oleh jaringan lemak yang banyak.

III.4.1. Komponen LINAC

Terdapat beberapa komponen utama dari akselerator linear diantaranya adalah tabung sinar X, *focusing coil*, target, serta kolimator. Perbedaan potensial pada tabung sinar X akan mengeluarkan elektron yang kemudian dapat ditingkatkan energinya oleh *focusing coil*. Elektron tersebut kemudian akan menabrak target yang biasanya merupakan unsur yang memiliki nomor atom tinggi. Target tersebut kemudian akan menciptakan radiasi bremsstrahlung yang kemudian akan dimanfaatkan untuk mengiradiasi kanker.

III.4.2. Kalibrasi Dosis Keluaran LINAC

Terdapat dua jenis kalibrasi yang dilakukan pada pengoperasian LINAC, yaitu kalibrasi relatif dan kalibrasi absolut.

III.4.2.1. Kalibrasi Relatif

Kalibrasi relatif adalah proses perbandingan pengukuran dosis radiasi pada suatu titik tertentu dengan pengukuran dosis pada titik referensi yang telah diketahui nilainya. Kalibrasi relatif pada LINAC bertujuan untuk memastikan bahwa dosis radiasi yang dihasilkan oleh mesin ini akurat dan konsisten. Proses kalibrasi ini dimulai dengan menentukan titik referensi. Titik referensi yang dipakai pada LINAC adalah dosis pada nilai maksimum dari *depth dose* nya. Pengukuran dosis dilakukan agar kemudian dapat dibandingkan dengan titik referensi. Dari sinilah faktor koreksi pada profil LINAC bisa dibuat.

III.4.3. Kalibrasi Absolut

Kalibrasi Absolut adalah kalibrasi pada LINAC dilakukan melalui *ionization chamber* yang berada pada kepala LINAC. Pada kalibrasi tersebu, nilai *Monitor Unit* (MU) akan dikalibrasikan agar setara dengan 1 cGy pada kedalaman dengan dosis output maksimum [21]. Nilai MU merupakan besaran dosis yang dihasilkan oleh LINAC. Nilai MU ini senantiasa akan diperiksa uniformitasnya serta kesetaraannya agar selalu +- 1% dari nilai terhadap cGy. Hal ini sesuai dengan pedoman yang disediakan oleh *American Association of Physicists in Medicine Task Group* 40 (AAPM TG)[18].

Terdapat beberapa usulan metode lain dalam nilai MU ini. Salah satunya adalah penyetaraan bukan pada output maksimum, melainkan pada kedalaman tertentu seperti 10 cm dari jarak target terhadap pasien[22]. Pada jurnal yang ditulis oleh F. Heuvel, beliau memberi pendapat bahwa nilai ini akan lebih resistan terhadap kesalahan distribusi pada pasien. Akan tetapi pada jurnal yang sama, beliau juga menyatakan bahwa argumen ini tidak cukup kuat apalagi untuk mengubah definisi yang sudah terimplementasi pada regulasi pembuatan fasilitas bunker radioterapi.

III.5. Monte Carlo

Monte Carlo adalah metode yang menggunakan sampel dari suatu fenomena untuk memperkirakan rata-rata populasi atas fenomena tersebut [18]. Prinsip kalkulasi serta *database* yang digunakan sebagai data probabilitas interaksi partikel akan dijelaskan sebagai berikut .

III.5.1. Prinsip Kalkulasi Monte Carlo

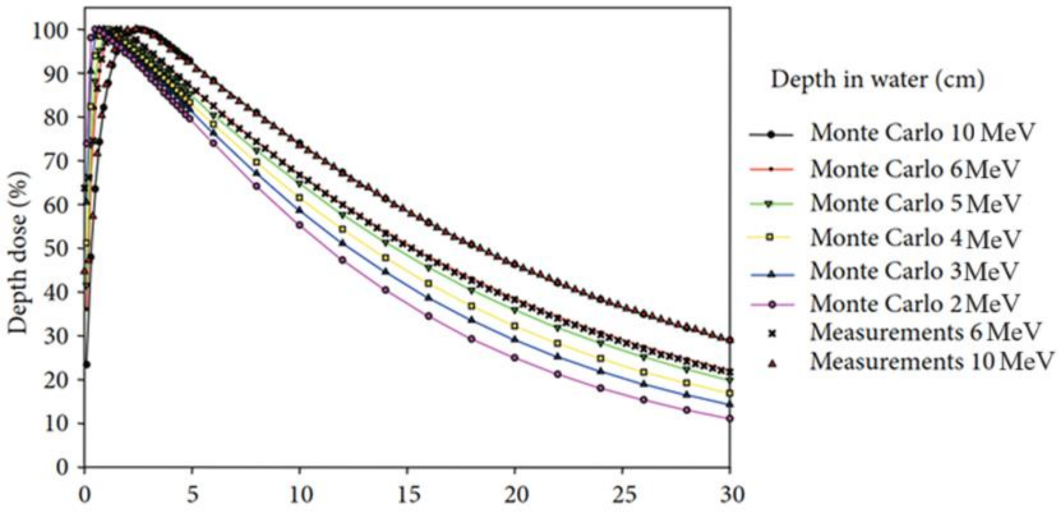
Data Monte carlo didasari oleh dua prinsip, yaitu *the law of large number* dan *central limit theorem*. Prinsip *the law of large number* adalah ketika rata-rata sampel semakin mendekati rata-rata populasi seiring dengan semakin besarnya jumlah sampel. Sementara prinsip *central limit theorem* menjelaskan bahwa semakin banyak jumlah sampel, distribusi akan semakin mendekati distribusi normal.

Kedua prinsip tersebut akan tetap terjadi meskipun sampel yang dimiliki pada awalnya memiliki distribusi yang tidak seperti distribusi normal. Hal ini dapat terlihat apabila sejumlah sampel diambil dari distribusi awal, kemudian sampel-sampel tersebut ditampilkan sebagai distribusi. Maka semakin banyak sampel yang diambil, distribusinya akan semakin mendekati distribusi normal.

III.5.2. *Database* Empiris Simulasi Partikel Monte Carlo

Basis dari probabilitas interaksi partikel didapatkan dengan menggunakan hasil empiris yang dihasilkan oleh eksperimen yang dilakukan oleh IAEA ataupun instansi kenukliran lain. Data *Evaluated Nuclear Data Library* (ENDF) dipublikasikan oleh *The Cross Section Evaluation Working Group* (CSEWG). Dokumen ini senantiasa diperbarui dan ditambahkan demi mendukung jenis-jenis interaksi partikel lain. Simulasi Monte Carlo memiliki peran yang semakin penting pada pemodelan hasil radiodiagnostik serta verifikasi perawatan pasien [15].

Penelitian sebelumnya telah membuktikan bahwa simulasi Monte Carlo telah sesuai dengan pengukuran eksperimen. Hasil ini konsisten pada rentang energi yang berbeda-beda. Penelitian tersebut membandingkan grafik *percentage depth dose* yang dihasilkan pada beberapa energi yang berbeda pada simulasi serta pengukuran langsung. Simulasi ini dijalankan pada energi 10 MeV, dosis maksimum D_{max} berada pada kedalaman 2,4 cm[23]. Distribusi tersebut dijabarkan melalui Gambar III.2. Hasil ini juga didukung oleh penelitian lain yang membandingkan *depth dose* antara linear akselerator buatan Elekta dan Siemens [24]. Perbandingan pengukuran nyata dengan metode simulasi Monte Carlo dapat direpresentasikan menggunakan Tabel III.1.



Gambar III.4 *Precentage Depth Dose water phantom* simulasi dan riil [23]

Tabel III.3 Kedalaman Kedalaman presentase dosis pada *water phantom* [23]

Dosis Relatif (%)	***Depth*/cm**			
	10 MeV		**6 MeV**	
	Simulasi	**Eksperimen**	**Simulasi**	**Eksperimen**
100	2,4	2,4	1,5	1,6
90	5,6	5,5	4,2	4,2
80	8,2	8,2	6,6	6,6
70	11,1	11,1	9,2	9,2
60	14,5	14,4	12,0	12,0
50	18,3	18,3	15,0	15,2

III.5.3. OpenMC

OpenMC diciptakan oleh *Computational Reactor Physics Group* (CRPG) di *Massachusetts Institute of Technology* (MIT)[20]. OpenMC menggunakan prinsip simulasi Monte Carlo pada transportasi partikel netral. Solusi numerikal persamaan transportasi atau disebut juga sebagai perhitungan deterministik akan mengdiskritkan waktu, energi, sudut yang kemudian masing-masing akan membuat teror sistematik [17]. Pembandingan hasil simulasi OpenMC terhadap kode transportasi partikel lain seperti PHITS dan MCNP juga telah dilakukan dan memberikan hasil yang positif[22]. OpenMC juga telah digunakan untuk mendesain modulator energi untuk kemudian diimplementasikan pada LINAC nyata[16].

III.5.3.1. Konversi Fluks ke Dosis Ekivalen

Konversi fluks menjadi dosis ekivalen telah tersedia pada *International Commission on Radiological Protection (ICRP) Publication* 116 yang telah didasari oleh perhitungan kalkulasi Monte Carlo pada program EGSnrc, FLUKA, GEANT4, MCNPX, dan PHITS [25]. OpenMC akan menghasilkan nilai fluks berupa particle-cm/sumber. Nilai ini kemudian diubah menjadi μSv cm^2 berdasarkan konversi yang disediakan oleh ICRP 116. Dengan menggunakan ukuran *tally* yang dipasang pada simulasi, nilai pSv kemudian akan ditemukan. Nilai ini kemudian disetarakan pada kalibrasi agar mendapatkan dosis perwaktu. Laju dosis ini kemudian diubah menjadi μSv/minggu yang digunakan sebagai pembatas dosis.

BAB IV
PELAKSANAAN PENELITIAN

IV.1. Alat dan Bahan Penelitian

Pada penilitian ini, aktivitas yang memerlukan alat penelitian adalah pada simulasi komputer. Simulasi komputer dilaksanakan dengan memanfaatkan server yang tersedia pada Departemen Teknik Nuklir dan Teknik Fisika (DTNTF) Universitas Gadjah Mada (UGM).

IV.1.1. Simulasi

Berikut adalah alat yang digunakan dalam penelitian ini :

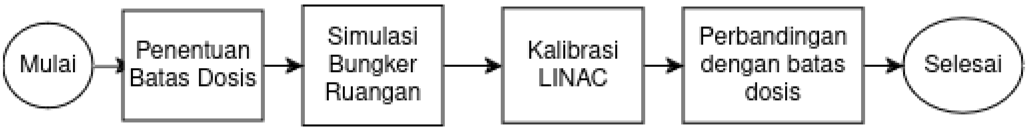
1. Laptop Zephyrus G14
   a. AMD Ryzen 7 5800HS @3.2 GHz
   b. 16 GB *Random Access Memory*
2. Server Intel(R) Xeon(R) Silver
   a. Intel Xeon Silver 4112 CPU @ 2.60GHz
   b. 32 GB *Random Access Memory*

Kedua komputer ini digunakan untuk melakukan simulasi serta membaca hasil yang didapatkan dari simulasi tersebut. Server disediakan oleh DTNTF UGM. Kemudian program yang digunakan memiliki lisensi *open source*, diantaranya :

1. OpenMC 0.13.1
2. Python 3.8.10
3. matplotlib: 3.7.5

IV.2. Tata Laksana Penelitian

Gambaran umum penelitian ini bisa dideskripsikan sebagai diagram alir di bawah ini:



Gambar IV.1 Ilustrasi *workflow* penelitian

IV.2.1.1. Pemodelan OpenMC

Pembuatan model OpenMC dilakukan berdasarkan desain awal yang merupakan kondisi ketika pengukuran langsung dilakukan. *Tally* dibuat tanpa menirukan geometri detektor. Nilai keluaran yang didapatkan dari OpenMC sebenarnya bukanlah dosis berupa µSv/jam melainkan jumlah partikel yang melewati *tally* yang dibuat. Maka dari itu sebelum dilakukan simulasi ruangan, simulasi kalibrasi terlebih dahulu dilakukan untuk mendapatkan faktor konversi agar keluaran fluks dari OpenMC dapat dikonversi menjadi µSv/jam.

Data *cross sections* pertama-tama perlu di unduh dari laman pada situs OpenMC. Terdapat 2 data yang tersedia yaitu ENDF serta JEFF. Penelitian ini akan dilakukan dengan menggunakan data ENDF/B-VII.1. Terdapat beragam data empiris penelitian yang tersedia seperti neutron, elektron, foton, deuteron dan lain-lain. Data ini kemudian di *import* melalui terminal linux.

IV.2.1.2. Simulasi Kalibrasi Absolut

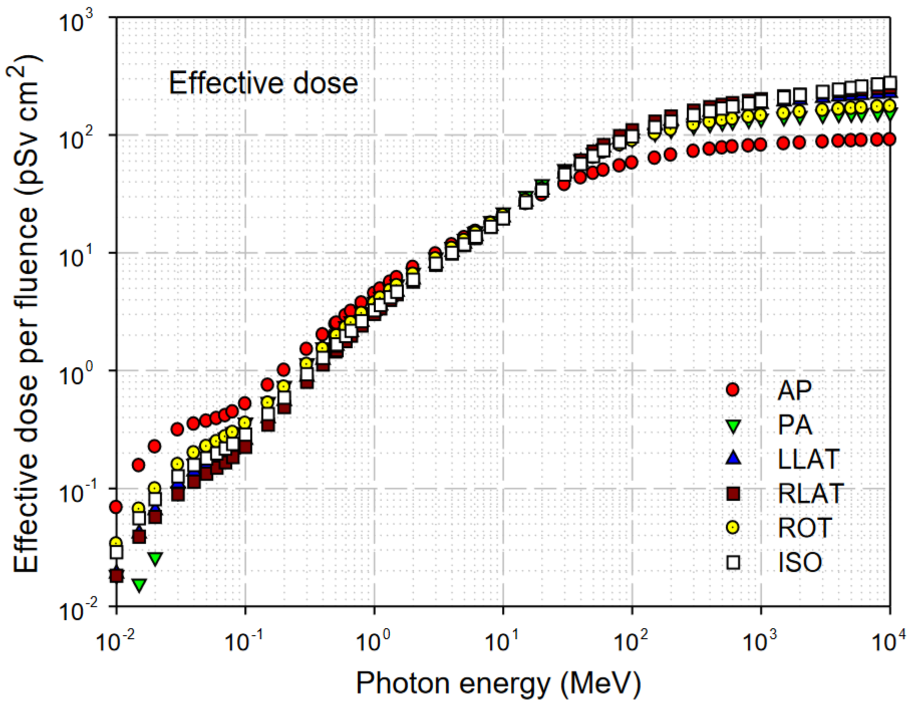
Simulasi akan menghasilkan jenis partikel beserta energi yang dimiliki oleh partikel tersebut. Berdasarkan hasil percobaan yang dilakukan menggunakan fantom organ oleh *International Commission on Radiological Protection* (ICRP). Hubungan energi partikel dosis perfluens direpresentasikan sebagai Gambar IV.2. Hubungan antara energi, fluks serta dosis perfluens dapat dituliskan sebagai Persamanan 4.1 [26]. Interpolasi dilakukan menggunakan data pada Gambar IV.2 agar partikel energinya tidak terdapat pada tabel dapat dikonversikan juga.

$$D_a = (E)\phi \tag{4.1}$$

D_a = Dosis per fluens (pSv cm^2)

E = Energi Partikel (MeV)

Φ = fluks (partikel – cm / sumber)



Gambar IV.2 Konversi nilai energi partikel ke dosis efektif per fluens menggunakan fantom organ pada arah iradiasi [25].

Pada Gambar IV.2, posisi arah iradiasi terhadap fantom dideskripsikan menggunakan istilah AP, *antero-posterior*; PA,*postero-anterior*; LLAT, *left lateral*; RLAT, *right lateral*; ROT, *rotational*; ISO, *isotropic*.

Setelah nilai dosis per fluens ditemukan, nilai fluks serta dosis keluaran diperlukan untuk dapat mengetahui laju dosis yang terjadi pada simulasi tersebut. Hasil ini direpresentasikan sebagai Persamanaan 4.2 [26].

$$Dv_{(E,\phi)} = \frac{Da_{(E)} \times \phi \times SR \times t}{V} \tag{4.2}$$

Dv = Dosis total (pSv)

SR = sumber / detik

t = waktu (detik)

V = volume (cm^3)

Penyederhanaan Persamaan 4.2 dilakukan agar dapat mempermudah perhitungan laju dosis pada detektor-detektor *tally* yang dipasang pada geometri ruangan. Penyederhanaan tersebut dilakukan dengan menggunakan nilai S yang mewakili hubungan dosis per fluens dengan flux dan *source Rate.* Untuk mempermudah perhitungan pada simulasi, nilai S tersebut juga digunakan untuk mengkonversi pSv/s untuk menjadi (µSv/jam).

$$\frac{Dv}{t} = \frac{Da \times S}{V} = \dot{D} \tag{4.3}$$

$\dot{D}$ = Laju dosis (µSv/jam)

S = Faktor Konversi (partikel- $cm^3 \frac{\mu Sv \times detik}{jam \times pSv}$)

Pada sumber radionuklida, nilai *Source Rate* diberikan oleh *data sheet* yang tersedia oleh menufakturnya. Karena pada LINAC data yang diketahui adalah berupa *Dose Rate*, kalibrasi dosis kemudian dilakukan untuk dilakukan untuk mengetahui *source rate* yang diperlukan untuk menghasilkan *Dose Rate* pada LINAC yaitu 360 MU/min atau setara dengan 360 Gy per jam.

Dengan perubahan formula seperti yang ditunjukkan pada Persamaan 4.3 dan penggunaan faktor konversi, persamaan dapat disederhanakan sebagai Persamaan 4.4.

$$\dot{D} = \frac{D_O S}{V} \tag{4.4}$$

Dengan nilai dosis keluaran simulasi masih berupa pSv cm^3/sumber, maka nilai keluaran masih perlu dibagi oleh volume dari sel kalibrasi itu sendiri.

$$\dot{D}_{maks} = \frac{D_{Osk} S}{V_{sk}} \tag{4.5}$$

$$S = \frac{\dot{D}_{maks} V_{sk}}{D_{osk}} \tag{4.6}$$

$\dot{D}_{maks}$ = Laju Dosis Maksimum (µSv/jam)

D_{Osk} = Dosis keluaran sel kalibrasi (pSv)

S = Faktor Konversi

V_{sk} = volume *tally* kalibrasi (cm^3)

Faktor koefisien kemudian digunakan untuk mencari laju dosis pada masin-masing detektor yang ada pada ruangan bunker. Hasil nilai fluks beserta energinya dihasilkan oleh simulasi OpenMC. Kemudian fungsi *openmc.dose_coefficient* dipakai untuk mengubah energi tersebut menjadi dosis. Dosis ini dikonversikan ke laju dosis menggunakan nilai konversi yang ditemukan pada kalibrasi OpenMC. Nilai yang dihasilkan masih tersebar pada volume *tally*. Oleh karena itu, pembagian dengan volume *tally* yang dipakai perlu dilakukan untuk mendapatkan nilai laju dosis pada titik-titik pengukuran.

Pada kondisi nyata, dosis pada *water phantom* akan dipengaruhi oleh hamburan dosis dari sekitar, *setup* pengukuran kualitas dan energi penyinaran serta kondisi lingkungan lain seperti suhu, tekanan dan kelembaban. Akan tetapi hal tersebut diabaikan karena simulasi dilakukan dalam keadaan ideal.

IV.2.1.3. Simulasi Ruangan pada 4 sudut *gantry*

Simulasi dilakukan pada empat sudut yang berbeda dengan rentang 90. Nilai inilah yang kemudian akan digunakan untuk memperkirakan total dosis yang diterima pada tiap ruangan tiap minggunya.

Program sederhana ini digunakan agar sudut rotasi dapat dipilih sebelum simulasi dijalankan. Hal ini memungkinkan eksekusi OpenMC lebih mudah untuk dieksekusi. Kemudian penggunaan sumber adalah replikasi dari penyinaran LINAC. *Box source* digunakan dan menyebar hingga mencapai *FIELD SIZE* 40 x 40 cm.

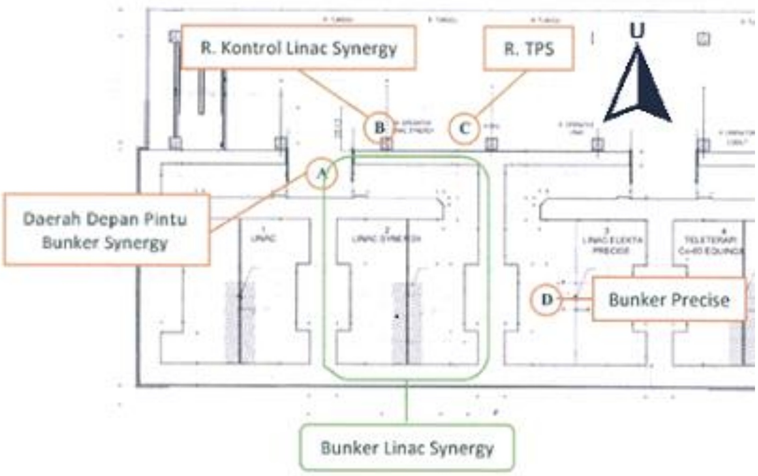
Energi dari sumber foton tersebar secara diskrit dengan energi 10 MeV. Kemudian persebaran dilakukan secara diskrit dan menyebar secara merata dengan volume bola. Persebaran tersebut hanya terjadi terhadap luas yang dideskripsikan sebagai luas *Field Size*. Kemudian posisi arah penyinaran juga diformulasikan agar dapat berada pada 100 cm dari isosenter sesuai dengan sudut rotasi penyinarannya. Hal tersebut dilakukan dengan fungsi pada lampiran 1 berupa fungsi sposi yang pada Lampiran A.

Pemasangan *Tally* fluks dilakukan dengan filter energi. Jenis *Tally* ini diperlukan untuk mengetahui jenis partikel serta energinya agar kemudian dapat dikonversikan menjadi dosis per fluens sesuai dengan ICRP 116.

Kemudian beberapa titik dengan variasi jarak terhadap dinding akan digunakan sebagai lokasi pengukuran laju dosis. Titik-titik inilah yang kemudian akan dikaji untuk dibandingkan dengan pembatas dosis yang berlaku.

Penampilan geometri kemudian dilakukana pada 3 variasi sumbu agar dapat diperiksa apakah sudah sesuai dengan desain bunker atau belum. Geometri ini juga dapat digunakan untuk menampilkan visualisasi persebaran dosis. Persebaran dosis didapatkan dengan membuat *grid tally* yang menyebar ke seluruh ruangan.

Kemudian sumber didefinisikan melalui fungsi *source.space, source.angle* , *source.energy* dan *source.particle*. Source.space dibuat sebagai sumber berbentuk kotak pada posisi yang dihasilkan oleh fungsi rotasi. Kemudian source.angle didefinisikan sebagai polar azimuthal. Persebaran *angle* ini akan terjadi dengan membentuk seperti permukaan bola hanya ke *Field Size* terluas, yaitu sebesar 40x40 cm. Persebaran energi terjadi secara diskrit pada energi 10 MeV. Hal ini dilakukan dengan deskripsi pada fungsi *source.energy. source.particle* mendeskripsikan jenis yang dipancarkan sumber. Pada penelitian ini, variabel tersebut diatur sebagai foton. Pada Gambar IV.3, denah bangunan digunakan dalam menggambarkan titik-titik pengukuran menggunakan simulasi.



Gambar IV.3 Denah titik pengukuran laju dosis ruang LINAC Synergy [19]

IV.3. Analisis Hasil Penelitian

Setelah nilai konversi didapatkan, simulasi pada 4 sudut *gantry* dilakukan kemudian dosis keluarannya dikonversi menjadi µSv/jam yang merupakan besaran laju dosis yang lebih lazim digunakan pada detektor radiasi. Konversi menggunakan faktor kalibrasi serta volume dari *tally* detektor dituliskan secara sederhana melalui Persamaan 4.4.

Setelah laju dosis pada LINAC diketahui, Nilai ini akan digunakan untuk mengetahui laju dosis dari titik-titik pengukuran pada tiap sudut penyinaran. Sebelum hal tersebut dilakukan, lama waktu penyinaran LINAC perlu dihitung agar kemudian dapat dikalikan dengan laju dosis. Waktu total pengoperasian dicari dengan asumsi yang digunakan pada dokumen konstruksi serta asumsi bahwa setiap penyinaran dilakukan dengan menggunakan DR maksimum. Waktu tersebut ditemukan dengan mencari tahu berapa lama waktu yang diperlukan untuk mencapai beban kerja dengan DR yang digunakan.

$$t_W = \frac{W}{D} \tag{4.7}$$

$$D_U = \dot{D} \times t_W \times U \tag{4.8}$$

t_W = Waktu pengoperasian LINAC untuk mencapai beban kerja dengan DR maksimum (jam)

W = *Workload* / beban kerja pengoperasian LINAC tiap minggu (Gy/minggu)

$\dot{D}$ = *Dose Rate* pengoperasian LINAC (MU/menit)

D_U = Dosis berdasarkan utilitas sudut spesifik dan beban kerja LINAC (µSv/minggu)

U = Faktor Utilitas Presentase LINAC menyinari sudut tersebut dibandingkan sudut *gantry* lainnya (%)

Karena beban kerja terpenuhi dengan memvariasikan sudut penyinaran LINAC, Faktor utilitas yang dikeluarkan oleh NCRP 151 digunakan. Faktor tersebut merepresentasikan penggunaan penyinaran LINAC pada tiap rentang 90° Laju dosis simulasi pada setiap sudutnya kemudian akan dikalikan dengan presentase tersebut. Dosis utilitas pada masing-masing sudut LINAC kemudian dijumlahkan agar ekspektasi laju dosis pada tiap titiknya pada tiap minggu pengoperasian LINAC dapat ditemukan. Karena titik-titik pengukuran dilakukan pada ruangan yang berbeda-beda, waktu yang dihabiskan oleh pekerja radiasi pada titik-titik tersebut akan berbeda-beda. Hal tersebut dijelaskan dan digunakan dalam perhitungan analitik pada pendesainan ketebalan minimum bunker. Relasi antara laju dosis dan okupansi ini kemudian dapat direpresentasikan melalui Persamaan 4.8.

$$D_{TO} = (D_{U0} + D_{U90} + D_{U180} + D_{U270}) \times T \tag{4.8}$$

Faktor Okupansi (T) = Nilai ini merepresentasikan frekuensi orang pada titik tersebut pada pengoperasian LINAC.

D_{TO} = Dosis Total Okupansi pada titik pengukuran (μSv/minggu)

W = *Workload* / beban kerja pengoperasian LINAC tiap minggu (Gy/minggu)

$\dot{D}$ = *Dose Rate* pengoperasian LINAC (MU/menit)

Persamaan 4.8 merupakan ekspektasi laju dosis yang diterima di satu titik pengukuran pada tiap minggunya. Contohnya pada ruang operator, Dosis total (D_{TO}) berasal dari laju dosis ketika pengoperasian LINAC mengarah ke bawah (D_{U0}) dan sudut-sudut penyinaran lainnya yang kemudian dikalikan dengan faktor okupansi (T) dari ruang operator.

IV.3.1. Interpretasi Persebaran Geometri Radiasi pada Simulasi

Nilai yang dihasilkan simulasi memiliki kelebihan berupa persebarannya dapat divisualisasikan. Hal ini akan dapat membantu interpretasi persebaran dosis yang terjadi pada simulasi ruangan LINAC. Visualisasi ini akan merepresentasikan persebaran laju dosis yang terjadi pada ruangan tersebut.

BAB V
HASIL DAN PEMBAHASAN

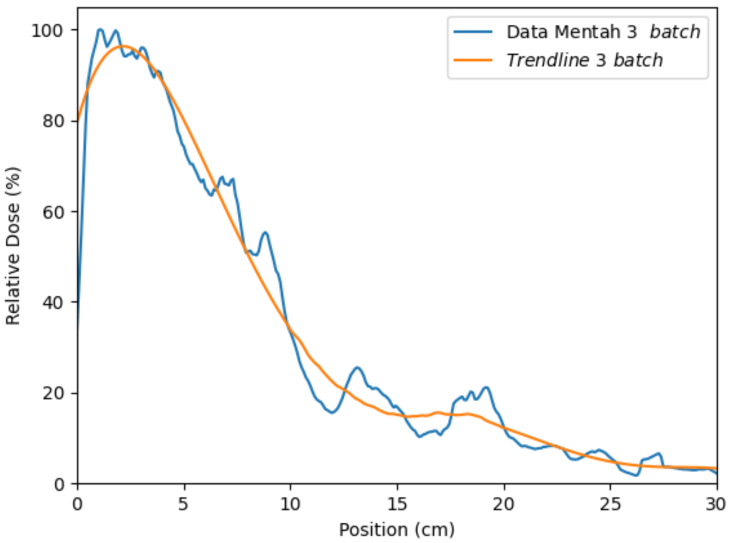
V.1. Hasil Simulasi OpenMC

Mula-mula kalibrasi dilakukan dengan menggunakan replika *water phantom*. Nilai D_{max} dari simulasi tersebut kemudian dipakai sebagai konversi dosis keluaran OpenMC menjadi dosis yang dapat dibandingkan dengan pembatas dosis.

V.1.1. Kalibrasi Dosis ke Laju Dosis Simulasi

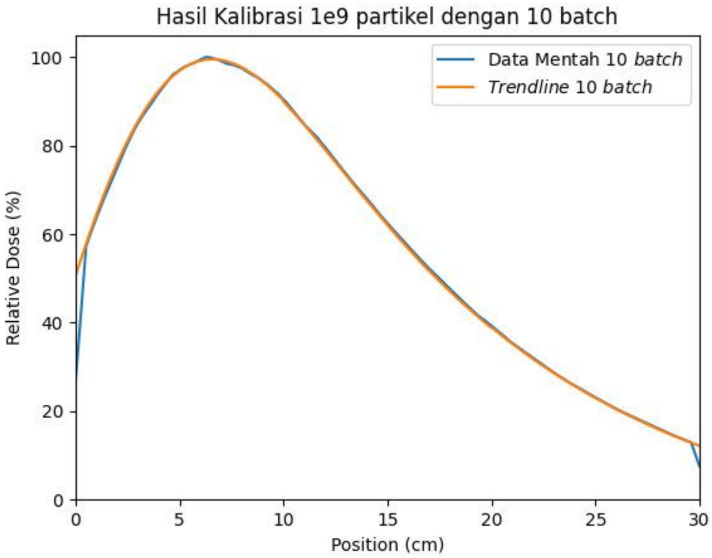
Kalibrasi dilakukan dengan menggunakan replika *water phantom* yang berjarak 100 cm terhadap *source*. *Field size* dari simulasi kalibrasi dosis adalah 30 cm X 30 cm.

Nilai D_{max} pada kalibrasi kemudian digunakan sebagai penyetaraan pada penyinaran LINAC yang memiliki nilai 600 MU/menit atau setara dengan 360 µSv/jam yang dapat direpresentasikan melalui Persamanaan 4.4.



Gambar V.1 *Percent Depth Dose* (PDD) Simulasi Kalibrasi *Water Phantom*

Grafik PDD memiliki perbedaan yang cukup signifikan dibandingkan bentuk pada simulasi maupun pengukuran nyata dikarenakan sumber yang digunakan adalah sumber sederhana. Sumber ini tidak membuat komponen-komponen seperti Primary Collimator *Flattening Filter, Monitor Chamber* serta *Jaw* layaknya pada kepala LINAC sesungguhnya. Simulasi ini juga tidak menggunakan berkas *phase-space* berupa .phsp. *Batch* dari simulasi ini berjumlah sama dengan *batch* pada simulasi ruangan yaitu sejumlah 3 *batch* dengan partikel berjumlah 1.000.000.000. Dapat terlihat bahwa puncak dosis berada pada 1,1 dan sekitar 1,9 cm di dalam *water phantom* dengan puncak 2,1 cm pada grafik yang trendline yang dibuat dengan cara dihaluskan.



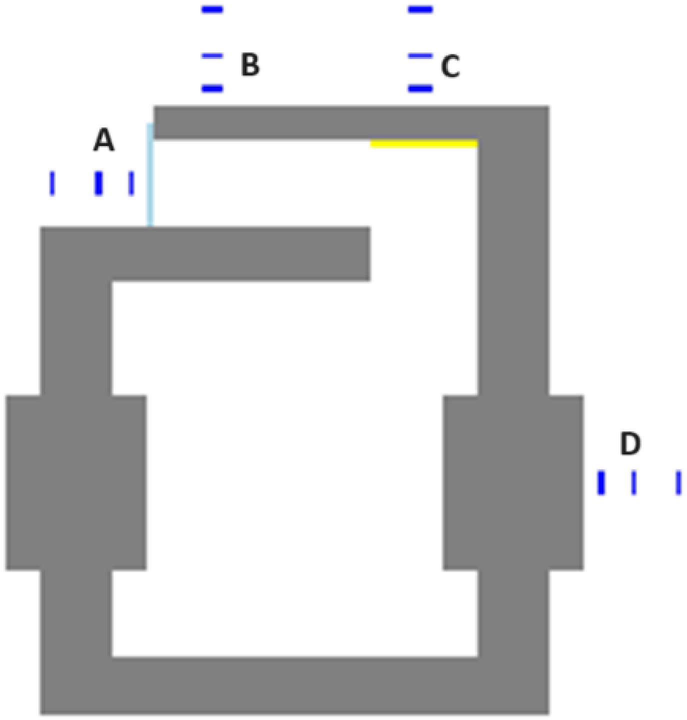
Gambar V.2 *Percent Depth Dose* (PDD) Simulasi Kalibrasi *Water Phantom* pada 10 *batch*

Pada pengukuran kalibrasi dengan menggunakan *batch* yang lebih besar, grafik tersebut menunjukkan dosis puncak pada jarak yang berbeda. Hal ini terjadi karena geometri yang digunakan untuk memunculkan sinar LINAC tidak menyerupai geometri LINAC nyata. Puncak pada batch kurang lebih berada pada kedalaman 6,1 cm. Hal ini cukup jauh apabila dibandingkan dengan hasil eksperimen maupun simulasi Monte Carlo dengan geometri asli.

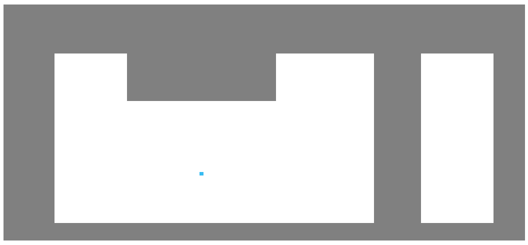
V.1.2. Geometri

Geometri dibuat dengan menggunakan fungsi yang tersedia pada python, geometri diciptakan pada sumbu xy, yz dan xz. Secara sederhana, dapat dideskripsikan bahwa ruangan terdiri dari dinding primer pada kedua sisi samping dan lorong pada satu sisi lainnya. Hal ini sesuai dengan denah yang tersedia pada Gambar V.3. Pada geometri tersebut, warna abu-abu berarti dinding beton dari ruangan tersebut. Kemudian warna kuning mewakilkan lapisan besi pada dinding. Selain itu ada juga detektor *tally* yang diilustrasikan sebagai warna biru tua.

Kemudian Warna biru muda adalah *box source* yang digunakan untuk memunculkan radiasi foton.



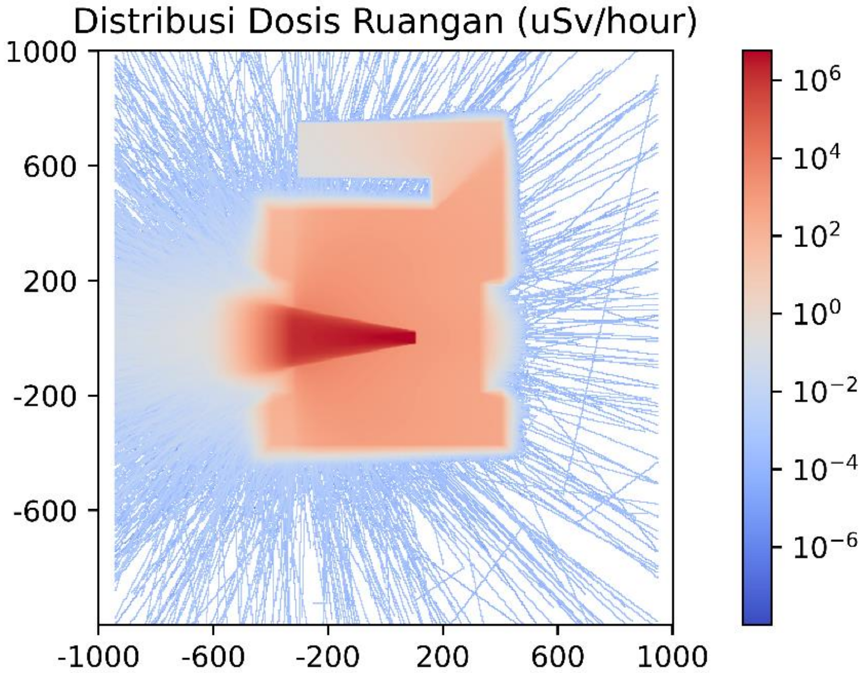
Gambar V.3 Geometri ruang LINAC simulasi pada persepsi XY



Gambar V.4 Geometri ruang LINAC simulasi pada persepsi YZ

Gambar V.5 Geometri ruang LINAC simulasi pada persepsi XZ

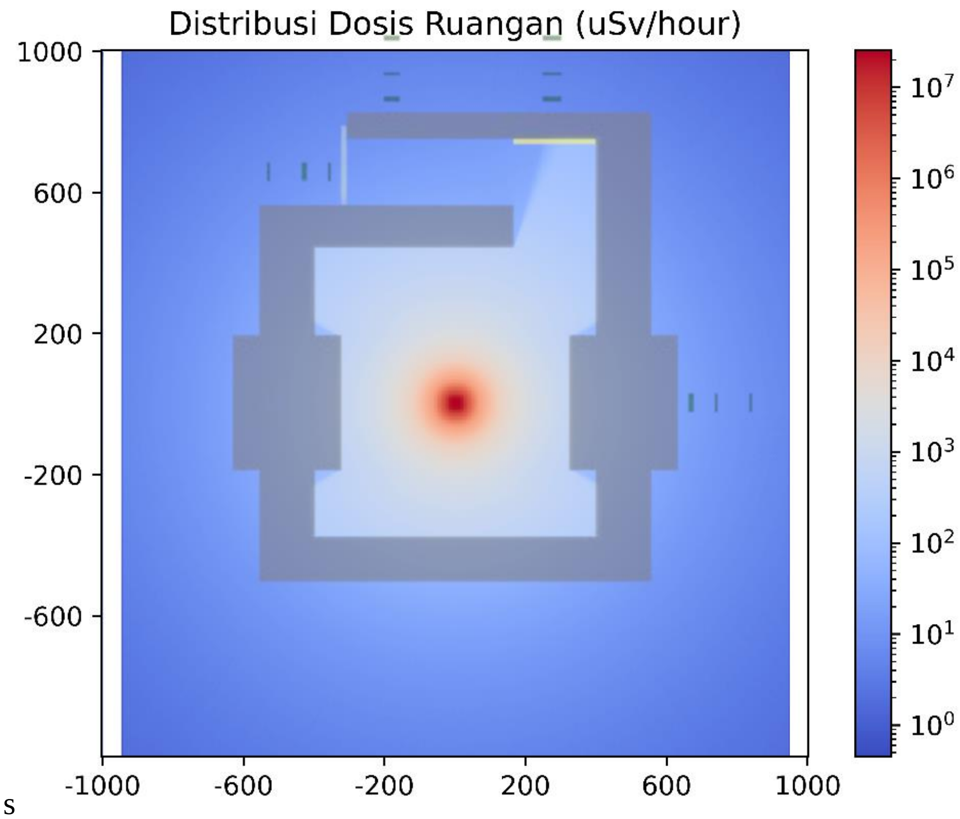
V.1.3. Peta Sebaran Dosis Persudut



Gambar V.6 *Heatmap* Distribusi Dosis Hasil Simulasi *default*

Dapat dilihat bahwa pada keluaran asli *heatmap* pada Gambar V.6, tidak terdapat radiasi yang signifikan pada sekitar ruangan kecuali dari dinding primer yang dikenai radiasi LINAC secara langsung. Persebaran dosis ini dibentuk menggunakan fungsi grid yang tersebar pada bunker tersebut. Dapat terlihat bahwa dosis yang signifikan terjadi sesuai dengan arah penyinaran kepala LINAC. Sedangkan pada sekitarnya dosis yang terdeteksi tidak terlalu signifikan. Simulasi dijalankan sebanyak 1.000.000.000 sejumlah 3 batch. Warna putih diruangan berarti jumlah partikel pada simulasi tersebut masih kurang dan perlu ditambah. Hal ini dikarenakan jumlah partikel yang lebih banyak akan dapat memunculkan hasil yang lebih akurat pada area ruangan yang memiliki dosis terlalu kecil. Pada Gambar V.6 sampai V-9, geometri digabung dengan *heatmap* agar interpretasi lebih mudah dilakukan.

Dengan menggunakan geometri yang sebelumnya dibuat dan *heatmap* dosis pada simulasi, visualisasi dari persebaran dosis bunker dapat dibuat. Pada dosis sudut 0° dengan arah sinar LINAC ke lantai, pantulan radiasi dari menyebar ke sekitar ruangan. Hal ini membuat laju dosis pada sudut ini lebih besar secara signifikan dibandingkan dengan sudut-sudut penyinaran lainnya. Dosis pada sudut ini juga lebih akurat karena setiap titik mendapatkan laju dosis dari berbagai sudut hingga memenuhi semua titik dari sekitar bunker.



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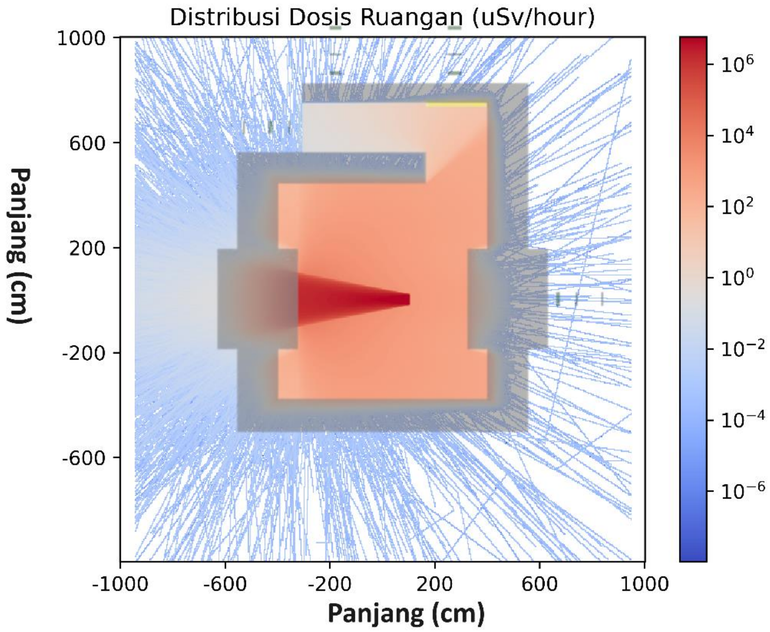
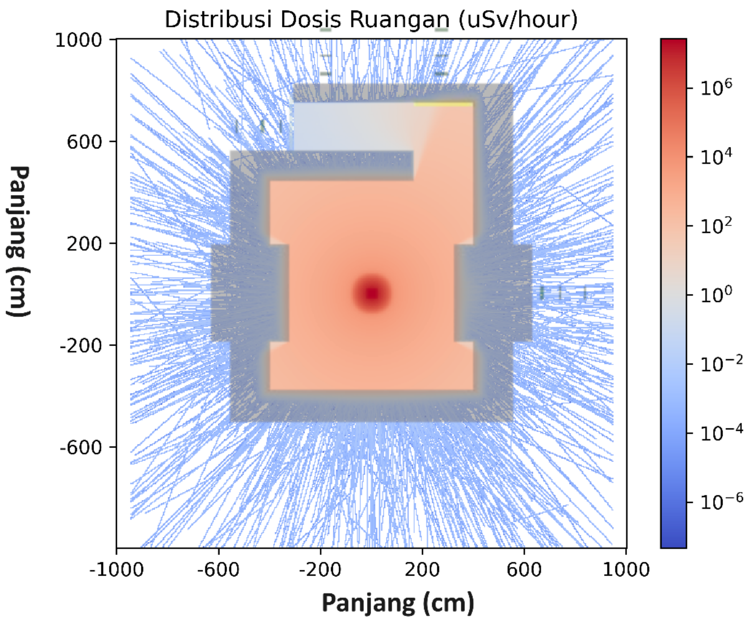
Gambar V.7 Distribusi Dosis pada *gantry* 0° (Bawah)

Hal yang sama tidak bisa dikatakan pada penyinaran sudut-sudut lainnya. Dapat dilihat bahwa dosis pada penyinaran sudut-sudut lain belum cukup menyeluruh hingga memenuhi *grid tally* pada simulasi sudut tersebut. Perhitungan analitik sederhana dapat membuktikan bahwa setiap titik dari sekitar bunker seharusnya memiliki dosis radiasi. Sedangkan berdasarkan gambar-gambar penyinaran ini, dapat terlihat bahwa terdapat *tally-tally* yang tidak mendapatkan dosis sama sekali. Peningkatan jumlah partikel sangat diperlukan pada sudut-sudut yang lain. Laju dosis pada sudut 270° dan 90° hanya signifikan ke arah bunker lain, sedangkan pada sudut 180° sebaran mengarah ke atap yang tidak bisa diakses.

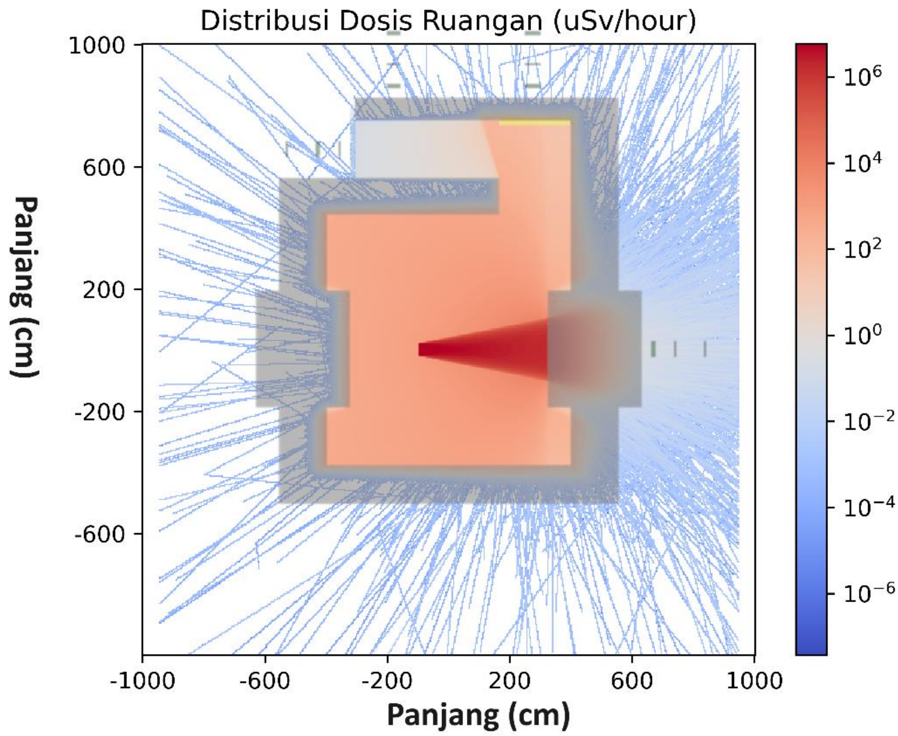


Gambar V.8 Distribusi Dosis pada *gantry* 90°



Gambar V.9 Distribusi Dosis pada *gantry* 180° (Atas)

Laju dosis pada sudut-sudut selain 0° menunjukkan titik-titik kosong pada simulasi tersebut. Hal ini dapat ditanggulangi dengan peningkatan partikel sejumlah penurunan orde pada laju dosis di sekitar ruangan bunker. Laju dosis disekitar ruangan pada sudut-sudut tersebut ada di orde -4, sedangkan pada orde -3 radiasi sudah dapat memenuhi visualisasi simulasi. Dari sini dapat diketahui bahwa peningkatan jumlah partikel yang diperlukan adalah sejumlah puluhan kali. Peningkatan ini bisa menjadi $1x10^{10}$ hingga mencapai $9x10^{10}$ partikel. Dengan menggunakan spesifikasi komputer pada penelitian ini, hal tersebut dapat memakan waktu 20 hingga 180 hari. Maka dari itu peningkatan spesifikasi komputer perlu dilakukan.



Gambar V.10 Distribusi Dosis pada *gantry* 270°

V.2. Perbandingan Batas Dosis dan Hasil Simulasi

Laju dosis yang didapatkan pada *tally-tally* yang dipasang mendapatkan hasil yang ditampilkan pada Tabel V.1. Dapat dilihat bahwa dosis lebih besar secara signifikan pada sudut 0°. Hal inilah yang membuat *grid tally* dapat terlihat penuh pada sudut tersebut.

Tabel V.1 Tabel Hasil Laju Dosis

Titik	Lokasi	Jarak ke Dinding (cm)	Laju Dosis Simulasi (μSv/jam)			
			0°	**90°**	**180°**	**270°**
A	Depan Pintu Bunker	30	$5,71\times10^{-3}$	$1,31\times10^{-3}$	$2,15\times10^{-3}$	$2,00\times10^{-4}$
		100	$3,35\times10^{-3}$	$9,10\times10^{-4}$	0,00	$6,37\times10^{-4}$
		200	$1,20\times10^{-2}$	$3,83\times10^{-4}$	$7,86\times10^{-4}$	$5,63\times10^{-4}$
B	R. Kontrol LINAC	30	$2,07\times10^{-2}$	$6,64\times10^{-4}$	$1,67\times10^{-3}$	$6,80\times10^{-4}$
		100	$4,00\times10^{-2}$	$3,55\times10^{-4}$	$1,93\times10^{-4}$	$2,40\times10^{-4}$
		200	$5,32\times10^{-2}$	$2,68\times10^{-4}$	$1,93\times10^{-4}$	$2,04\times10^{-4}$
C	R.TPS	30	$1,87\times10^{-2}$	$6,64\times10^{-4}$	$1,55\times10^{-4}$	0,00
		100	$2,90\times10^{-2}$	$3,49\times10^{-4}$	$1,22\times10^{-3}$	$7,00\times10^{-4}$
		200	$4,81\times10^{-2}$	$5,66\times10^{-4}$	$1,23\times10^{-3}$	$6,93\times10^{-4}$
D	Bunker Linac *Precise*	30	$5,47\times10^{-2}$	$7,58\times10^{-4}$	$2,51\times10^{-3}$	$3,20\times10^{-1}$
		100	$6,37\times10^{-2}$	$6,70\times10^{-4}$	$3,30\times10^{-2}$	$2,48\times10^{-1}$
		200	$9,93\times10^{-2}$	$3,76\times10^{-4}$	$1,81\times10^{-4}$	$2,20\times10^{-1}$

Berdasarkan pengukuran laju dosis pada Tabel V.1, dapat dilihat bahwa dosis radiasi tidak selalu mengecil semakin jauh deteksi dilakukan. Hal ini dikarenakan radiasi yang melewati detektor terjauh dapat dilakukan oleh radiasi yang bukan datang secara tegak lurus terhadap dinding. Kemudian pada sudut 180° serta 270°, terdapat detektor yang tidak dilewati oleh radiasi sama sekali.

Dosis yang dihasilkan sudah sangat rendah mengingat LINAC hanya dioperasikan beberapa kali tiap harinya. Dengan menggunakan asumsi beban kerja yang dipakai untuk menghitung ketebalan LINAC, total lama LINAC beroperasi dapat dihitung. Perhitungan ini dilakukan dengan asumsi bahwa laju dosis penyinaran maksimum digunakan. Perhitungan beban kerja dapat dihitung secara komprehensif berdasarkan Tabel V.2. Perhitungan beban kerja pada perhitungan analitik pada dinding sekunder dilakukan dengan mempertimbangkan faktor koefisien Q. Sedangkan hal ini berbeda dengan perhitungan analitik primer yang tidak menggunakan faktor tersebut.

Tabel V.2 Perhitungan Beban Kerja Mingguan

Jenis Penyinaran	Jumlah Pasien (orang)	Dosis per pasien (Gy/pasien)	Hari beroperasi (hari/minggu)	Total Dosis (Gy/minggu)
Konvensional	20	4	5	400
IMRT	50	4	5	1000
VMAT	10	4	5	200
QA	12	2	6	144
Total Beban Kerja (Gy/minggu)				1744
Lama LINAC Beroperasi (jam/minggu)				4,84

Pada setiap dinding tersebut, batas dosis yang berlaku adalah pembatas dosis pekerja. Hal ini dikarenakan setiap ruangan di sekitar bunker merupakan ruangan pekerja dan tidak dapat dimasuki oleh masyarakat umum. Pembatas dosis pada pekerja radiasi menurut Perka BAPETEN no.4 tahun 2013 adalah sebesar 200 μSv/minggu.

Tabel V.3 Laju Dosis pada Penyinaran LINAC 0°

Titik	Lokasi	Jarak ke Dinding (cm)	Laju Dosis (μSv/jam)	Laju Dosis per minggu (μSv/minggu)	Laju Dosis Utilitas (μSv/minggu)
A	Depan Pintu Bunker	30	$5,71\times10^{-3}$	$2,77\times10^{-2}$	$8,58\times10^{-3}$
		100	$3,35\times10^{-3}$	$1,62\times10^{-2}$	$5,03\times10^{-3}$
		200	$1,20\times10^{-2}$	$5,83\times10^{-2}$	$1,81\times10^{-2}$
B	R. Kontrol LINAC	30	$2,07\times10^{-2}$	$1,00\times10^{-1}$	$3,11\times10^{-2}$
		100	$4,00\times10^{-2}$	$1,94\times10^{-1}$	$6,00\times10^{-2}$
		200	$5,32\times10^{-2}$	$2,58\times10^{-1}$	$8,00\times10^{-2}$
C	R.TPS	30	$1,87\times10^{-2}$	$9,07\times10^{-2}$	$2,81\times10^{-2}$
		100	$2,90\times10^{-2}$	$1,40\times10^{-1}$	$4,35\times10^{-2}$
		200	$4,81\times10^{-2}$	$2,33\times10^{-1}$	$7,23\times10^{-2}$
D	Bunker Linac *Precise*	30	$5,47\times10^{-2}$	$2,65\times10^{-1}$	$8,21\times10^{-2}$
		100	$6,37\times10^{-2}$	$3,08\times10^{-1}$	$9,56\times10^{-2}$
		200	$9,93\times10^{-2}$	$4,81\times10^{-1}$	$1,49\times10^{-1}$

Tabel V.4 Laju Dosis pada Penyinaran LINAC 90°

Titik	**Lokasi**	**Jarak ke Dinding (cm)**	**Laju Dosis (µSv/jam)**	**Dosis per minggu (µSv/minggu)**	**Dosis Utilitas (µSv/minggu)**
A	Depan Pintu Bunker	30	$1{,}31\times10^{-3}$	$6{,}34\times10^{-3}$	$1{,}35\times10^{-3}$
		100	$9{,}10\times10^{-4}$	$4{,}41\times10^{-3}$	$9{,}39\times10^{-4}$
		200	$3{,}83\times10^{-4}$	$1{,}85\times10^{-3}$	$3{,}95\times10^{-4}$
B	R. Kontrol LINAC	30	$6{,}64\times10^{-4}$	$3{,}22\times10^{-3}$	$6{,}86\times10^{-4}$
		100	$3{,}55\times10^{-4}$	$1{,}72\times10^{-3}$	$3{,}67\times10^{-4}$
		200	$2{,}68\times10^{-4}$	$1{,}30\times10^{-3}$	$2{,}76\times10^{-4}$
C	R.TPS	30	$6{,}64\times10^{-4}$	$3{,}22\times10^{-3}$	$6{,}86\times10^{-4}$
		100	$3{,}49\times10^{-4}$	$1{,}69\times10^{-3}$	$3{,}60\times10^{-4}$
		200	$5{,}66\times10^{-4}$	$2{,}74\times10^{-3}$	$5{,}84\times10^{-4}$
D	Bunker Linac *Precise*	30	$7{,}58\times10^{-4}$	$3{,}67\times10^{-3}$	$7{,}82\times10^{-4}$
		100	$6{,}70\times10^{-4}$	$3{,}24\times10^{-3}$	$6{,}91\times10^{-4}$
		200	$3{,}76\times10^{-4}$	$1{,}82\times10^{-3}$	$3{,}88\times10^{-4}$

Tabel V.5 Laju Dosis pada Penyinaran LINAC 180° (Atas)

Titik	**Lokasi**	**Jarak ke Dinding (cm)**	**Laju Dosis (µSv/jam)**	**Dosis per minggu (µSv/minggu)**	**Dosis Utilitas (µSv/minggu)**
A	Depan Pintu Bunker	30	$2{,}15\times10^{-3}$	$1{,}04\times10^{-2}$	$2{,}74\times10^{-3}$
		100	0,00	0,00	0,00
		200	$7{,}86\times10^{-4}$	$3{,}81\times10^{-3}$	$1{,}00\times10^{-3}$
B	R. Kontrol LINAC	30	$1{,}67\times10^{-3}$	$8{,}09\times10^{-3}$	$2{,}13\times10^{-3}$
		100	$1{,}93\times10^{-4}$	$9{,}33\times10^{-4}$	$2{,}45\times10^{-4}$
		200	$1{,}93\times10^{-4}$	$9{,}33\times10^{-4}$	$2{,}45\times10^{-4}$
C	R.TPS	30	$1{,}55\times10^{-4}$	$7{,}51\times10^{-4}$	$1{,}97\times10^{-4}$
		100	$1{,}22\times10^{-3}$	$5{,}92\times10^{-3}$	$1{,}56\times10^{-3}$
		200	$1{,}23\times10^{-3}$	$5{,}95\times10^{-3}$	$1{,}56\times10^{-3}$
D	Bunker Linac *Precise*	30	$2{,}51\times10^{-3}$	$1{,}21\times10^{-2}$	$3{,}19\times10^{-3}$
		100	$3{,}30\times10^{-2}$	$1{,}60\times10^{-1}$	$4{,}20\times10^{-2}$
		200	$1{,}81\times10^{-4}$	$8{,}76\times10^{-4}$	$2{,}30\times10^{-4}$

Tabel V.6 Laju Dosis pada Penyinaran LINAC 270°

Titik	Lokasi	Jarak ke Dinding (cm)	Laju Dosis (μSv/jam)	Dosis per minggu (μSv/minggu)	Dosis Utilitas (μSv/minggu)
A	Depan Pintu Bunker	30	$2{,}00\times10^{-4}$	$9{,}67\times10^{-4}$	$2{,}06\times10^{-4}$
		100	$6{,}37\times10^{-4}$	$3{,}09\times10^{-3}$	$6{,}57\times10^{-4}$
		200	$5{,}63\times10^{-4}$	$2{,}73\times10^{-3}$	$5{,}81\times10^{-4}$
B	R. Kontrol LINAC	30	$6{,}80\times10^{-4}$	$3{,}29\times10^{-3}$	$7{,}02\times10^{-4}$
		100	$2{,}40\times10^{-4}$	$1{,}16\times10^{-3}$	$2{,}48\times10^{-4}$
		200	$2{,}04\times10^{-4}$	$9{,}88\times10^{-4}$	$2{,}11\times10^{-4}$
C	R.TPS	30	0,00	0,00	0,00
		100	$7{,}00\times10^{-4}$	$3{,}39\times10^{-3}$	$7{,}23\times10^{-4}$
		200	$6{,}93\times10^{-4}$	$3{,}36\times10^{-3}$	$7{,}16\times10^{-4}$
D	Bunker Linac *Precise*	30	$3{,}20\times10^{-1}$	1,55	$3{,}30\times10^{-1}$
		100	$2{,}48\times10^{-1}$	1,20	$2{,}56\times10^{-1}$
		200	$2{,}20\times10^{-1}$	1,07	$2{,}27\times10^{-1}$

Dengan lama waktu operasional LINAC perminggu serta batas dosis sesuai dengan ketentuan NCRP 151 serta SRS 47, keselamatan desain bunker dapat dikaji kesesuaiannya. Nilai tersebut didapatkan dengan mengkalikan total laju dosis terhadap faktor utilitas dari masing-masing sudut penyinaran. Berdasarkan Tabel III.2, utilitas senilai 31% pada 0°, 21,3% pada 90° dan 270°, serta 26,3 pada 180°. Laju dosis dari simulasi tiap sudut kemudian akan dikalikan dengan utilitasnya masing-masing agar kemudian dapat dijumlahkan untuk mengetahui ekspektatsi dosis pekerja di setiap titik yang dikaji. Nilai inilah yang pada akhirnya dapat dibandingkan untuk mengetahui seberapa kecil dosis yang diterima pekerja di titik-titik tersebut dibandingkan dengan pembatas dosis yang berlaku. Hal tersebut dijelaskan melalui Persamaan 4.7.

Laju dosis pada hasil pengukuran masing-masing sudut akan dijumlahkan untuk menemukan laju dosis pada pekerja. Total dari pengoperasian LINAC pada tiap minggunya serta penggunaan faktor okupansi untuk memperkirakan dosis yang diterima pada pekerja berdasarkan okupansi ditunjukkan pada Tabel V.7.

Tabel V.7 Total Ekspektasi Laju Dosis pada Pekerja di Ruangan

Titik	Lokasi	Faktor Okupansi (T)	Jarak ke Dinding (cm)	Total Dosis (μSv/minggu)	Total Dosis Okupansi (μSv/minggu)
A	Depan Pintu Bunker	0,125	30	$1{,}29\times10^{-2}$	$1{,}61\times10^{-3}$
			100	$6{,}62\times10^{-3}$	$8{,}28\times10^{-4}$
			200	$2{,}01\times10^{-2}$	$2{,}51\times10^{-3}$
B	R. Kontrol LINAC	1	30	$3{,}46\times10^{-2}$	$3{,}46\times10^{-2}$
			100	$6{,}09\times10^{-2}$	$6{,}09\times10^{-2}$
			200	$8{,}07\times10^{-2}$	$8{,}07\times10^{-2}$
C	R.TPS	1	30	$2{,}90\times10^{-2}$	$2{,}90\times10^{-2}$
			100	$4{,}62\times10^{-2}$	$4{,}62\times10^{-2}$
			200	$7{,}52\times10^{-2}$	$7{,}52\times10^{-2}$
D	Bunker Linac *Precise*	0,5	30	$4{,}16\times10^{-1}$	$2{,}08\times10^{-1}$
			100	$3{,}94\times10^{-1}$	$1{,}97\times10^{-1}$
			200	$3{,}77\times10^{-1}$	$1{,}88\times10^{-1}$

Dosis yang diberikan LINAC pada titik-titik tersebut jauh lebih kecil dibandingkan nilai batas dosisnya yaitu 200 μSv/minggu. Nilai ini 961 hingga 242.000 kali lebih kecil dibandingkan NBD yang berlaku. Perlu diperhatikan target dosis pada setiap bunker adalah pada dinding bunker tersebut, sedangkan kajian dosis dilakukan pada jarak 30 cm, 100 serta 200 cm. Lokasi ini adalah lokasi yang lebih mungkin dihabiskan oleh para pekerja radiasi pada saat penyinaran dilakukan.

Pada pendesainan ruang bunker, penambahan ketebalan senilai *half value layer* (HVL) juga dilakukan setelah ketebalan dinding minimum didapatkan. Hal tersebut kemudian juga dilanjutkan dengan pembulatan ukuran pada dinding beton. Maka dari itu, *Inverse square law* serta faktor atenuasi berperan besar pada laju dosis yang berkali-kali lipat dibandingkan nilai batas dosisnya.

Pada simulasi ini, karakteristik LINAC merupakan penyederhanaan dari penyinaran LINAC nyata. Keakuratan hasil simulasi dapat ditingkatkan dengan penambahan *flattening filter* (FF), penggunaan *phase space file* serta peningkatan jumlah partikel.

BAB VI
KESIMPULAN DAN SARAN

VI.1. Kesimpulan

1. Laju dosis hasil simulasi di sekitar bunker LINAC berada pada rentang 8,28 $\times 10^{-3}$ hingga 0,208 μSv/minggu. Nilai ini jauh lebih kecil dibandingkan dengan pembatas dosis yang diberlakukan oleh Perka BAPETEN no.3 tahun 2013. Hal ini berarti keselamatan bunker sudah sesuai dengan standar yang berlaku.
2. Didapatkan visualisasi persebaran laju dosis pada ruangan bunker untuk membantu interpretasi dosis yang terjadi pada masing-masing titik pengukuran laju dosis pada sekitar bunker.

VI.2. Saran

Terdapat hal-hal yang dapat dikembangkan dalam pelaksanaan penelitian ini. Peningkatan keakuratan hasil simulasi dapat dilakukan dengan penambahan *flattening filter* (FF), penggunaan *phase space file* serta peningkatan jumlah partikel. Penggunaan komputer dengan spesifikasi yang lebih tinggi juga dapat digunakan agar dapat mensimulasikan jumlah partikel yang lebih banyak.

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LAMPIRAN A

Listing Program OpenMC

Pemanggilan *cross section* ENDF

```
Export          OPENMC_CROSS_SECTIONS='/home/sena/Code/endfb-vii.1
hdf5/cross_sections.xml'

# export OPENMC_CROSS_SECTIONS=$var/cross_sections.xml

export HDF5_DISABLE_VERSION_CHECK=1
```

Program Kalibrasi

```
import openmc

# vim:foldmethod=marker

import openmc

import matplotlib.pyplot as plt

import openmc.stats, openmc.data

from math import cos, atan2, pi


batches = 10

inactive = 10

particles = int(input('Enter number of particle (It was 1e8)\n= '))
#1_000_000_000



# Material

# {{{

air=openmc.Material(name='Air')

air.set_density('g/cm3',0.001205)

air.add_nuclide('N14',0.7)

air.add_nuclide('O16',0.3)

#air.add_s_alpha_beta('c_H_in_Air')

air.add_element('C',0.002)

air.add_element('Fe',0.001)

air.add_element('Si',0.001)

air.add_element('Mn',0.001)
```

```
water=openmc.Material(name='Water')
water.set_density('g/cm3',1.0)
water.add_nuclide('H1',2.0)
water.add_nuclide('O16',1.0)
water.add_s_alpha_beta('c_H_in_H2O')

water2=openmc.Material(name='Water')
water2.set_density('g/cm3',1.0)
water2.add_nuclide('H1',2.0)
water2.add_nuclide('O16',1.0)
water2.add_s_alpha_beta('c_H_in_H2O')

materials = openmc.Materials( [air,water,water2])
materials.export_to_xml()
# }}}

SSD = 100.0 #Source to Skin Distance
PHANTOM_WIDTH = 40
ld= 0.1# panjang dan lebar WP
d = 30.0 #kedalaman WP
padd = 10.0 #padding terhadap source dan detektor

# {{{

n = 300
phantom_cells = []
dx = d/n
for i in range(n):
  x0 = SSD + i*dx
  x1 = SSD +(i+1)*dx
  r_x = +openmc.XPlane(x0) & -openmc.XPlane(x1)
```

```
  r_y = +openmc.YPlane(-ld/2.0) & -openmc.YPlane(ld/2.0)
  r_z = +openmc.ZPlane(-ld/2.0) & -openmc.ZPlane(ld/2.0)

  cell = openmc.Cell(region=r_x & r_y & r_z)
  cell.fill = water2
  phantom_cells.append(cell)

r_x = +openmc.XPlane(SSD) & -openmc.XPlane(SSD+d)
r_y = +openmc.YPlane(-ld/2.0) & -openmc.YPlane(ld/2.0)
r_z = +openmc.ZPlane(-ld/2.0) & -openmc.ZPlane(ld/2.0)
r_phantally = r_x & r_y & r_z

r_y_wp= +openmc.YPlane(-PHANTOM_WIDTH/2.0)\
  & -openmc.YPlane(PHANTOM_WIDTH/2.0)
r_z_wp= +openmc.ZPlane(-PHANTOM_WIDTH/2.0)\
  & -openmc.ZPlane(PHANTOM_WIDTH/2.0)
r_wp= r_x & r_y_wp & r_z_wp
c_wp= openmc.Cell(region=r_wp& ~r_phantally)
c_wp.fill = water

r_x_air = +openmc.XPlane(-padd, boundary_type='vacuum')\
  & -openmc.XPlane(SSD+d+padd, boundary_type='vacuum')
r_y_air = +openmc.YPlane(-PHANTOM_WIDTH/2.0-padd,
boundary_type='vacuum')\
  & -openmc.YPlane(PHANTOM_WIDTH/2.0+padd, boundary_type='vacuum')
r_z_air = +openmc.ZPlane(-PHANTOM_WIDTH/2.0-padd,
boundary_type='vacuum')\
  & -openmc.ZPlane(PHANTOM_WIDTH/2.0+padd, boundary_type='vacuum')
r_air = r_x_air & r_y_air & r_z_air
c_air = openmc.Cell(region=r_air & ~r_wp)
c_air.fill = air

universe = openmc.Universe(cells= [c_air]+phantom_cells+ [c_wp])
```

```
geometry = openmc.Geometry()
geometry.root_universe = universe
geometry.export_to_xml()
# }}}
plot= openmc.Plot()
plot.basis = 'yz'
plot.filename='yz_cal'
plot.origin = (110, 0, 0)
plot.width = (50., 50.)
plot.pixels=(1000,1000)
plot.color_by='material'
plot.colors={
  water:'blue',
  air:'green',
  water2:'black'
}
plot_file=openmc.Plots( [plot])
plot_file.export_to_xml()
openmc.plot_geometry()
plot.to_ipython_image()

plot.basis='xz'
plot.filename='xz_cal'
plot.width=(100,100)
plot.pixels=(1000,1000)
plot.origin=(100,0,0)
plot_file=openmc.Plots( [plot])
plot_file.export_to_xml()
openmc.plot_geometry()
plot.to_ipython_image()
```

```
# {{{
batches= batches
inactive = inactive
particles = particles

particle_filter = openmc.ParticleFilter('photon')
# energy, dose = openmc.data.dose_coefficients('photon', 'AP')
energy, dose = openmc.data.dose_coefficients('photon', 'RLAT')
dose_filter = openmc.EnergyFunctionFilter(energy, dose)

#######    TALLIES    #########
tallies_file = openmc.Tallies()
cell_filter = openmc.CellFilter(phantom_cells)
tally = openmc.Tally(name='tally')
tally.filters = [cell_filter, particle_filter, dose_filter]
tally.scores = ['flux']

"""
#######    MESH TALLY    #########
tallysize=0.1
mesh = openmc.Mesh()
mesh.dimension = [1, 1, 400]
mesh.lower_left = [SSD, -tallysize/2, -FIELD_SIZE/2]#type: ignore
mesh.upper_right = [SSD+tallysize, tallysize/2, FIELD_SIZE/2]#type:
ignore
mesh_filter = openmc.MeshFilter(mesh)

tally01 = openmc.Tally(name="0 depth tally")
tally01.filters = [mesh_filter, particle_filter, dose_filter]
tally01.scores = ['flux']

meshdpp01 = openmc.Mesh()
```

```
meshdpp01.dimension = [300, 1, 1]

meshdpp01.lower_left = [SSD, -tallysize/2, -tallysize/2]#type: ignore

meshdpp01.upper_right = [SSD+d, tallysize/2, tallysize/2]#type: ignore

meshdpp01_filter = openmc.MeshFilter(meshdpp01)


tallydpp01 = openmc.Tally(name="profile depth dose tally")

tallydpp01.filters = [meshdpp01_filter, particle_filter, dose_filter]

tallydpp01.scores = ['flux']


meshdpp5 = openmc.Mesh()

meshdpp5.dimension = [300, 1, 1]

meshdpp5.lower_left = [SSD, -tallysize/2, -tallysize/2]#type: ignore

meshdpp5.upper_right = [SSD+d, tallysize/2, tallysize/2]#type: ignore

meshdpp5_filter = openmc.MeshFilter(meshdpp5)


tallydpp5 = openmc.Tally(name="profile depth dose tally")

tallydpp5.filters = [meshdpp5_filter, particle_filter, dose_filter]

tallydpp5.scores = ['flux']


tallies_file.append(tallydpp01)

tallies_file.append(tallydpp5)

tallies_file.append(tally01)

"""

tallies_file.append(tally)

tallies_file.export_to_xml()


source = openmc.Source() #type: ignore

source.space = openmc.stats.Point((0,0,0))

phi = openmc.stats.Uniform(0, 2*pi)
```

```
#mu = openmc.stats.Uniform(cos(atan2(1/2, SSD)), 1)
mu = openmc.stats.Uniform(cos(atan2(40/2, SSD)), 1)
source.angle                                                                  =
openmc.stats.PolarAzimuthal(mu,phi,reference_uvw=(1,0,0))
source.energy = openmc.stats.Discrete( [10e6], [1]) #10MeV
source.particle = 'photon'



settings = openmc.Settings()
settings.batches = batches
settings.inactive = inactive
settings.particles = particles
settings.run_mode = 'fixed source'
settings.photon_transport = True
settings.export_to_xml()
# }}}

openmc.run()
```

Program Pembaca Kalibrasi

```
import openmc
import matplotlib.pyplot as plt
import numpy as np
from scipy.signal import savgol_filter
import matplotlib.ticker as mtick
##

sp = openmc.StatePoint('statepoint.5.h5')
##
print(sp.n_particles)

n = 300
d = 30
```

```
tal = sp.tallies [1]
datay = tal.mean
# datax is linspace between 0 and 50
datax = np.linspace(0,d,n)
datay.shape = (n,)
# smooth datay
datay = np.convolve(datay, np.ones(10)/10, mode='same')
# find the index of the maximum value in datay
max_index = np.argmax(datay)
x_max = datax [max_index]
y_max = datay [datay.argmax()]
print(x_max,max_index,y_max)

y_max=datay [max_index]
plt.axvline(x=x_max, color='r')
plt.legend(loc='best')
plt.ylabel("Relative Dose (%)")
plt.xlabel("Position (cm)")
plt.xlim(0,30)
plt.ylim(0,105)
datay=datay/y_max*100
plt.plot(datax, datay)

tal = sp.tallies [1]
datay = tal.mean
datay=datay/y_max*100
n = len(datay)

x_smooth=datax
y_smooth = np.interp(x_smooth, datax, datay)
# print (f"datay =\n {datay}")
# print (f"ysmooth=\n{y_smooth}")
```

```
# Apply Savitzky-Golay filter for smoothing
window_size = 200
poly_order =5
y_smooth = savgol_filter(y_smooth, window_size, poly_order)

max_index=np.argmax(y_smooth)
x_maxs = x_smooth [max_index]
y_maxs=y_smooth [max_index]
plt.plot(x_smooth, y_smooth)
plt.title(f'x_max = {x_max} cm \n xsmooth_max = {x_maxs} cm')
plt.savefig('pdd.png')
plt.show()
plt.savefig('pdd.png')
"""
tal = sp.tallies [2]
datay = tal.mean
# datax is linspace between 0 and 50
datax = np.linspace(0,d,n)
datay.shape = (n,)
# smooth datay
datay = np.convolve(datay, np.ones(10)/10, mode='same')
# find the index of the maximum value in datay
max_index = np.argmax(datay)
x_max = datax [max_index]
y_max = datay [datay.argmax()]
print(x_max,max_index,y_max)

y_max=datay [max_index]
#plt.axvline(x=x_max, color='r')
plt.legend(loc='best')
plt.ylabel("Relative Dose (%)")
```

```
plt.xlabel("Position (cm)")
plt.xlim(-0,30)
plt.ylim(0,105)
datay=datay/y_max*100
plt.plot(datax, datay)

tal = sp.tallies [1]
datay = tal.mean
datay=datay/y_max*100
n = len(datay)

x_smooth=datax
y_smooth = np.interp(x_smooth, datax, datay)
# print (f"datay =\n {datay}")
# print (f"ysmooth=\n{y_smooth}")

# Apply Savitzky-Golay filter for smoothing
window_size = 200
poly_order =5
y_smooth = savgol_filter(y_smooth, window_size, poly_order)

max_index=np.argmax(y_smooth)
x_maxs = x_smooth [max_index]
y_maxs=y_smooth [max_index]
plt.plot(x_smooth, y_smooth)
# draw line at x_max
#plt.axvline(x=x_maxs, color='r')
# write x_max value on plot
plt.title(f'x_max = {x_max} cm \n xsmooth_max = {x_maxs} cm')
#print(f"Y Value on xmax={_max}")
plt.show()
plt.savefig('2.5 depth.png')
```

```
def rounde(unrounded):
  roundede="{:.4e}".format(unrounded)
  return roundede

print(f"\nDosis = {rounde(y_max)}pSvcm3/src")
print(f"Dosis smoothen = {rounde(y_maxs)}pSvcm3/src\n")

print(f"s_rate=mu*volume_wp/phandosevalues") # mu=
mu=1e11#600 MU/min = 10 cGy/s = 1e11 pSv/s
volume_wp=(d/n)*6*6
wpdose=y_max
s_rate=mu*volume_wp/wpdose

print(f"s_rate={mu}*{volume_wp}/{rounde(wpdose)}={s_rate}")
#600cGy/min=36000cGy/h=36uSv/h
print(f"600 cgy/min = 36uSv/h")
print(f"\n36uSv/h = k * {y_max} pSvcm3/src/{volume_wp}cm3\n")
k=volume_wp/y_max*36
print(f"k * {y_max} pSvcm3/src/{volume_wp}cm3 = {k*y_max/volume_wp}uSv/h")
print(f"36 uSv/h = {k*y_max/volume_wp} uSv/h")
print(f"k = {k}")



print(f"k smoothen = {volume_wp/y_maxs*36}")

tally = sp.tallies [3]#2.5 depth {{{
import matplotlib.pyplot as plt

flux = tally.get_slice(scores= ['flux'])
data = tally.mean.flatten()
```

```
x = np.linspace(-25, 25, len(data))

max_index=np.argmax(data)
x_max = x [max_index]
y_max = np.max(data)
print("ymax = ", y_max)
plt.axvline(x=x_max,color='r',label='peak')
yflat=data/y_max*100
#print(yflat)

fmt='%.0f%%' #changing into format
xticks = mtick.FormatStrFormatter(fmt)

#plt.yflat.set_major_formatter(FuncFormatter(lambda      y,      _:
'{:.0%}'.format(yflat)))
plt.plot(x, yflat,label="raw")

ysmooth=np.interp(x,x,yflat)
window_size=300
poly_order=10
start=100
end=400
area_tofilter=yflat [start:end]
filtered=savgol_filter(area_tofilter,window_size,poly_order)
ysmooth [start:end]=filtered
plt.plot(x,ysmooth,label="savgol filter")
plt.legend(loc='best')
plt.ylabel("Relative Dose (%)")
plt.xlabel("Position (cm)")
plt.xlim(-25,25)
plt.ylim(0,105)
```

```
plt.title('2.5 depth.png')
plt.savefig('2.5 depth.png')
plt.show()
"""
```

Program Ruangan

```
import openmc #type: ignore
import matplotlib.pyplot as plt
from math import * #type: ignore

particle=int(input('Particle number (\'twas 1e7)\n= '))
SOURCE_SIZE=20
FIELD_SIZE=40
SSD=100

air=openmc.Material(name='Air')
air.set_density('g/cm3',0.001205)
air.add_nuclide('N14',0.7)
air.add_nuclide('O16',0.3)
#air.add_s_alpha_beta('c_H_in_Air')
air.add_element('C',0.002)
air.add_element('Fe',0.001)
air.add_element('Si',0.001)
air.add_element('Mn',0.001)

air2=openmc.Material(name='Air')
air2.set_density('g/cm3',0.001205)
air2.add_nuclide('N14',0.7)
air2.add_nuclide('O16',0.3)
#air.add_s_alpha_beta('c_H_in_Air')
air2.add_element('C',0.002)
```

```
air2.add_element('Fe',0.001)
air2.add_element('Si',0.001)
air2.add_element('Mn',0.001)

water=openmc.Material(name='Water')
water.set_density('g/cm3',1.0)
water.add_nuclide('H1',2.0)
water.add_nuclide('O16',1.0)
water.add_s_alpha_beta('c_H_in_H2O')

soft=openmc.Material(name='Soft Tissue')
soft.set_density('g/cm3',1.0)
soft.add_element('H', 10.4472, percent_type='ao')#1
soft.add_element('C', 23.219, percent_type='ao')#6
soft.add_element('N', 2.388, percent_type='ao')#7
soft.add_element('O', 63.0238, percent_type='ao')#8
soft.add_element('Na', 0.113, percent_type='ao')#11
soft.add_element('Mg', 0.013, percent_type='ao')#12
soft.add_element('S', 0.199, percent_type='ao')#16
soft.add_element('Cl', 0.134, percent_type='ao')#17
soft.add_element('K', 0.199, percent_type='ao')#19
soft.add_element('Ca', 0.023, percent_type='ao')#20
#https://physics.nist.gov/cgi-bin/Star/compos.pl?matno=261

bpe=openmc.Material(name='Borate Polyethylene')
bpe.set_density('g/cm3',1.0)
bpe.add_element('H', 0.111, percent_type='ao')#1
bpe.add_element('C', 0.856, percent_type='ao')#6
bpe.add_element('B', 0.143, percent_type='ao')#5
bpe.add_element('O', 0.889, percent_type='ao')#8

lead=openmc.Material(name='Lead')
```

```
lead.set_density('g/cm3',11.35)

lead.add_element('Pb',1.0)


concrete=openmc.Material(name='Concrete')

concrete.set_density('g/cm3',2.3)  #Harus  disesuaikan  dengan  uji
beton

concrete.add_element('H', 0.01, percent_type='ao')#1

concrete.add_element('C', 0.01, percent_type='ao')#6

concrete.add_element('O', 0.52, percent_type='ao')#8

concrete.add_element('Na', 0.01, percent_type='ao')#11

concrete.add_element('Mg', 0.01, percent_type='ao')#12

concrete.add_element('Al', 0.01, percent_type='ao')#13

concrete.add_element('Si', 0.25, percent_type='ao')#14

concrete.add_element('S', 0.01, percent_type='ao')#16

concrete.add_element('K', 0.01, percent_type='ao')#19

concrete.add_element('Ca', 0.01, percent_type='ao')#20

concrete.add_element('Fe', 0.01, percent_type='ao')#26

concrete.add_element('Pb', 0.01, percent_type='ao')#82


iron = openmc.Material(name="Iron")

iron.set_density('g/cm3',7.87) #Harus disesuaikan dengan uji beton

iron.add_element('Fe', 1.0)#1


materials=openmc.Materials(
[air,air2,air3,water,soft,bpe,lead,concrete,iron])

materials.export_to_xml()


################################################

rotationDegree=int(input("Harap masukkan sudut rotasi LINAC\n= "))

############# Geometry #######################

#x

t1=openmc.XPlane(632)

t2=openmc.XPlane(632-76.5)
```

```
t3=openmc.XPlane(632-76.5-155)

t3fe=openmc.XPlane(632-76.5-155-235)

t4=openmc.XPlane(632-76.5-155-76.5)

t5=openmc.XPlane(632-76.5-155-76.5)

t6=openmc.XPlane(632-76.5-155-235)


b1=openmc.XPlane(-632)

b2=openmc.XPlane(-632+76.5)

b3=openmc.XPlane(-632+76.5+155.0)

b4=openmc.XPlane(-632+76.5+155.0+76.5)

b5=openmc.XPlane(-632+76.5+250.5)


#y #Di berkas 125, tapi ga make sense jadi diubah ke 1200 biar masuk
angkanya

u1=openmc.YPlane(190.0+250.0+120.0+185.0+81.0)

u2=openmc.YPlane(190.0+250.0+120.0+185.0)

ufe=openmc.YPlane(190.0+250.0+120.0+185.0-10)

u3=openmc.YPlane(190.0+250.0+120.0)

u4=openmc.YPlane(190.0+250.0)

u5=openmc.YPlane(190.0)


s1=openmc.YPlane(-190.0-185.0-128.0)

s2=openmc.YPlane(-190.0-185.0)

s3=openmc.YPlane(-190.0)


#z total tinggi = 6000, lantai 1 setebal 480


zm1=openmc.ZPlane(-300.0)

zmax=openmc.ZPlane(300.0)

z3=openmc.ZPlane(-300.0+48.0+124.0+186.0+117.0+125.0)#1250     ATO
2500???

z2=openmc.ZPlane(-300.0+48.0+124.0+186.0+117.0)

z1=openmc.ZPlane(-300.0+48.0+124.0+186.0)
```

```
z0=openmc.ZPlane(-300.0+48.0)


#pintu utara, pintu barat, pintu selatan geometri nya

pu=openmc.YPlane(190.0+250.0+120.0+185.0+40.0)

#                   ^^^ asumsi pintu lebih lebar 40cm dibandingkan lubang pintunya

pb0=b5

pb1=openmc.XPlane(-632.0+76.5+250.5-1) #Angkanya ini masih ngarang karena gatau tebal pintu, ada kemungkinan formulanya di RHSPintu.py salah

pb2=openmc.XPlane(-632.0+76.5+250.5-1-15)

pb3=openmc.XPlane(-632.0+76.5+250.5-1-15-1)

ps=u3



###############################################

dt1 = -t1 & +t2 & +s3 & -u5 & +z0 & -z2

dt2 = -t2 & +t3 & +s1 & -u1 & +z0 & -z2

dt3 = -t3 & +t4 & +s3 & -u5 & +z0 & -z2


db1 = +b1 & -b2 & +s3 & -u5 & +z0 & -z2

db2 = +b2 & -b3 & +s1 & -u3 & +z0 & -z2

db3 = +b3 & -b4 & +s3 & -u5 & +z0 & -z2


du1 = +b5 & -t3 & -u1 & +u2 & +z0 & -z2

du2 = +b3 & -t6 & -u3 & +u4 & +z0 & -z2


ds1 = +b3 & -t3 & +s1 & -s2 & +z0 & -z2


fe1 = -t3 & +t3fe & -u1 & +ufe & +z0 & -z2

datas= +b1 & -t1 & +s1 & -u1 & +z2 & -z3 #celing

datte= +b4 & -t4 & -u5 & +s3 & +z1 & -z2 #linac's middle wall

dbaw = +b1 & -t1 & +s1 & -u1 & +zm1 & -z0 #flooring
```

```
################################################
#pintu
ppb = -pu & +ps & -pb0 & +pb1 & +z0 & -z2#pintu Pb
pbpe= -pu & +ps & -pb1 & +pb2 & +z0 & -z2#pintu BPE
ppb2= -pu & +ps & -pb2 & +pb3 & +z0 & -z2#pintu Pb

#Udara
#void1= -dt1 & +dt2 & -dt3 & +db1 & -db2 & +db3 & -du1 & +du2 & -
ds1 & +ppb & -pbpe & +ppb2 & +datas & -dbaw
#void1cell=openmc.Cell(fill=air,region=void1)

#Cell =

dt1cell=openmc.Cell(fill=concrete,region=dt1)
dt2cell=openmc.Cell(fill=concrete,region=dt2)
dt3cell=openmc.Cell(fill=concrete,region=dt3)
db1cell=openmc.Cell(fill=concrete,region=db1)
db2cell=openmc.Cell(fill=concrete,region=db2)
db3cell=openmc.Cell(fill=concrete,region=db3)
du1cell=openmc.Cell(fill=concrete,region=du1)
du2cell=openmc.Cell(fill=concrete,region=du2)
ds1cell=openmc.Cell(fill=concrete,region=ds1)

ppbcell=openmc.Cell(fill=lead,region=ppb)
pbpecell=openmc.Cell(fill=bpe,region=pbpe)
ppb2cell=openmc.Cell(fill=lead,region=ppb2)

datascell=openmc.Cell(fill=concrete,region=datas)
dattecell=openmc.Cell(fill=concrete,region=datte)
dbawcell=openmc.Cell(fill=concrete,region=dbaw)

fecell=openmc.Cell(fill=iron, region=fe1)
```

```
###############################################
#        Detektor/Tally        #

detd= 4.5
detde=2
#             Utara       #
deu1=openmc.YPlane (190.0+250.0+120.0+185.0+81.0+30.0)
deu1t=openmc.YPlane(190.0+250.0+120.0+185.0+81.0+30.0+10.8)
#+tebal detektor
deu1ts=openmc.YPlane(190.0+250.0+120.0+185.0+81.0+30.0+detde)
#+tebal detektor
deu2=openmc.YPlane (190.0+250.0+120.0+185.0+81.0+100.0)
deu2t=openmc.YPlane(190.0+250.0+120.0+185.0+81.0+100.0+10.8)
b5=openmc.XPlane(-632+76.5+250.5)
deu2ts=openmc.YPlane(190.0+250.0+120.0+185.0+81.0+100.0+detde)
deu3=openmc.YPlane (190.0+250.0+120.0+185.0+81.0+200.0)
deu3t=openmc.YPlane(190.0+250.0+120.0+185.0+81.0+200.0+10.8)
deu3ts=openmc.YPlane(190.0+250.0+120.0+185.0+81.0+200.0+detde)

deuz0=openmc.ZPlane(-300.0+48.0+100.0)#Tinggi  detektor,  default
untuk semua detektor kecuali atas
deuz1=openmc.ZPlane(-300.0+48.0+100.0+200.0)
deuz0s=openmc.ZPlane(-300.0+48.0+100.0-detd/2)#Tinggi    detektor,
default untuk semua detektor kecuali atas
deuz1s=openmc.ZPlane(-300.0+48.0+100.0+detd/2)
#deuz0s=openmc.ZPlane(-300.0+48.0+100.0-(150/2))#Tinggi  detektor,
default untuk semua detektor kecuali atas
#deuz1s=openmc.ZPlane(-300.0+48.0+100.0+(150/2))

deubb=openmc.XPlane(-632.0+76.5+250.5+100.0) #Koordinat x nya masih
ngasal
deubt=openmc.XPlane(-632.0+76.5+250.5+100.0+50.0)
deubbs=openmc.XPlane(-632.0+76.5+250.5+100.0-detd/2)  #Koordinat  x
nya masih ngasal
```

```
deubts=openmc.XPlane(-632.0+76.5+250.5+100.0+detd/2)

deucb=openmc.XPlane(250.0) #koordinat x nya masih ngasal
deuct=openmc.XPlane(250.0+50.0)
deucbs=openmc.XPlane(250.0-detd/2) #koordinat x nya masih ngasal
deucts=openmc.XPlane(250.0+detd/2)

detb1= +deu1 & -deu1t & +deuz0 & -deuz1 & +deubb & -deubt #detektor
utara barat, x nya ngasal
detb2= +deu2 & -deu2t & +deuz0 & -deuz1 & +deubb & -deubt
detb3= +deu3 & -deu3t & +deuz0 & -deuz1 & +deubb & -deubt
dett1= +deu1 & -deu1t & +deuz0 & -deuz1 & +deucb & -deuct #detektor
utara timur, x nya ngasal
dett2= +deu2 & -deu2t & +deuz0 & -deuz1 & +deucb & -deuct
dett3= +deu3 & -deu3t & +deuz0 & -deuz1 & +deucb & -deuct

detb1s= +deu1 & -deu1ts & +deuz0s & -deuz1s & +deubbs & -deubts
#detektor utara barat, x nya ngasal
detb2s= +deu2 & -deu2ts & +deuz0s & -deuz1s & +deubbs & -deubts
detb3s= +deu3 & -deu3ts & +deuz0s & -deuz1s & +deubbs & -deubts
dett1s= +deu1 & -deu1ts & +deuz0s & -deuz1s & +deucbs & -deucts
#detektor utara timur, x nya ngasal
dett2s= +deu2 & -deu2ts & +deuz0s & -deuz1s & +deucbs & -deucts
dett3s= +deu3 & -deu3ts & +deuz0s & -deuz1s & +deucbs & -deucts

#Detektor
detub1cell=openmc.Cell(fill=air2,region=detb1) #sel detektor barat
1
detub2cell=openmc.Cell(fill=air2,region=detb2)
detub3cell=openmc.Cell(fill=air2,region=detb3)
detut1cell=openmc.Cell(fill=air2,region=dett1)
detut2cell=openmc.Cell(fill=air2,region=dett2)
detut3cell=openmc.Cell(fill=air2,region=dett3)
```

```
detub1scell=openmc.Cell(fill=air2,region=detb1s)    #sel   detektor
barat 1

detub2scell=openmc.Cell(fill=air2,region=detb2s)

detub3scell=openmc.Cell(fill=air2,region=detb3s)

detut1scell=openmc.Cell(fill=air2,region=dett1s)

detut2scell=openmc.Cell(fill=air2,region=dett2s)

detut3scell=openmc.Cell(fill=air2,region=dett3s)



#             Timur        #

det1=openmc.XPlane(632.0+30.0)

det1t=openmc.XPlane(632.0+30.0+10.8)

det1ts=openmc.XPlane(632.0+30.0+detde)

det2=openmc.XPlane(632.0+100.0)

det2t=openmc.XPlane(632.0+100.0+10.8)

det2ts=openmc.XPlane(632.0+100.0+detde)

det3=openmc.XPlane(632.0+200.0)

det3t=openmc.XPlane(632.0+200.0+10.8)

det3ts=openmc.XPlane(632.0+200.0+detde)


detu=openmc.YPlane(25.0)

dets=openmc.YPlane(-25.0)

detus=openmc.YPlane(detd/2)

detss=openmc.YPlane(-detd/2)


dett1= +det1 & -det1t & +deuz0 & -deuz1 & -detu & +dets

dett2= +det2 & -det2t & +deuz0 & -deuz1 & -detu & +dets

dett3= +det3 & -det3t & +deuz0 & -deuz1 & -detu & +dets

dett3s= +det3 & -det3t & +deuz0 & -deuz1 & -detu & +dets

dett1s= +det1 & -det1ts & +deuz0s & -deuz1s & -detus & +detss

dett2s= +det2 & -det2ts & +deuz0s & -deuz1s & -detus & +detss

dett3s= +det3 & -det3ts & +deuz0s & -deuz1s & -detus & +detss
```

```
dett1cell=openmc.Cell(fill=air2,region=dett1)
dett2cell=openmc.Cell(fill=air2,region=dett2)
dett3cell=openmc.Cell(fill=air2,region=dett3)
dett1scell=openmc.Cell(fill=air2,region=dett1)
dett2scell=openmc.Cell(fill=air2,region=dett2)
dett3scell=openmc.Cell(fill=air2,region=dett3)

#              Barat       #

deb1=openmc.XPlane(-632.0+76.5+250.5-1-15-30.0)
deb1t=openmc.XPlane(-632.0+76.5+250.5-1-15-30.0-10.8)
deb1ts=openmc.XPlane(-632.0+76.5+250.5-1-15-30.0-detde)
deb2=openmc.XPlane(-632.0+76.5+250.5-1-15-100.0)
deb2t=openmc.XPlane(-632.0+76.5+250.5-1-15-100.0-10.8)
deb2ts=openmc.XPlane(-632.0+76.5+250.5-1-15-100.0-detd)
deb3=openmc.XPlane(-632.0+76.5+250.5-1-15-200.0)
deb3t=openmc.XPlane(-632.0+76.5+250.5-1-15-200.0-10.8)
deb3ts=openmc.XPlane(-632.0+76.5+250.5-1-15-200.0-detde)

debu=openmc.YPlane(190.0+250.0+120.0+185.0-67.5)
debs=openmc.YPlane(190.0+250.0+120.0+185.0-67.5-50.0)
debus=openmc.YPlane(190.0+250.0+120.0+185.0-(67.5+detd/2))
debss=openmc.YPlane(190.0+250.0+120.0+185.0-(67.5-detd/2))

detb1= -deb1 & +deb1t & +deuz0 & -deuz1 & -debu & +debs
detb2= -deb2 & +deb2t & +deuz0 & -deuz1 & -debu & +debs
detb3= -deb3 & +deb3t & +deuz0 & -deuz1 & -debu & +debs
detb1s= -deb1 & +deb1ts & +deuz0s & -deuz1s & -debus & +debss
detb2s= -deb2 & +deb2ts & +deuz0s & -deuz1s & -debus & +debss
detb3s= -deb3 & +deb3ts & +deuz0s & -deuz1s & -debus & +debss
```

```
detb1cell=openmc.Cell(fill=air2,region=detb1)
detb2cell=openmc.Cell(fill=air2,region=detb2)
detb3cell=openmc.Cell(fill=air2,region=detb3)
detb1scell=openmc.Cell(fill=air2,region=detb1s)
detb2scell=openmc.Cell(fill=air2,region=detb2s)
detb3scell=openmc.Cell(fill=air2,region=detb3s)

#             Atas        #
deau=openmc.XPlane(25.0)
deas=openmc.XPlane(-25.0)
deaus=openmc.XPlane(detd/2)
deass=openmc.XPlane(-detd/2)

deat=openmc.YPlane(25.0)
deab=openmc.YPlane(-25.0)
deats=openmc.YPlane(detd/2)
deabs=openmc.YPlane(-detd/2)

dea1=openmc.ZPlane(300.0+50.0) #1250 ATO 2500???
dea1t=openmc.ZPlane(300.0+50.0+10.8) #1250 ATO 2500???
dea1ts=openmc.ZPlane(300.0+30.0+detde) #1250 ATO 2500???
dea2=openmc.ZPlane(300.0+100.0)
dea2t=openmc.ZPlane(300.0+100.0+10.8)
dea2ts=openmc.ZPlane(300.0+100.0+detde)
dea3=openmc.ZPlane(300.0+200.0)
dea3t=openmc.ZPlane(300.0+200.0+10.8)
dea3ts=openmc.ZPlane(300.0+200.0+detde)

deta1= +dea1 & -dea1t & -deau & +deas & +deab & -deat
deta2= +dea2 & -dea2t & -deau & +deas & +deab & -deat
deta3= +dea3 & -dea3t & -deau & +deas & +deab & -deat
deta1s= +dea1 & -dea1ts & -deaus & +deass & +deabs & -deats
```

```
deta2s= +dea2 & -dea2ts & -deaus & +deass & +deabs & -deats

deta3s= +dea3 & -dea3ts & -deaus & +deass & +deabs & -deats


deta1cell=openmc.Cell(fill=air2,region=deta1)

deta2cell=openmc.Cell(fill=air2,region=deta2)

deta3cell=openmc.Cell(fill=air2,region=deta3)

deta1scell=openmc.Cell(fill=air2,region=deta1s)

deta2scell=openmc.Cell(fill=air2,region=deta2s)

deta3scell=openmc.Cell(fill=air2,region=deta3s)


#Water Phantom 10x10x5

#TODO: Water phantom dpp and lateral tallies, to get flux value and
relative comparison to the dose.

#make tally, search for the dose, search the corresponding flux,
use it as the conversion rate for searching dose(sv/h)


phantom_rotation=270 #phantom rotation

pr=phantom_rotation

detaxu=openmc.YPlane(5)

detaxs=openmc.YPlane(-5)


if pr==0 or pr==180:

  detaxt=openmc.XPlane(5)

  detaxb=openmc.XPlane(5)

  detaxza=openmc.ZPlane(-128+100+2.5)

  detaxzb=openmc.ZPlane(-128+100-2.5)

if pr==180:

  detaxt=openmc.XPlane(5)

  detaxb=openmc.XPlane(-5)

  detaxza=openmc.ZPlane(-128-100+2.5)

  detaxzb=openmc.ZPlane(-128-100-2.5)

elif pr==90 or pr==270:

  detaxza=openmc.XPlane(2.5)
```

```
  detaxzb=openmc.XPlane(-2.5)
  detaxt=openmc.ZPlane(-128+5)
  detaxb=openmc.ZPlane(-128-5)

else:
  print('Phantom Rotation Error, benarkan kode pr')

detax= -detaxu & +detaxs & -detaxt & +detaxb & -detaxza & +detaxzb
# type: ignore

detaxcell=openmc.Cell(fill=water,region=detax)

#Kotak Udara Pembatas
ymax=openmc.YPlane(1100,boundary_type='vacuum')
ymin=openmc.YPlane(-1100,boundary_type='vacuum')
xmax=openmc.XPlane(945,boundary_type='vacuum')
xmin=openmc.XPlane(-945,boundary_type='vacuum')
zmin=openmc.ZPlane(-550,boundary_type='vacuum')
zmaxx=openmc.ZPlane(550,boundary_type='vacuum')

#void1cell = openmc.Cell(fill=air, region= (-datascell.region) & (-
dt1cell.region) & (-dt2cell.region) & (-dt3cell.region) & (-
db1cell.region) & (-db2cell.region) & (-db3cell.region) & (-
du1cell.region) & (-du2cell.region) & (-ds1cell.region))

void1= +zmin & -zmaxx \
  & +ymin & -ymax & +xmin & -xmax\
  & ~dt1cell.region & ~dt2cell.region & ~dt3cell.region \
    & ~db1cell.region & ~db2cell.region & ~db3cell.region \
      & ~du1cell.region & ~du2cell.region & ~ds1cell.region\
      & ~ppbcell.region & ~pbpecell.region & ~ppb2cell.region\
      & ~datascell.region & ~dbawcell.region & ~dattecell.region\
      & ~detb1cell.region & ~detb2cell.region & ~detb3cell.region\
```

```
      &     ~detub1cell.region     &     ~detub2cell.region     &
~detub3cell.region\
      &     ~detut1cell.region     &     ~detut2cell.region     &
~detut3cell.region\
      & ~dett1cell.region & ~dett2cell.region & ~dett3cell.region\
      & ~deta1cell.region & ~deta2cell.region & ~deta3cell.region\

void1cell = openmc.Cell(fill=air, region=void1)

univ=openmc.Universe(cells= [dt1cell,dt2cell,dt3cell,
              db1cell,db2cell,db3cell,
              du1cell,du2cell,ds1cell,
              ppbcell,pbpecell,ppb2cell,
              datascell,dbawcell,dattecell,
              void1cell,
              detb1cell,detb2cell,detb3cell,
              detub1cell,detub2cell,detub3cell,
              detut1cell,detut2cell,detut3cell,
              dett1cell,dett2cell,dett3cell,
              deta1cell,deta2cell,deta3cell,
              detb1scell,detb2scell,detb3scell,
              detub1scell,detub2scell,detub3scell,
              detut1scell,detut2scell,detut3scell,
              dett1scell,dett2scell,dett3scell,
              deta1scell,deta2scell,deta3scell,
              detaxcell,fecell])
geometry=openmc.Geometry(univ)
geometry.export_to_xml()

colors= {}
colors [lead]='black'
colors [bpe]='lightblue'
colors [concrete]='grey'
```

```
colors [air]='white'

colors [air2]='blue'

ssh=SOURCE_SIZE/2

###############################################

#        Rotation

###############################################

def sposi(d,rot):

  u= (-sin(radians(rot)))

  v= 0

  w= (-cos(radians(rot))) #source position

  uvw = (u,v,w)

  xyz = ( d*(sin(radians(rot))) ), 0, -128+ ( d*(cos(radians(rot)) ))

  xyz = xyz

  if rot==0 or rot==180:

    xyz1= ( d*(sin(radians(rot))) )+ssh, ssh, -128+ ( d*(cos(radians(rot)) )+0.1)

    xyz2= ( d*(sin(radians(rot))) )-ssh, -ssh, -128+ ( d*(cos(radians(rot)) ))

  else:

    xyz1= ( d*(sin(radians(rot))) )+0.1, ssh, -128+ ( d*(cos(radians(rot)) )+ssh)

    xyz2= ( d*(sin(radians(rot))) ), -ssh, -128+ ( d*(cos(radians(rot)) )-ssh)

  return uvw, xyz,xyz1,xyz2

  #asumsi tinggi pasien 75

###############################################

#    Input (linac distance,rotation)   #

linacuvw, linacxyz,linacxyzn1,linacxyzn2=sposi(SSD,rotationDegree)

###############################################

print("linacuvw,linacxyz,linacuvwn1,linacuvwn2",linacuvw,
linacxyz,linacxyzn1,linacxyzn2)


###############################################
```

```
#      Penampil Geometri        #
##############################################
plot=openmc.Plot()
plot.basis='xy'
plot.origin=(0,200,0)
plot.width=(2000,2000)
plot.pixels=(2000,2000)
plot.filename='xy_room'
plot.colors={
  lead:'black',
  bpe:'lightblue',
  concrete:'grey',
  air:'white',
  air2:'blue'
}
plot.color_by='material'
plot_file=openmc.Plots( [plot])
plot_file.export_to_xml()
openmc.plot_geometry()
#!convert something.ppm something.png

plot.to_ipython_image()


plot.basis='yz'
plot.filename='yz_room'
plot.width=(2000,2000)
plot_file=openmc.Plots( [plot])
plot_file.export_to_xml()
openmc.plot_geometry()
plot.to_ipython_image()

plot.basis='xz'
```

```
plot.filename='xz_room'

plot.pixels=(2000,2000)

plot.width=(2000,2000)

plot.origin=(0,0,0)

plot_file=openmc.Plots( [plot])

plot_file.export_to_xml()

openmc.plot_geometry()

plot.to_ipython_image()

"""

plt.rcParams.update({'font.size': 5})

univ.plot(width=(2500,2700),basis='xy',color_by='material',colors=
colors)

plt.savefig('xyRSHS.png',dpi=500, bbox_inches='tight')

plt.show()

univ.plot(width=(1400,1040),basis='xz',color_by='material',colors=
colors)

plt.savefig('xzRSHS.png',dpi=500, bbox_inches='tight')

plt.show()

univ.plot(width=(1800,1040),basis='yz',color_by='material',colors=
colors)

plt.savefig('yzRSHS.png',dpi=500, bbox_inches='tight')

plt.show()

"""


#Tidak terdapat library matplotlib pada server, sehingga penampil
geometri harus dimatikan untuk running server


###############################################

#          Plot Grid          #

###############################################

#ax.set_title('Distribusi Dosis Ruangan (uSv/hour)') #type: ignore

#univ.plot(width=(2000,2000),basis='xy',color_by='material',colors
=colors)
```

```
#plt.savefig('DoseDistributionMap.png',dpi=500,
bbox_inches='tight')


###############################################

#         Setting           #

###############################################

settings=openmc.Settings()

source =openmc.Source()

"""

#source.space=openmc.stats.Points(xyz=)

source.space=openmc.stats.Point(xyz=linacxyz) # type: ignore

#phi2=openmc.stats.Isotropic() #isotropic ato uniform?

#phi1=openmc.stats.Monodirectional((0,0,1))

phi     =openmc.stats.Uniform(0.0,2*pi)     #     type:     ignore
#phi=distribution of the azimuthal angle in radians

#mu= distribution of the cosine of the polar angle


#tan theta = r/SAD=20/1000; theta = atan(20/100)=0.19739555984988;
cos theta=0.98058

mu=openmc.stats.Uniform(0.98058,1) # type: ignore #mu= distribution
of the cosine of the polar angle


source.angle                                                    =
openmc.stats.PolarAzimuthal(mu,phi,reference_uvw=linacuvw) # type:
ignore

source.energy = openmc.stats.Discrete( [10e6], [1]) #10MeV # type:
ignore

"""

#source.space=openmc.stats.Point(xyz=linacxyz) # type: ignore

#source.space = openmc.stats.Box((-FIELD_SIZE/2-d, -SOURCE_SIZE/2,
-SOURCE_SIZE/2),         (-FIELD_SIZE/2-d-t,         SOURCE_SIZE/2,
SOURCE_SIZE/2))

phi     =openmc.stats.Uniform(0.0,2*pi)     #     type:     ignore
#phi=distribution of the azimuthal angle in radians

mu          =          openmc.stats.Uniform(cos(atan2(((FIELD_SIZE-
SOURCE_SIZE)/2),SSD)), 1)
```

```
source.space = openmc.stats.Box((linacxyzn1), (linacxyzn2))

##source.angle = openmc.stats.Monodirectional(linacuvw)

source.angle                                                    =
openmc.stats.PolarAzimuthal(mu,phi,reference_uvw=linacuvw) # type:
ignore

source.energy = openmc.stats.Discrete( [10e6], [1])


source.particle = 'photon'

#source.particle = 'neutron'

settings.source = source

settings.batches= 3

settings.particles = particle

#Asumsi     36e7     partikel     pada     mula,     pada
600MU=600cGy/m=6Gy/m=6Sv/m=6e6uSv/m=360e6uSv/h

settings.run_mode = 'fixed source'

settings.photon_transport = True

settings.export_to_xml()


###############################################

#         Tallies           #

###############################################

tally = openmc.Tallies()

#TODO: Change Tally size to smaller actual tally size value

#Tally Dose Distribution

mesh = openmc.RegularMesh() # type: ignore

mesh.dimension = [500, 500]

xlen = 2000;ylen = 2000

mesh.lower_left = [-xlen/2, -xlen/2]

mesh.upper_right = [ylen/2, ylen/2]

mesh_filter = openmc.MeshFilter(mesh)


tally1 = openmc.Tally(name = 'Room Dose Distribution')

tally1.scores = ['flux']
```

```
particle1 = openmc.ParticleFilter('photon')


energy,  dose  =  openmc.data.dose_coefficients('photon',  'RLAT') 
#Data konve # type: ignore

dose_filter  =  openmc.EnergyFunctionFilter(energy,  dose)  #konvert 
partikel  energi  tertentu  ke  deskripsi  icrp116  partikelcm/src-
>pSvcm3/srcwphantom_cell,particle3,dose_filter


tally1.filters= [mesh_filter,particle1,dose_filter]

tally.append(tally1)


#Tally Detektor

filter_cell = openmc.CellFilter((detb1cell,detb2cell,detb3cell,\

              detub1cell,detub2cell,detub3cell,\

              detut1cell,detut2cell,detut3cell,\

              dett1cell,dett2cell,dett3cell,\

              deta1cell,deta2cell,deta3cell))

tally2 = openmc.Tally(name = 'flux')

particle2 = openmc.ParticleFilter('photon')

tally2.filters = [filter_cell, particle2, dose_filter]

tally2.scores = ['flux']

tally.append(tally2)


filter_cell_small                                                  =
openmc.CellFilter((detb1scell,detb2scell,detb3scell,\

              detub1scell,detub2scell,detub3scell,\

              detut1scell,detut2scell,detut3scell,\

              dett1scell,dett2scell,dett3scell,\

              deta1scell,deta2scell,deta3scell))

tally4 = openmc.Tally(name = 'flux detektor small')

tally4.filters = [filter_cell_small, particle2, dose_filter]

tally4.scores = ['flux']

tally.append(tally4)
```

```
energy2, dose2 = openmc.data.dose_coefficients('neutron', 'AP') #Data konve # type: ignore

dose_filter = openmc.EnergyFunctionFilter(energy2, dose2) #konvert partikel energi tertentu ke deskripsi icrp116 partikelcm/src->pSvcm3/srcwphantom_cell,particle3,dose_filter

particle3=openmc.ParticleFilter('neutron')

neutroncell = openmc.CellFilter((detb1cell,detb2cell,detb3cell))

tally3= openmc.Tally(name='neutron')

tally3.filters= [neutroncell,particle3,dose_filter]

tally3.scores= ['flux']

tally.append(tally3)


"""

#Tally Water Phantom

wphantom_cell=openmc.CellFilter(detaxcell)

tally3=openmc.Tally(name='wphantom')

particle3= openmc.ParticleFilter('photon')

#tally3.filters= [wphantom_cell,particle3]

#Energy_filter = openmc.EnergyFilter( [1e-3, 1e13])

tally3.filters= [wphantom_cell,particle3,dose_filter] #output pSvcm3/src

tally3.scores = ['flux']

tally.append(tally3)

"""


tally.export_to_xml()

openmc.run()
```

Program Pembaca Ruangan

```
import openmc # type: ignore

import matplotlib.pyplot as plt

from matplotlib.colors import LogNorm
```

```
import pandas as pd

sp = openmc.StatePoint('./statepoint.3.h5')

f=open("output.txt","w")
f.write(str(sp.tallies))
f.close()
print(sp.n_particles)
print(sp.tallies)
meshtally = sp.tallies [1]
print(sp.tallies [1])
print(sp.tallies [2])
#print(sp.tallies [3])

x=500#harus sama dengan resolusi pada file utama
y=500

conversion = 16368352439.27938
# dose * flux * src_rate * t / V = [pSv cm²] [p-cm/src] [src/sec]
[sec]/ [cm³]= [pSv]
# dose * cfac *t/V=pSv
# dose*cfac*60/V


v=(2000/x)*(2000/y)*5 #volume of room dose distribution

# {{{ ######## Pembuatan Visualisasi Distribusi Dosis
#################
dosevalues=meshtally.get_values() #Nilai perpixel dari grid
500x500
dosevalues.shape=(x,y)
#pSvcm3/src*(src/s)/cm3=pSv/s
```

```
dosevalues = dosevalues*conversion/v #pSv/s = (pSv*cm3/src) *src/s
/cm3

dose=(dosevalues/1_000_000)*3600 #pSv/s -> uSv/hour

dose=dose [::-1,:]


GD=pd.DataFrame(dose)


GD.to_csv("gridTallyDose.csv")


fig, ax = plt.subplots()

cs = ax.imshow(dose, cmap='coolwarm', norm=LogNorm()) # type:
ignore

cb = plt.colorbar(cs)

ax.set_title('Distribusi Dosis Ruangan (uSv/hour)') #type: ignore

plt.savefig('RoomDoseDistribution.png',dpi=900 )

plt.axis('off')

#################################################################

# }}}


celltally = sp.tallies [2]

celldosevalues = celltally.get_values() #pSvcm3/src;dosevolume per
source

celldosestddev = celltally.std_dev

print(celltally)

celldosevalues.shape = celldosevalues.shape [0]

celldosestddev.shape = celldosestddev.shape [0]


smalltally = sp.tallies [2]

smalltallydose = smalltally.get_values()

smalltallydose.shape = smalltallydose.shape [0]

print("===========",smalltallydose,"===========")

vcell=10.8*50*200 #cm3

#conversion = 4092088109819844.5
```

```
"""
dose_psvs=celldosevalues * conversion/ vcell
usvh_dose=(dose_psvs/1e6)*3600

usvh_stddev=(stddev_psvs/1e6)*3600

dose=usvh_dose
dosestddev=usvh_stddev
"""
dose= celldosevalues*conversion/vcell
dosestddev=celldosestddev*conversion/vcell

DF=pd.DataFrame(dose,dosestddev)

DF.to_csv("cellTallyDose.csv")

#dose=(dose/1e6)*3600 #pSv/s -> uSv/h 3.6e9
#dosestddev = (celldosestddev * s_rate / vcell) *(1e6/3600)
#k=2457897.1324533457
#dose=k*celldosevalues*600/vcell
#dosestddev = (k*celldosevalues*600/vcell) *(1e6/3600)

print(dose)
for v,s in zip(dose,dosestddev):
  f=open("output.txt","a")
  print(f'{v:.7e} +- {s:.7e}uSv/h')
  f.write(str(f'\n{v} +- {s} uSv/h'))
  f.close()

plt.show()
#dosevalues = dosevalues*s_rate/v #picosieverts/s
```

```
"""
# plt.show()
plt.savefig('figlogdoseplot.png')
plt.show()
# plt.close()
# plt.clf()
"""

# plt.clf()
```

LAMPIRAN B

Hasil Pengukuran Laju Dosis Commisioning

A. Hasil Pengukuran Radiasi Gamma (µSv/jam)

Titik	Lokasi	Jarak ke Dinding (cm)	Sudut Gantry				Keterangan
			0°	90°	180°	270°	
A	Depan Pintu Bunker (**Area Pekerja**)	30	0,635	0,524	0,579	0,825	Aman bagi Pekerja Radiasi,
		100	0,501	0,458	0,430	0,546	
		200	0,296	0,290	0,257	0,391	
B	R. Kontrol Linac Synergy (**Area Pekerja**)	30	0,039	0,022	0,028	0,134	Aman bagi Pekerja Radiasi
		100	0,078	0,039	0,067	0,045	
		200	0,033	0,011	0,056	0,033	
C	R. TPS (**Area Pekerja**)	30	1,723	0,407	2,153	4,918	Aman bagi Pekerja Radiasi
		100	1,366	0,246	1,210	2,930	
		200	0,953	0,184	1,020	1,606	
D	Bunker Linac Precise (**Area Pekerja**)	30	0,000	0,251	0,045	0,795	Aman bagi Pekerja Radiasi
		100	0,039	0,162	0,022	0,011	
		200	0,006	0,117	0,006	0,039	
E	Atap Bunker (**Area Pekerja**)	50	0,112	0,078	6,490	0,050	Merupakan area tanpa akses yang sudah diberi pagar permanen (foto terlampir) Tidak Aman bagi Pekerja Radiasi

Dosis *Background* : 0,095 µSv/jam

LAMPIRAN C

Perhitungan Analitik Desain Bunker

Dosis maksimum per minggu (P) sesuai dengan Perka Bapeten No 3/2013 pasal 41

- NBD untuk Pekerja Radiasi adalah :
  10 mSv (sepuluh milisievert) per tahun atau
  0,2 mSv (nol koma dua milisievert) per minggu.

- NBD untuk anggota masyarakat adalah :
  0,5 mSv (nol koma lima milisievert) per tahun atau
  0,01 mSv (nol koma nol satu milisievert) per minggu

Beban kerja (W) :

W_{konv} = 20 pasien/hari x 4 Gy/pasien x 5 hari/minggu = 400 Gy/minggu

W_{IMRT} = 50 pasien/hari x 4 Gy/pasien x 5 hari/minggu = 1000 Gy/minggu

W_{VMAT} = 10 pasien/hari x 4 Gy/pasien x 5 hari/minggu = 200 Gy/minggu

W_{QA} = 12 beam/hari x 2 Gy/beam x 6 hari/minggu = 144 Gy/minggu

Berdasarkan NCRP 151 Beban kerja pesawat Linac adalah:

- Beban kerja radiasi primer (Wprimer)

  Wprimer = (W_{konv} + W_{IMRT} + W_{VMAT} + W_{QA})
  = (400 + 1000 + 200 + 144) Gy/minggu
  = **1744 Gy/minggu**
  = **1744000 mSv/minggu**

- Beban kerja radiasi bocor (Wbocor)

Faktor C1 untuk IMRT :

ΣField = 8

MU/Field = 450

ΣMU = 8 x 450 = 3600

Dose = 4 Gy

MU IMRT = $\frac{3600}{4}$ = 900

MU Konv = 200

C1 = $\frac{MU\ IMRT}{MU\ Konv} = \frac{900}{200}$

C1 = 4,5

Faktor C2 untuk VMAT :

ΣMU = 3600

Dose = 4 Gy

MU VMAT = $\frac{3600}{4}$ = 900

MU Konv = 200

C2 = $\frac{MU\ VMAT}{MU\ Konv} = \frac{900}{200}$

C2 = 4,5

Faktor C4 untuk QA : C4 = 1

Wbocor = $W_{konv} + C1.W_{IMRT} + C2.W_{VMAT} + C4.W_{QA}$

= (400 + 4,5 . 1000 + 4.5 . 200 + 1 . 144) Gy/minggu

= 5944 Gy/minggu

= 5944000 mSv/minggu

Tenth Value Layer (TVL) data tabel 4. dari SRS 47 IAEA :

Untuk energi 10 MV

	Concrete Densitas 2350 kg/m3	**Besi** Densitas 7800 kg/m3	**Timbal** Densitas 11360 kg/m3
***Primary* TVL**	389 mm	105 mm	56 mm
***Leakage* TVL**	305 mm	85 mm	46 mm

Sehingga diperoleh nilai HVL sebagai berikut :

	Concrete Densitas 2350 kg/m3	**Besi** Densitas 7800 kg/m3	**Timbal** Densitas 11360 kg/m3
***Primary* HVL**	117,10 mm	31,61 mm	16,9 mm
***Leakage* HVL**	91,81 mm	25,6 mm	13,8 mm

- *Perisai Primer*

$$B = 10^{-\left(\frac{x}{Tvl}\right)}$$

$$P = \frac{B\,WUT}{(d + SAD)^2}$$

Titik	Area Dibalik Dinding	Kategori Area	Ketebalan Dinding (cm)			Dose Rate (Sv/Week		Keterangan
			Hasil Perhitungan (cm Beton)	Dinding Terbangun (Beton)	Penambahan Shielding	Hasil Perhitungan	Limit	
P1	Bunker Linac 3	Pekerja	194,35	308,00	0,00	1,20E-07	2,00E-04	Aman
P2	Bunker Linac 1	Pekerja	194,35	308,00	0,00	1,20E-07	2,00E-04	Aman
P3	Atap (Kosong)	Pekerja	141,37	242,00	0,00	2,59E-07	2,00E-04	Aman

- *Perisai Sekunder*

$$B_{LX} = \frac{1000 \cdot P \cdot (dsec)^2)}{W \cdot T} = \left(\frac{1}{2}\right)^{\frac{X\ beton\ bocor}{HVL\ beton\ 10MV}}$$

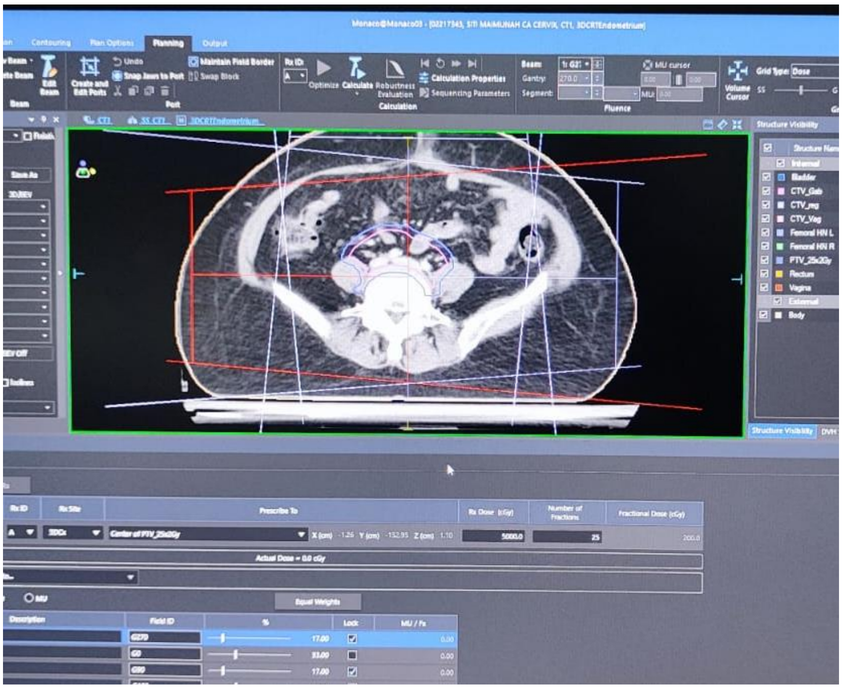
$$\left(\frac{1}{2}\right)^{\frac{X\ beton\ bocor}{HVL\ beton\ 10MV}} = \frac{1000 \cdot P \cdot (dsec)^2)}{W \cdot T}$$

$$P = \frac{\left(\frac{1}{2}\right)^{\frac{Xbetonbocor}{HVLbeton10MV}} \cdot W \cdot T}{1000 \cdot (dsec)^2)}$$

Titik	Area Dibalik Dinding	Kategori Area	Ketebalan Dinding (cm)			Dose Rate (Sv/Week		Keterangan
			Hasil Perhitungan (cm Beton)	Dinding Terbangun (Beton)	Penambahan Shielding	Hasil Perhitungan	Limit	
S1	Maze	Pekerja	89,18	125,00	0,00	6,65E-06	2,00E-04	Aman
S2	R. KN	Pekerja	92,10	128,00	0,00	6,65E-06	2,00E-04	Aman
S3	R.KN	Pekerja	86,16	128,00	0,00	4,25E-06	2,00E-04	Aman
S4	Bunker	Pekerja	95,06	155,00	0,00	1,08E-06	2,00E-04	Aman
S5	Bunker	Pekerja	95,06	155,00	0,00	1,08E-06	2,00E-04	Aman
S6	Koridor Pekerja	Pekerja	71,99	125,00	0,00	1,83E-06	2,00E-04	Aman
S7	R. Operator	Pekerja	87,34	206,00	0,00	1,29E-08	2,00E-04	Aman
S8	R. TPS	Pekerja	95.23	81,00	+ 10 cm Besi (setara 30 cm beton) = 80,00 + 30,00 = 111,00 Beton	1,79E-05	2,00E-04	Aman
S9	Bunker	Pekerja	87,76	155,00	0,00	6,24E-07	2,00E-04	Aman
S10	Bunker	Pekerja	95,06	155,00	0,00	1,08E-06	2,00E-04	Aman
S11	Bunker	Pekerja	95,06	155,00	0,00	1,08E-06	2,00E-04	Aman
S12	R. KN	Pekerja	87,79	128,00	0,00	4,18E-06	2,00E-04	Aman
S13	Atap	Pekerja	46,67	125,00	0,00	2,70E-07	2,00E-04	Aman
S14	Atap	Pekerja	46,67	125,00	0,00	2,70E-07	2,00E-04	Aman
S15	Pintu	Pekerja	64,44	130,00	0,00	7,09E-07	2,00E-04	Aman

LAMPIRAN D

Contoh Penggunaan Variasi Sudut *Gantry* LINAC



LAMPIRAN E

Program Pengubahan Fluks Ke Dosis

```
from pathlib import Path

import numpy as np

_FILES = (
  ('electron', 'electrons.txt'),
  ('helium', 'helium_ions.txt'),
  ('mu-', 'negative_muons.txt'),
  ('pi-', 'negative_pions.txt'),
  ('neutron', 'neutrons.txt'),
  ('photon', 'photons.txt'),
  ('photon kerma', 'photons_kerma.txt'),
  ('mu+', 'positive_muons.txt'),
  ('pi+', 'positive_pions.txt'),
  ('positron', 'positrons.txt'),
  ('proton', 'protons.txt')
)

_DOSE_ICRP116 = {}


def _load_dose_icrp116():
  """Load effective dose tables from text files"""
  for particle, filename in _FILES:
    path = Path(__file__).parent / filename
    data = np.loadtxt(path, skiprows=3, encoding='utf-8')
    data [:, 0]*= 1e6  # Change energies to eV
    _DOSE_ICRP116 [particle]= data
```

```
[docs]def dose_coefficients(particle, geometry='AP'):
  """Return effective dose conversion coefficients from ICRP-116

  This function provides fluence (and air kerma) to effective dose
conversion
  coefficients for various types of external exposures based on
values in
  `ICRP Publication 116
<https://doi.org/10.1016/j.icrp.2011.10.001>`_.
  Corrected values found in a correigendum are used rather than the
values in
  theoriginal report.

  Parameters
  ----------
  particle : {'neutron', 'photon', 'photon kerma', 'electron',
'positron'}
    Incident particle
  geometry : {'AP', 'PA', 'LLAT', 'RLAT', 'ROT', 'ISO'}
    Irradiation geometry assumed. Refer to ICRP-116 (Section 3.2)
for the
    meaning of the options here.

  Returns
  -------
  energy : numpy.ndarray
    Energies at which dose conversion coefficients are given
  dose_coeffs : numpy.ndarray
    Effective dose coefficients in [pSv cm^2]at provided energies.
For
    'photon kerma', the coefficients are given in [Sv/Gy].

  """
  if not _DOSE_ICRP116:
```

```
    _load_dose_icrp116()

  # Get all data for selected particle
  data = _DOSE_ICRP116.get(particle)
  if data is None:
    raise ValueError(f"{particle} has no effective dose data")

  # Determine index for selected geometry
  if particle in ('neutron', 'photon', 'proton', 'photon kerma'):
    index    =    ('AP',    'PA',    'LLAT',    'RLAT',    'ROT',
'ISO').index(geometry)
  else:
    index = ('AP', 'PA', 'ISO').index(geometry)

  # Pull out energy and dose from table
  energy = data [:, 0].copy()
  dose_coeffs = data [:, index + 1].copy()
  return energy, dose_coeffs
```

LAMPIRAN F

Contoh Tabel Konversi Fluks Dosis ICRP 116

Table A.1. Photons: effective dose per fluence, in units of pSv cm^2, for mono-energetic particles incident in various geometries.

Energy (MeV)	AP	PA	LLAT	RLAT	ROT	ISO
0.01	0.0685	0.0184	0.0189	0.0182	0.0337	0.0288
0.015	0.156	0.0155	0.0416	0.0390	0.0664	0.0560
0.02	0.225	0.0260	0.0655	0.0573	0.0986	0.0812
0.03	0.313	0.0940	0.110	0.0891	0.158	0.127
0.04	0.351	0.161	0.140	0.114	0.199	0.158
0.05	0.370	0.208	0.160	0.133	0.226	0.180
0.06	0.390	0.242	0.177	0.150	0.248	0.199
0.07	0.413	0.271	0.194	0.167	0.273	0.218
0.08	0.444	0.301	0.214	0.185	0.297	0.239
0.1	0.519	0.361	0.259	0.225	0.355	0.287
0.15	0.748	0.541	0.395	0.348	0.528	0.429
0.2	1.00	0.741	0.552	0.492	0.721	0.589
0.3	1.51	1.16	0.888	0.802	1.12	0.932
0.4	2.00	1.57	1.24	1.13	1.52	1.28
0.5	2.47	1.98	1.58	1.45	1.92	1.63
0.511	2.52	2.03	1.62	1.49	1.96	1.67
0.6	2.91	2.38	1.93	1.78	2.30	1.97
0.662	3.17	2.62	2.14	1.98	2.54	2.17
0.8	3.73	3.13	2.59	2.41	3.04	2.62
1.0	4.49	3.83	3.23	3.03	3.72	3.25
1.117	4.90	4.22	3.58	3.37	4.10	3.60
1.33	5.59	4.89	4.20	3.98	4.75	4.20
1.5	6.12	5.39	4.68	4.45	5.24	4.66
2.0	7.48	6.75	5.96	5.70	6.55	5.90
3.0	9.75	9.12	8.21	7.90	8.84	8.08
4.0	11.7	11.2	10.2	9.86	10.8	10.0
5.0	13.4	13.1	12.0	11.7	12.7	11.8
6.0	15.0	15.0	13.7	13.4	14.4	13.5
6.129	15.1	15.2	13.9	13.6	14.6	13.7
8.0	17.8	18.6	17.0	16.6	17.6	16.6
10.0	20.5	22.0	20.1	19.7	20.6	19.6
15.0	26.1	30.3	27.4	27.1	27.7	26.8
20.0	30.8	38.2	34.4	34.4	34.4	33.8
30.0	37.9	51.4	47.4	48.1	46.1	46.1
40.0	43.1	62.0	59.2	60.9	56.0	56.9
50.0	47.1	70.4	69.5	72.2	64.4	66.2
60.0	50.1	76.9	78.3	82.0	71.2	74.1
80.0	54.5	86.6	92.4	97.9	82.0	87.2
100	57.8	93.2	103	110	89.7	97.5
150	63.3	104	121	130	102	116
200	67.3	111	133	143	111	130
300	72.3	119	148	161	121	147
400	75.5	124	158	172	128	159
500	77.5	128	165	180	133	168
600	78.9	131	170	186	136	174
800	80.5	135	178	195	142	185
1000	81.7	138	183	201	145	193
1500	83.8	142	193	212	152	208
2000	85.2	145	198	220	156	218
3000	86.9	148	206	229	161	232
4000	88.1	150	212	235	165	243
5000	88.9	152	216	240	168	251
6000	89.5	153	219	244	170	258
8000	90.2	155	224	251	172	268
10,000	90.7	155	228	255	175	276

AP, antero-posterior; PA, postero-anterior; LLAT, left lateral; RLAT, right lateral; ROT, rotational; ISO, isotropic.

LAMPIRAN G

Program Pengubahan Fluks Ke Dosis

No.	NSUB Name	Sublibrary Name	Short	VII.1	VII.0	VI.8
1	0	Photonuclear	g	163	163	-
2	3	Photo-atomic	photo	100	100	100
3	4	Radioactive decay	decay	3817	3838	979
4	5	Spont. fis. yields	s/fpy	9	9	9
5	6	Atomic relaxation	ard	100	100	100
6	10	Neutron	n	423	393	328
7	11	Neutron fis.yields	n/fpy	31	31	31
8	12	Thermal scattering	tsl	21	20	15
9	19	Standards	std	8	8	8
10	113	Electro-atomic	e	100	100	100
11	10010	Proton	p	48	48	35
12	10020	Deuteron	d	5	5	2
13	10030	Triton	t	3	3	1
14	20030	^{3}He	he3	2	2	1

1. The photonuclear sublibrary was carried over unchanged from ENDF/B-VII.0. It contains evaluated cross sections for 163 materials (all isotopes) mostly up to 140 MeV. The sublibrary was supplied by Los Alamos National Laboratory (LANL) and it is largely based on the IAEA-coordinated collaboration completed in 2000.
2. The photo-atomic sublibrary was taken over from ENDF/B-VII.0=ENDF/B-VI.8. It contains data for photons from 10 eV up to 100 GeV interacting with atoms for 100 materials (all elements). The sublibrary was supplied by Lawrence Livermore National Laboratory (LLNL).
3. The decay data sublibrary has been re-evaluated and considerably improved by the Brookhaven National Laboratory (BNL).

4. The spontaneous fission yields were taken over from ENDF/B-VII.0=ENDF/B-VI.8. The data were supplied by LANL.
5. The atomic relaxation sublibrary was taken over from ENDF/B-VII.0=ENDF/B-VI.8. It contains data for 100 materials (all elements) supplied by LLNL.
6. The neutron reaction sublibrary represents the heart of the ENDF/B-VII.1 library. The sublibrary has been updated and extended, it contains 423 materials, including 422 isotopic and 1 elemental evaluation.. Altogether 234 materials have been changed in ENDF/B-VII.1 compared to ENDF/B-VII.0.
7. Neutron fission yields were reevaluated for ^{239}Pu (fast and 14 MeV) by Los Alamos; others were taken over from ENDF/B-VII.0=ENDF/B-VI.8, having been supplied by LANL.
8. The thermal neutron scattering sublibrary was carried over unchanged from ENDF/B-VII.0. It contains thermal scattering-law data, largely supplied by LANL, with the addition of SiO_2 in ENDF/B-VII.1 from ORNL.
9. The neutron cross section standards sublibrary was carried over from ENDF/B-VII.0 unchanged. Therefore, as for VII.0, the VII.0 standard cross sections were completely adopted by the VII.1 neutron reaction sublibrary except for the thermal cross section for ^{235}U(n,f) where a slight difference occurs to satisfy thermal data testing, and some very small differences for ^{235}U(n,f) and ^{238}U(n,γ) in the keV–MeV region.
10. The electro-atomic sublibrary was taken over from ENDF/B-VII.0=ENDF/B-VI.8. It contains data for 100 materials (all elements) supplied by LLNL.
11. The proton-induced reactions were carried over from ENDF/B-VII.0 with only minor corrections. These were supplied by LANL and the data being mostly to 150 MeV.
12. The deuteron-induced reactions were supplied by LANL, carried over unchanged from ENDF/B-VII.0. This sublibrary contains 5 evaluations.
13. The triton-induced reactions were supplied by LANL, carried over unchanged from ENDF/B-VII.0. This sublibrary contains 3 evaluations.
14. Reactions induced with ^{3}He were supplied by LANL, carried over unchanged from ENDF/B-VII.0. This sublibrary contains 2 evaluations.

https://www.nndc.bnl.gov/endf-b7.1/

https://openmc.org/official-data-libraries/

Lampiran: Histori alur persetujuan

No	Jabatan	Nama	Jenis	Tanggal Disetujui
1	Dosen Pembimbing	Ir. Anung Muharini, M.T., IPM.	Paraf	Selasa, 30 Juli 2024 15:02
2	Ketua Program Studi Sarjana Teknik Nuklir	Dr. Ing. Ir. Sihana	Paraf	Selasa, 30 Juli 2024 16:27
3	Ketua Departemen Teknik Nuklir dan Teknik Fisika	Dr. Ir. Alexander Agung, S.T., M.Sc., IPU.	Tanda Tangan	Selasa, 30 Juli 2024 17:59

Diajukan oleh Sukiyat

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